

Estimating a Process of Health Formation in Older Adults: The Roles of Physical Functioning, Cognition, and Mental Health.

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Abstract

Health in older adults evolves along multiple, interdependent dimensions. This paper develops and estimates a dynamic model of health formation that distinguishes physical, cognitive, and mental domains. The model is estimated using a comprehensive set of health measures from the Health and Retirement Study. The results reveal novel patterns of complementarity across the health domains. Physical health plays a central role, serving as a significant determinant of mental health as well as of employment, health investment, and mortality. Mental health is an important determinant of employment and mortality, but it is found to be more transitory and to have little feedback on the other health domains. The model is used to identify the periods in which targeted health domains have the largest impacts and generate the greatest spillovers across domains. Finally, we map the latent domains to commonly used measures such as self-reported health, clarifying which aspects of health drive variation in these widely used indicators.

1 Introduction

A well-established gradient exists between health and economic well-being near the end of working life and into retirement, with causal relationships operating in both directions (Cutler et al., 2008). However, the

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relative importance of the more nuanced mechanisms driving this relationship is less well understood. Health is a complex, dynamic, and multidimensional concept whose implications for work, retirement, and disability policies vary considerably depending on its manifestation. Furthermore, estimates of the effects of health are sensitive to both the measurements available and the empirical methods used, both of which can vary greatly across empirical studies (French and Jones, 2017; Blundell et al., 2023). Despite extensive research on the causes and consequences of health, there remains no unified consensus on how best to measure and model it in economic studies.

In this paper, we develop a tractable yet flexible model of health formation that captures the complexity of interactions across health conditions by disaggregating health into a sparse set of latent domains. The framework maps a large set of observed health measures into this smaller, interpretable set of domains. We model the dynamic interdependence across the health domains, identifying both own- and cross-complementarities in the formation of future health. The latent domains both shape and are shaped by individual choices, including employment status and health investment, and vary with persistent individual heterogeneity

We apply the model to better understand the determinants of health formation and the role of health in driving economic behavior near the end of working life. The estimated structural parameters provide insights into the mechanisms linking health to labor market outcomes, with implications for policies aimed at reducing health-related disparities as individuals transition out of the labor force. Using these estimates, we characterize average marginal effects across health domains and over the life cycle, revealing when improvements in one domain generate spillovers to others and the extent to which these effects vary across individuals.

We choose the health domains *ex ante* with the intent to capture dimensions of health that are most relevant to work capacity and that have garnered interest in both health and labor studies.¹ The health domains are categorized as physical functioning, cognitive functioning, and mental health. Physical functioning encompasses the ability to perform essential tasks in work and daily life, such as getting out of bed, walking, and lifting objects. Cognitive functioning refers to capabilities related to attention, memory, problem solving, planning, and reasoning. Mental health reflects an individual’s emotional and psychological well-being, influencing how one thinks, feels, and handles stress. Our choice of health domains has an intuitive appeal. First, physical and cognitive health represent two common dimensions of functional capacity with grounded connections to work and retirement decisions (Wen, 2022; Blundell et al., 2023; Millard, 2025). Second, there is an emerging economic literature documenting the considerable consequences of mental health

¹Some recent examples of these studies include Cunha et al. (2010); Lise and Postel-Vinay (2020); Jolivet and Postel-Vinay (2020); Guvenen et al. (2020); Darden (2022); Wen (2022); Blundell et al. (2023); Capatina and Keane (2023); Millard (2025).

challenges in the labor market (Wang et al., 2023). Health deterioration in these dimensions can significantly affect work productivity, financial management, and the navigation of complex medical treatment plans. Further, although recent research has documented the economic consequences of mental health, the economic determinants of its formation remain insufficiently understood.

We estimate our model using data from the Health and Retirement Study (HRS), which provides panel data on a rich set of objective and subjective health measures, as well as relevant demographic and labor market variables. We map the many observed indicators into a smaller, interpretable set of latent health domains and estimate the health formation technology from their joint distribution over time. This approach reduces reliance on any single measure or fixed set of measures and mitigates well-known sources of measurement error that are prevalent in subjective health data (Stern, 1989; Bound, 1991; Blundell et al., 2023). Our model further allows us to characterize the mapping between these latent domains and a set of health measures that are commonly used in economic research such as self-reported health status and the prevalence of chronic conditions. This mapping provides mechanistic insight into the dimensions of health underlying these measures and informs the interpretation of findings from the broader empirical literature that relies on them.

We specify a flexible translog production technology to separately estimate and analyze the formation of physical, cognitive, and mental health at older ages. The fully parameterized model captures two sets of relationships, complementarities across health domains and feedbacks between health and individual behavior. This specification formalizes, for instance, the channels through which physical health affects mental health and vice versa, and identifies which health domains are most consequential for outcomes such as employment and mortality.

The estimated formation process reveals several patterns in the evolution of the three health domains. Cognitive and physical health exhibit strong persistence, with 80–90% of any shock carrying forward each period, suggesting that early deficits in these domains are difficult to reverse. Mental health, by contrast, exhibits substantially weaker persistence, implying that shocks dissipate more quickly and that mental health trajectories more closely revert to their long-run trend. The estimates further reveal an asymmetric pattern of dependence across domains. Physical health is a sizable determinant of mental health, highlighting its role as a precursor to mental well-being. The spillover effects from physical to mental health are positive and nonlinear, with the largest returns accruing to individuals in better physical health, suggesting it is easier to maintain mental well-being when in good physical health. Cognitive health positively affects the production of physical health, while physical health negatively affects cognitive health formation. We provide suggestive

evidence that this pattern reflects substitutability in health-preserving effort. Individual heterogeneity, reflecting an individual’s health productivity type, is positively associated with health preservation across all domains, indicating that certain demographic groups, such as those with higher education or socioeconomic status growing up, are better able to maintain good health as they age. Finally, correlated shocks to health formation generate dependence across domains, with the strongest correlations observed between shocks to mental and cognitive health and between mental and physical health.

The production technology also incorporates individual behaviors. Employment is found to promote health formation, consistent with findings in the related literature (Rohwedder and Willis, 2010; Fitzpatrick and Moore, 2018; Black et al., 2017). A composite measure of healthy investment, which combines smoking, drinking, and exercise, does not directly affect cognitive or mental health. Instead, health investment operates through physical health, with the largest gains accruing to individuals with low physical health stocks.

Using the estimated model, we calculate average marginal effects across health domains to identify the ages at which improvements in health have the largest downstream effects and whether cross-domain spillovers are sufficiently large and concentrated at particular ages. These results inform the targeting and timing of health interventions by indicating when interventions are likely to generate the greatest direct and spillover effects across domains. We highlight three important policy implications. First, health interventions are most effective within their own domain. The most meaningful positive cross-domain spillovers operate from physical to mental health and from cognitive to physical health. Second, the effectiveness of indirect policy channels varies across socioeconomic groups, as spillovers are generally less effective for the economically disadvantaged. Third, the years surrounding retirement represent a critical window for health intervention, as health improvements at this stage are reinforced by labor supply responses, with combined effects peaking around ages 63–65.

Counterfactual shock experiments expand on the marginal effects analysis by tracing how a shock propagates through the model to all future ages. Physical health operates as a central hub in that shocks to physical health propagate broadly across domains, reducing employment and increasing mortality. Shocks to cognitive health generate meaningful spillovers primarily through their effects on physical health, while shocks to mental health are more localized and transitory, with comparatively smaller downstream consequences. From a modeling perspective, the dependence between cognitive and physical health as well as the dependence of mental health on physical health are of first-order importance. By contrast, the effects of mental health on the other domains are of secondary importance for capturing the overall dependence structure.

Finally, we use the framework to map latent health domains to commonly used health measures, such

as self-reported health and objective health conditions. Self-reported health is significantly generated by a combination of latent factors, with physical health as the primary determinant but mental and cognitive health also contributing meaningfully. Decomposing self-reported health in this way enables more precise identification of the health domains underlying observed relationships between health and economic outcomes in studies that measure health using self-reports. For example, two individuals may report the same overall health status, yet one may suffer from poor physical health while the other from poor mental health. The estimated effect of self-reported health on any outcome will then reflect a weighted average across dimensions, possibly attenuating the effect of the relevant component. An analogous decomposition for objective health conditions characterizes which latent domains drive variation across categories of chronic disease. Physical health is the dominant factor for most chronic conditions, while mental health emerges as the primary domain for psychiatric conditions. Such decompositions inform researchers about the mechanisms underlying estimated effects of specific conditions and aid interpretation across studies that rely on different condition-based measures.

This paper makes several novel contributions to the related literature. We fit into a large literature aimed at better understanding how individuals arrive at older ages with mental, cognitive, and physical health and the consequences of this for their economic behavior. We frame our contribution in four broad areas.

First, analyzing micro-level relationships between health and economic behavior is inherently complex due to challenges in measuring health itself.² Our paper presents a comprehensive representation of health that reduces the dependence on a single or small set of health indicators. The health economics literature has long explored subjective measures (Butler et al., 1987; Benítez-Silva et al., 2004; French, 2005; Kreider and Pepper, 2007; Meyer and Mok, 2019), objective measures (Bartel and Taubman, 1979; Bound, 1989; Smith, 2004), and their combinations (Stern, 1989; Blundell et al., 2023). Consequently, the estimated effects of health on employment and related outcomes vary across studies, depending on which measures are used or available. Our framework accommodates both subjective and objective measures, yielding a tractable yet flexible representation of individual health.

More specifically, our approach aligns with studies that employ dynamic factor models to condense a large number of measures into a more sparse and interpretable set of latent variables (Cunha et al., 2010; Bound et al., 2010; Iskhakov, 2010; Poterba et al., 2017; Cunha et al., 2021; Blundell et al., 2023). Numerous studies use unidimensional health indices to investigate health-related disparities, behaviors, or predictive outcomes (French, 2005; Bound et al., 2010; Hosseini et al., 2022, 2021; Danesh et al., 2024; De Nardi et al., 2024).

²For further discussion, see Currie and Madrian (1999), O'Donnell et al. (2015), French and Jones (2017), and Blundell et al. (2023).

While these measures have proven useful, such approaches are limited in isolating the specific dimensions of health that drive observed effects because they compress all heterogeneity into a single dimension.³ Our model fits among more recent multidimensional frameworks (Conti et al., 2010; Amengual et al., 2021; Wen, 2022; Cozzi et al., 2024) but advances this literature by modeling health as an evolving, richly structured latent process with explicitly dynamic interdependence across the dimensions of health.

Second, this study addresses a notable gap in the economic literature concerning the endogenous formation of health. A large body of research in health and retirement treats health formation as exogenous to individual decision making (French and Jones, 2017).⁴ More similar to our framework are studies employing dynamic models of health, where health formation depends on individual decisions and state variables (Grossman, 1972; Gilleskie, 1998; Yogo, 2016; Michaud and Wiczer, 2018; Strulik, 2022; Darden, 2022). We advance this line of research by allowing the influence of economic factors on health to vary between its domains and modeling the interdependencies across the domains. For example, we model the response of mental health to the deterioration of physical health, thereby offering a more comprehensive analysis of the interrelationships within an individual’s overall health. Additionally, by modeling the cross-dependencies among health domains, our framework mitigates an omitted-variable bias present when narrowing one’s focus to estimate the effect of one domain in isolation.

Third, we provide interpretation of the mechanisms through which commonly used health measures and specific conditions affect economic outcomes. A large literature studies the economic consequences of health and health shocks at older ages using diverse indicators and specific diagnoses.⁵ Our framework isolates and compares the contributions of distinct latent dimensions of health that underlie these measures. In particular, we project self-reported health and a set of objective diagnoses (e.g., cancer, diabetes) onto the latent domains, yielding intuition about which health domains drive their estimated effects. This helps clarify how concrete health events translate into economic behavior through changes in the multidimensional latent health vector.

Finally, this study advances our understanding of the economic determinants and implications of mental health. Previous research has examined the influence of health dynamics on economic inequalities

³Univariate measures (e.g., self-reported health) have documented empirical content in that they correlate with objective outcomes, such as mortality, and covary with economic behaviors, such as employment, in expected ways. They are also widely available in survey data and relatively less costly to implement in structural models (De Nardi et al., 2024).

⁴This simplification is often justified on the grounds that, for older individuals, who generally have established health histories, long-standing habits, and broad access to public health insurance, health is largely predetermined. Moreover, studies that adopt this assumption often report small behavioral effects (De Nardi et al., 2016). However, this assumption may be overly restrictive for certain aspects of health, particularly mental health, which might be more sensitive to changing economic conditions and individual choices. For instance, continued work beyond the traditional retirement age may impose psychic costs that deteriorate mental health.

⁵For instance, see Bartel and Taubman (1979), Kahn (1998), Moran et al. (2011), and Stephens et al. (2018).

over the life cycle; however, the endogenous nature of mental health and its interactions with other health dimensions are still not well understood. This paper jointly examines both the determinants and effects of mental health within a unified framework. Mental health is increasingly recognized as a critical factor that influences economic outcomes, particularly in the labor market (Jolivet and Postel-Vinay, 2020; Biasi et al., 2021). Our analysis complements this perspective by investigating the economic environments that aggravate mental health challenges (Adhvaryu et al., 2019; Frank and Glied, 2023). By explicitly modeling endogenous interactions between mental health, physical health, and economic circumstances, our study uncovers mechanisms that have been underexplored in the economics literature, shedding new light on how these dynamics jointly shape economic and health trajectories in later life.

The remainder of the paper is structured as follows. Section 2 details the empirical model, estimation, and identification. Section 3 describes the data used for the analysis. Section 4 reviews the main estimation results. Section 5 conducts counterfactual exercises using the model. Finally, Section 6 concludes.

2 Estimating the Technology of Health Formation

This section describes the framework for estimating a process of health formation near the end of working life. The health formation process described below follows from a standard dynamic model of health, consumption, and employment as outlined in Section A of the Appendix. For brevity, we suppress the individual subscript i throughout the model description, as all variables are defined at the individual level. At each age, t , health is represented as a vector of stocks, $x_t = (x_t^p, x_t^c, x_t^{mh})'$, where the superscripts $\{p, c, mh\}$ denote physical, cognitive, and mental health, respectively. A lower stock of health in any given dimension corresponds to “worse” health, for instance the presence of more severe physical limitations.

At first observation ($t = 0$), individuals are endowed with an initial stock of health, $x_0 = (x_0^p, x_0^c, x_0^{mh})'$. Individuals are modeled starting at the age of fifty-five, and their health at that time is the product of all previous behavior, early health shocks, and economic factors.⁶ Initial conditions depend on one’s latent heterogeneity, $b_0 \in \mathbb{R}$, which summarizes the accumulation of factors that affect the formation of one’s health from birth until the age at which they are first observed. Latent heterogeneity reflects factors that have been shown to be important determinants of health formation throughout the life cycle, such as education, childhood health, and socioeconomic status (Currie and Moretti, 2003; Case et al., 2005; Currie, 2009; Conti et al., 2010; Currie et al., 2010; Case and Paxson, 2010; Lundborg et al., 2014; Almond et al., 2018; Adhvaryu

⁶As such, the initial endowment of health is the result of an unobserved formation process that occurs before the individual is first observed in the data. This is the initial conditions problem as described by Heckman (1991) and Wooldridge (2005).

et al., 2019; De Nardi et al., 2024).

The stock of each distinct health domain evolves dynamically according to a specified production technology, which is a function of the previous period's health stocks, health inputs, and labor market decisions. With three dimensions of health, the system is described by state-space equations, where $t \in \{1, \dots, T\}$, denotes the discrete time index over which health dynamics are modeled. The dynamic formation of each health domain, x_t^k , is determined by

$$x_t^k = f_t^k(x_{t-1}, I_{t-1}, L_{t-1}, \eta_t^k). \quad (1)$$

The function f_t^k characterizes the health formation technology for health domain $k \in \{p, c, mh\}$, revealing how previous-periods health and decisions affect the current health stock. For each k , x_t^k is determined by $x_{t-1} = (x_{t-1}^p, x_{t-1}^c, x_{t-1}^{mh})'$, capturing both self- and cross-dependencies in health formation across the latent health domains. We assume that serial dependence in health formation follows a first-order Markov process.

The transition process depends on the previous period's investment into health I_{t-1} , which encompasses both healthy behaviors, such as exercise, and unhealthy behaviors, such as smoking. We summarize the set of health investment behaviors using a single variable I_t . Healthy behaviors represent deliberate choices made by individuals based on their current health status. We model health investment decisions as a function of latent health and initial heterogeneity,

$$I_t = I(x_t, b_0, t, W_t^I, \epsilon_t^I). \quad (2)$$

Additional controls, W_t^I , encompass observable exogenous determinants of health investment decisions and ϵ_t^I accounts for unobserved determinants of health investment.

Similarly, health transitions depend on the previous period's employment, L_{t-1} . On the one hand, employment can negatively impact health, especially when individuals are subjected to physical stressors, hazardous substances, or psychological pressures. On the other hand, the rate of health decline has been found to increase when exiting the labor force (Fitzpatrick and Moore, 2018; Black et al., 2018). We model employment as

$$L_t = L(x_t, b_0, t, W_t^L, \epsilon_t^L), \quad (3)$$

where the value of employment depends on the stocks of latent health, x_t , age, and latent heterogeneity, b_0 .

Furthermore, additional controls, W_t^L , encompass observable unmodeled determinants of employment and ϵ_t^L captures unobserved determinants of employment.

Finally, η_t^k , captures unobserved determinants of health production. In this model, an initial conditions problem arises as x_t depends on x_0 through successive substitution for x_{t-1} in Equation (1). Hence, there may be unobserved factors in η_t^k (i.e., relevant for determining x_t) that are correlated with x_{t-1} through x_0 . Moreover, unobserved factors in η_t^k may be correlated with individuals' decisions: work (L_{t-1}) and health investment (I_t). Ignoring this dependence introduces an endogeneity problem in the right-hand side of equation (1). To address this, we express the unobservable factors in equation (1) as

$$\eta_t^k = \eta^k(b_0, \epsilon_t^k), \quad k \in \{p, c, m\}. \quad (4)$$

The idea is that b_0 , which can be thought of as capturing an individual's health productivity type, summarizes the relevant factors that jointly affect initial conditions, individual's choices, and the productivity of health formation. Under the structural assumptions of the model, controlling for b_0 addresses the endogeneity problem. This structure allows for serial dependence in unobservables of the formation technology and is consistent with a fuller model of optimizing behavior.⁷

The idiosyncratic shocks to health formation, $\epsilon_t = (\epsilon_t^p, \epsilon_t^c, \epsilon_t^{mh})$, are potentially correlated with each other but are assumed to be exogenous to all other variables in equation (1). These shocks capture idiosyncratic factors that can affect several health domains simultaneously. For instance, an injury from an accident may have a physical and mental toll.

Finally, we examine how latent health stocks affect mortality risk. Mortality is modeled using a discrete-time hazard function that specifies the probability an individual dies in period t conditional on survival through period $t - 1$,

$$H(d_t = 1 | d_{t-1} = 0, x_t, b_0, t). \quad (5)$$

The hazard is generated by a latent-index model,

$$d_t^* = d(x_t, b_0, t) + \epsilon_t^d, \quad (6)$$

⁷The validity of this argument follows from the solution to the theoretical life-cycle consumption and labor supply model described in Section A of the Appendix. This model allows for latent heterogeneity in health formation, preferences, and productivity, all of which may be correlated. The model illustrates how health formation depends on endogenous health investment and employment decisions operating through b_0 .

where $d_t = 1$ if $d_t^* > 0$ and $d_t = 0$ otherwise.

Mortality is also of central policy relevance. It has direct welfare implications, underpins social insurance programs, and drives public expenditures through pensions, health care, and disability insurance. In dynamic models of health, death represents the absorbing state of health capital depreciation (à la Grossman (1972)), making it a natural outcome to study when modeling health trajectories. By linking mortality risk to specific latent components, our framework clarifies which dimensions of health drive mortality differences and, in turn, is useful to inform policies aimed at prevention, risk adjustment, and targeting.

The idiosyncratic shocks are allowed to be freely correlated across all structural equations within a period t . This captures the possibility that a single idiosyncratic event simultaneously affects multiple outcomes within a period, operating through channels beyond its effect on the latent health domains. For example, contracting an illness may force an individual to miss work, directly disrupting employment and health investment decisions in that period, even holding latent health fixed. Similarly, an accident may affect mortality risk while also generating contemporaneous disruptions to work and health-related behavior.

2.1 Measurement System

The primary challenge in estimating equation (1) arises from the latent nature of the underlying health domains. Instead, we observe a set of imperfect measures of health. Our approach builds on a substantial body of literature related to the estimation of production functions for latent skills that develops methodologies to estimate dynamic factor models (Cunha et al., 2010; Attanasio et al., 2020; Cunha et al., 2021; Del Bono et al., 2022; Agostinelli and Wiswall, 2025).

We assume that the measurement system has a dedicated factor structure so that, for each latent health domain, we have a set of measures that are generated exclusively by that domain. For each $k \in \{p, c, mh\}$ and period $t \in \{0, \dots, T_i\}$, we observe M^k noisy measures, $z_{t,m}^k$, $m \in \{1, \dots, M^k\}$, generated by

$$z_{t,m}^k = Z_m^k(x_t^k, v_{t,m}^k)$$

where $v_{t,m}^k$ is i.i.d. measurement error. Further, we assume the parameters of the measurement system are age-invariant (Agostinelli and Wiswall, 2025).⁸

⁸The health measures are explicitly designed to be administered consistently across survey waves, allowing researchers to track changes in health as individuals age. This design underlies the assumption that two individuals with the same underlying health vector, but differing in age, will on average record the same measured health. Similar assumptions appear in related work. For example, Bound et al. (2010) estimate a multi-indicator latent health factor and test whether factor loadings vary with age. Their results cannot reject equality of loadings across ages 50–70, leading them to impose constant loadings and treat the resulting health index as age-invariant.

Typically, data on health outcomes encompass three distinct types of measurements: continuous, categorical, and binary.

1. For continuous measures, we adhere to the prevailing literature and assume a semi-log linear mapping (Attanasio et al., 2020; Del Bono et al., 2022; Agostinelli and Wiswall, 2025). Continuous measure $z_{t,m}^k$ is generated by

$$z_{t,m}^k = \mu_m^k + \lambda_m^k \ln x_t^k + v_{t,m}^k.$$

The measurement parameters μ_m^k and λ_m^k indicate location and scale, respectively, while $\sigma_{v_m^k}^2$ denotes the variance of the measurement error.

2. For categorical measures (ordered), we assume they are generated by a latent index

$$\begin{aligned} z_{t,m}^{k*} &= \lambda_m^k \ln x_t^k + v_{t,m}^k, \\ z_{t,m}^k &= j \text{ if } \tau_{m,j-1}^k < z_{t,m}^{k*} \leq \tau_{m,j}^k. \end{aligned}$$

We normalize $\tau_{m,0}^k = -\infty$ and $\tau_{m,J}^k = \infty$, and the remaining thresholds on the interior and the scale parameter, λ_m^k , are estimated.

3. For binary measures, we assume they are generated by

$$z_{t,m}^k = \begin{cases} 1 & \text{if } z_{t,m}^{k*} = \lambda_m^k \ln x_t^k + v_{t,m}^k > \tau_m^k, \\ 0 & \text{otherwise} \end{cases}$$

where $z_{t,m}^{k*}$ is a latent index and the threshold, τ_m^k , and scale parameter, λ_m^k , are estimated.

Moreover, we have a set of observed measures, z_m^b , $m \in \{1, \dots, M^b\}$, which we assume are exclusively generated by latent initial heterogeneity, b_0 . These measures are based on individual characteristics prior to model entry, including their education and childhood health.

Last, we include additional measures that may be generated from a combination of the full set of latent health domains, encompassing both objective chronic health conditions and self-reported health. These measures may fall into any of the three types described earlier. A comprehensive description of the complete set of health measures is provided in Section 3 below.

2.2 Implementation

The model described above is written in general form. To bring the model to the data, we impose structure on the functional forms of the health formation process and the other behavioral equations in the model.

Initial Conditions

As individuals are first observed at age 55 and substantial health dynamics occur prior to model entry, we allow initial conditions to be jointly distributed. Let $w_0 = (\ln x_0^p, \ln x_0^c, \ln x_0^{mh}, b_0)'$ denote the vector of initial conditions. We approximate the distribution of w_0 using a two-component Gaussian mixture.⁹ Let F_{w_0} denote the joint distribution of all log latent factors in the model in period $t = 0$. Then,

$$F_{w_0} = \pi \Phi(\omega^1, \Sigma^1) + (1 - \pi) \Phi(\omega^2, \Sigma^2), \quad (7)$$

where $\pi \in (0, 1)$ is a common mixture weight, $\Phi(\omega^j, \Sigma^j)$ is a CDF of a normal distribution with mean vector ω^j and Σ^j is a 4x4 covariance matrix.

The location and scale of latent factors is not identified without further normalization on either F_{w_0} or the measurement system.¹⁰ First, we assume the vector of initial conditions is mean zero, $E(w_0) = 0$. Second, we assume that the diagonal of the unconditional variance of each latent factor is one, $Var(\ln x_0^k) = 1$, $k \in \{p, c, mh\}$ and $Var(\ln b_0) = 1$. These normalizations anchor the location and scale of the latent variables and eliminate observational equivalence between latent variances and measurement loadings. The component-specific means and component-specific variances are free parameters to be estimated. The off-diagonal elements of the component-specific covariance matrices are left unrestricted and capture cross-domain dependence at baseline, subject to ensuring positive definiteness.

Health Formation Technology

We consider a flexible translog specification of the health production technology, which allows for nonlinearities, interactions across health domains, and dynamic complementarities in health production. In addition, the translog specification has desirable properties for identifying treatment effects in dynamic factor models of the type considered here (Del Bono et al., 2022). Define the vector $w_t = (\ln x_t^p, \ln x_t^c, \ln x_t^{mh}, b_0, I_t)'$.

⁹Gaussian mixtures provide a flexible approximation to otherwise unrestricted distributions and are widely used in the empirical literature employing latent factor and dynamic structural models (Cunha et al., 2010; Attanasio et al., 2020; Del Bono et al., 2022; Agostinelli and Wiswall, 2025)

¹⁰An alternate approach can be to normalize parameters of the measurement system, as in Cunha et al. (2010).

Then, the transition equation for each health domain k is given by

$$\ln x_{t+1}^k = a_0^k + w_t' \beta^k + w_t' B^k w_t + c_e^k L_t + c_t^k t + \epsilon_{t+1}^k \quad (8)$$

where $\beta^k \in \mathbb{R}^5$ captures first order effects of the inputs, and $B^k \in \mathbb{R}^5 \times \mathbb{R}^5$ is a symmetric, positive definite matrix governing second-order effects from a set of interactions and quadratic terms.

Employment, Mortality, and Health Investment

We use a nonstructural approximating model for the health investment and employment policy functions. The employment decision is a discrete choice that is modeled using a latent variable framework. The latent value of work, L_t^* , is assumed linear in its arguments,

$$\begin{aligned} L_t^* &= \beta_0^L + \beta_1^L \ln x_t^p + \beta_2^L \ln x_t^c + \beta_3^L \ln x_t^{mh} + \beta_4^L b_0 + \beta_5^L t + \beta_6^L W^L + \epsilon_t^L, \\ \epsilon_t^L &\sim \mathcal{N}(0, 1), \\ L_t &= \mathbb{1}(L_t^* \geq 0). \end{aligned} \quad (9)$$

The value of employment varies by health status. For instance, health can makes it more costly to perform productive tasks, or lower productivity and the subsequent returns to working. Additional controls in W^L include total non-housing wealth, accumulated years of labor market experience, and marital status at first observation, as in Blundell et al. (2023).

Health investment decisions, I_t , reflect choices made conditional on health. The policy function for health investment is assumed linear in the health domains with

$$\begin{aligned} I_t &= \beta_0^I + \beta_1^I \ln x_t^p + \beta_2^I \ln x_t^c + \beta_3^I \ln x_t^{mh} + \beta_4^I b_0 + \beta_5^I t + \beta_6^I W_t^I + \epsilon_t^I, \\ \epsilon_t^I &\sim \mathcal{N}(0, \sigma_I^2). \end{aligned} \quad (10)$$

The signs of β_1^I , β_2^I , and β_3^I are unrestricted. For instance, poor physical health may reduce investment because it becomes more difficult for individuals to engage in health-producing activities, or it may increase investment if individuals respond by investing more to improve their health. Additional exogenous controls in W^I include total non-housing wealth and marital status at first observation.

In a similar manner, for individuals that have survived to $t - 1$, the probability they die in period t

($d_t = 1$) is determined by a latent variable, d_t^* , where

$$\begin{aligned} d_t^* &= \beta_0^d + \beta_1^d \ln x_t^p + \beta_2^d \ln x_t^c + \beta_3^d \ln x_t^{mh} + \beta_4^d b_0 + \beta_5^d t + \beta_6^d t^2 + \epsilon_t^d, \\ \epsilon_t^d &\sim \mathcal{N}(0, 1), \\ d_{it} &= \mathbb{1}\{d_t^* \geq 0\}. \end{aligned} \tag{11}$$

Finally, we estimate the model using an unbalanced panel, where sample attrition is prevalent. We model attrition as endogenous to one's health.¹¹ We denote the value of attriting from the sample in period t as a_t^* . Then, we model the decision to attrit in period t ($a_{it} = 1$), conditional on not having attrited earlier, as

$$\begin{aligned} a_t^* &= \beta_0^a + \beta_1^a \ln x_t^p + \beta_2^a \ln x_t^c + \beta_3^a \ln x_t^{mh} + \beta_4^a b_0 + \beta_5^a t + \epsilon_t^a, \\ \epsilon_t^a &\sim \mathcal{N}(0, 1), \\ a_{it} &= \mathbb{1}\{a_t^* \geq 0\}. \end{aligned} \tag{12}$$

Moreover, we accommodate temporary gaps and right-censoring in outcomes, treating all censoring as independent of the underlying process (uninformative).

Finally, Σ_ϵ denotes the covariance matrix of the idiosyncratic shocks, $(\epsilon_t^p, \epsilon_t^c, \epsilon_t^{mh}, \epsilon_t^I, \epsilon_t^L, \epsilon_t^d, \epsilon_t^a)$. The off diagonal elements, σ_{jk}^ϵ , denote the covariance of the idiosyncratic component determining outcome j and k .

2.3 Identification of Health Formation Technology

Identifying and estimating the health formation function is challenging, as both inputs and outputs are observed only through noisy proxies. Inputs may be endogenous, and unobserved components in the input equations may be correlated with unobservables in the technology function. To achieve identification of the model, we impose additional restrictions on the measurement system that follow common practice in the related literature (Cunha et al., 2021).

We impose the following restrictions on measurement errors, $v_{t,m}^k$, where k indexes health domains, t indexes time, and m indexes observed measures.

(V1) $v_{t,m}^k$ are normally distributed with mean zero for all $k \in \{p, c, mh, b\}$, $t \in \{1, \dots, T_i\}$, and $m \in \{1, \dots, M^k\}$. The variance of measurement errors for discrete measures is normalized to one.

¹¹For analysis on attrition in the HRS, see Banks et al. (2011).

(V2) $v_{t,m}^k$ is independent of $(x_\tau^p, x_\tau^c, x_\tau^{mh}, b_0)$ for all $t, \tau \in \{1, \dots, T_i\}$; $m \in \{1, \dots, M^k\}$; and $k \in \{p, c, mh, b\}$.

(V3) $v_{t,m}^k$ is independent of $v_{\tau,n}^l$ whenever $(k, m, t) \neq (l, n, \tau)$, for $t, \tau \in \{1, \dots, T_i\}$; $m \in \{1, \dots, M^k\}$; $n \in \{1, \dots, M^l\}$; and $k, l \in \{p, c, mh, b\}$.

Let θ denote the parameters to identify, which involve the distributional parameters of initial conditions $\{\omega^1, \Sigma^1, \omega^2, \Sigma^2, \pi\}$, the parameters of the measurement system $\{\mu_m^k, \lambda_m^k, \tau_{j,m}^k, \sigma_{v_m^k}\}_{k \in \{p, c, mh, b\}, m \in \{1, \dots, M_k\}}$, the parameters of the transmission process $\{a^k, \beta^k, B^k, c_e^k, c_t^k, \sigma_{\epsilon^k}^2\}_{k \in \{p, mh, c\}}$, parameters of the nonstructural equations for employment, mortality, attrition, and healthy behaviors, $\{\beta_j^I, \beta_j^L, \beta_j^d\}_{j=1, \dots, 6}$, $\{\beta_j^a\}_{j=1, \dots, 5}$, and parameters of the covariance matrix for idiosyncratic shocks $\{\sigma_{jk}^\epsilon\}_{j,k=(\epsilon^p, \epsilon^c, \epsilon^{mh}, d, a, e, I, j \neq k)}$. Identification of the model's parameters is described in two steps. First, we identify the distribution of the initial endowments and parameters of the measurement system. Given these, the second step recovers the remaining model parameters.

Initial Distributional Parameters and Measurement System

In the first step, we identify the joint distribution of initial latent endowments and the parameters of the measurement system using observations in the first period. Since measurement parameters are assumed age-invariant, the first observed period contains sufficient information to identify them. Discrete measures make up the majority of our measurement system, so we establish identification for the case in which the domains are measured exclusively by binary measures, which is the most restrictive case.¹²

Consider a single latent factor, x_0^k , $k \in \{p, c, mh, b\}$, that follows a two-component Gaussian mixture

$$\ln x_0^k \sim \pi \mathcal{N}(\omega_k^1, s_k^1) + (1 - \pi) \mathcal{N}(\omega_k^2, s_k^2).$$

In addition to the measurement-error restrictions (V1)–(V3), we impose:

(C1) *Relevance*. $\lambda_m^k \neq 0$ for all m , so every proxy is informative about the factor.

(C2) *Distinctness*. The pairs (λ_m^k, τ_m^k) are distinct across m , so no two measurement functions coincide.

(C3) *Sign normalization*. $\lambda_1^k > 0$, fixing the orientation of the latent state.

(C4) *Label order*. $\omega_k^1 < \omega_k^2$, resolving the label-swap ambiguity inherent to mixtures.

¹²Binary proxies reveal only whether a factor lies above or below a threshold, so each proxy is coarse and contains much less information than continuous measures. The majority of the related literature shows identification of such models using continuous measure. For instance, see Cunha and Heckman (2008); Cunha et al. (2010); Agostinelli and Wiswall (2025).

(C5) *Non-degeneracy.* $\pi \in (0, 1)$ and $(\omega_k^1, s_k^1) \neq (\omega_k^2, s_k^2)$, so the two components are genuinely distinct; otherwise the mixture collapses to a single Gaussian and the component-specific parameters are not separately recoverable.

The normalization restrictions, $E(\ln x_0^k) = 0$ and $Var(\ln x_0^k) = 1$, leave three free mixture parameters, (ω_k^1, s_k^1, π) . Suppose x_0^k is measured by $M \geq 4$ binary indicators generated exclusively by that factor. This sub-problem is to identify $2M$ parameters of the binary measures (scales and thresholds) and three distributional parameters, providing $2M + 3$ free parameters. Under conditional independence of measurement errors, all dependence across the M measures arises from the shared latent variable, $\ln x_0^k$.

The set of marginal probabilities, $Pr(z_{0,m}^k = 1)$ provide M moments linking the thresholds and mixture parameters, while the pairwise joint probabilities, $Pr(z_{0,m}^k = 1, z_{0,l}^k = 1)$, $m \neq l$, give an additional $\binom{M}{2}$ moments, and the trivariate probabilities $Pr(z_{0,m}^k = z_{0,l}^k = z_{0,j}^k = 1)$ provide $\binom{M}{3}$ moments.¹³ Identification exploits the fact that, for the pairwise joint probabilities of latent health domain k , the within-component pairwise correlation, ρ_{jl}^{ck} factorizes as $\rho_{jl}^{ck} = \varphi_j^{ck} \cdot \varphi_l^{ck}$, where φ_m^{ck} is the signal-to-noise ratio of measure m determined jointly by its loading and the mixture component variance.¹⁴

The marginal, bivariate, and trivariate moments constitute a system of $M + \binom{M}{2} + \binom{M}{3}$ equations in the $2M + 3$ unknowns. The order condition holds when $M \geq 4$ and identification follows from the argument below. By establishing the Jacobian matrix of the moment map has full rank at some parameter value, the Inverse Function Theorem guarantees that the moment map is locally invertible near that point, so the parameters are locally identified there. However, verifying full rank over the entire parameters space is analytically intractable. We circumvent this difficulty by establishing our Jacobian matrix is in the class of *real-analytic* functions. If a nonzero real-analytic function (such as the determinant of a Jacobian) is non-zero at one point, then its zero-set is a thin “exceptional” set of Lebesgue measure zero.¹⁵ We obtain generic identification via analyticity, conditions (C1)-(C5) remove the known rank-deficient areas of the domain, and we verify full rank of the Jacobian at the estimated parameter value to show we are not within the rank-deficient set.

The strategy of establishing generic local identification by combining a Jacobian rank condition with the real-analyticity of the moment map has precedent in econometrics. McManus (1992) is the closest in spirit

¹³Fourth-order moments provide additional identifying information but are unused in our argument, as moments of order one through three are sufficient.

¹⁴Specifically, $\varphi_m^{ck} = \lambda_m^k s_k^c / \sqrt{(\lambda_m^k)^2 (s_k^c)^2 + 1}$, where s_k^c is the standard deviation of factor k in mixture component c . The factorization $\rho_{jl}^{ck} = \varphi_j^{ck} \cdot \varphi_l^{ck}$ breaks within-component correlation into components depending only on j and l respectively. Since measurement errors are independent across indicators, all pairwise co-movement traces back to the shared latent factor, so the pairwise probabilities jointly pin down the loadings and within-component variances.

¹⁵A formal definition of real analyticity and a proof of generic identification are provided in Appendix B.1.

to what we do. He asks directly how “common” identification is in nonlinear parametric models and shows that, for broad analytic classes, identification holds generically, with failure confined to a thin (measure-zero) set. This is exactly the logical form of our claim. Identification is not verified parameter-by-parameter but established everywhere except on a negligible exceptional set ruled out by (C1)–(C5). Fox et al. (2012) prove that the identified set has Lebesgue measure one because the relevant mapping is real-analytic, so that non-identification is confined to a measure-zero set. Fox and Gandhi (2016) use real-analyticity in a related way. Komunjer and Ng (2011) and Qu and Tkachenko (2012) establish local identification from the rank of a finite-dimensional Jacobian of a moment or observational-equivalence system, but do not explicitly evoke our analyticity argument. These papers evaluate the rank condition at a point and either assume regularity or that the rank of the relevant Jacobian remains constant in a neighborhood of the point at which identification is established. We supply an argument that the real-analyticity of the determinant of the Jacobian matrix is what justifies inferring genericity from a single-point rank check, rather than positing constant rank as a regularity condition.

In our application, some latent factors are measured using a combination of binary, categorical, and continuous indicators. Categorical measures are treated as ordered threshold crossings and can be decomposed into binary comparisons, so the same identification logic applies. Once the mixture distribution of the latent factor is identified using discrete measures, parameters of continuous measures (locations, loadings, and error variances) are identified from means, variances, and covariances of the observed outcomes.

Finally, cross-domain co-movement in observed measures identifies the off-diagonal elements of the component covariance matrices Σ^j . Since measurement errors are assumed independent across domains conditional on w_0 , any cross-domain correlation in observed proxies is attributable to dependence between latent factors. With a sufficiently rich baseline measurement system, the full joint distribution of initial endowments and measures is therefore identified. Further detail is provided in Appendix B.

Identification of the Transition Process and Remaining Model Parameters

Conditional on the parameters of the measurement system and the joint distribution of initial latent endowments identified in Step 1, the model implies a well-defined joint distribution of latent health stocks and heterogeneity in every period. Because the measurement system is known and measurement errors are independent of the latent states, the evolution of the observed measures over time reflects only the transition dynamics of the latent factors and the stochastic shocks governing those transitions.

To exploit this structure, we summarize the information contained in the observed binary measures

using principal components constructed separately for each latent health domain, $k \in \{p, c, mh\}$, and for the time-invariant heterogeneity term, b_0 , using only measures that load exclusively on each domain. Principal component analysis (PCA) provides a simple approach for summarizing the shared variation within a large set of potentially highly correlated measures. Given the identified measurement thresholds and principal component analysis (PCA) weights, the first principal component, PC_t^k , provides a auxiliary statistic whose conditional expectation is a strictly increasing function of the corresponding latent factor,

$$E(PC_t^k | x_t^k) = h(x_t^k).$$

Although the mapping from the latent factor to the index is generally nonlinear due to the discrete proxies in the measurement system, the index preserves the ordering and dependence of the latent state and therefore serves as a monotone proxy for $\ln x_t^k$.¹⁶

The transition process is identified using an auxiliary regression that characterizes the conditional mean of PC_{t+1}^k as a function of current PCA indices and state variables, estimated jointly across domains. For each domain k , define the auxiliary transition equation

$$PC_{t+1}^k = \tilde{a}_0^k + \tilde{w}_t^\top \tilde{\beta}^k + \frac{1}{2} \tilde{w}_t^\top \tilde{B}^k \tilde{w}_t + \tilde{c}_e^k e_t + \tilde{\nu}_{t+1}^k,$$

where $\tilde{w}_t = (PC_t^p, PC_t^c, PC_t^{mh}, PC_0^b, I_t)'$. Estimation of this auxiliary model yields auxiliary coefficients $\beta^A(\theta)$ that depend on the model parameters governing the latent transition process, θ . Model parameters governing the location, persistence, and innovation variance affect distinct observable features of the PCA indices; their mean, serial correlation, and dispersion, so that different parameter values generate different auxiliary coefficients.

Further, as several latent health domains are observed exclusively through discrete proxies, we require additional moments to separate the latent component mean and variance. To illustrate, for latent component $\ln x_0^k$, suppose the first step estimated the threshold parameters as $\tau_m^k < 0$, $\forall m \in \{1, \dots, M_k\}$, implying $Pr(z_{0,m}^k = 1) > 0.5$, $\forall m \in \{1, \dots, M_k\}$, which is the case for $k = mh$ and $k = p$ in our data. Then, a decline in $Pr(z_{t,m}^k = 1)$ over t in the observed data can be equally explained by a declining mean or an increasing variance of $\ln x_t^k$. To separately identify the dynamics of latent factor means and variances, we supplement

¹⁶Each binary measure, z_m^k , is a noisy thresholded signal of $\ln x_0^k$. As $\ln x_0^k$ increases, the probability that $z_m^k = 1$ also increases, implying that the expected number of positive indicators rises with the latent factor. Since the covariance among the indicators is generated by their shared dependence on $\ln x_0^k$, the first principal component extracts this common source of variation. Consequently, the first principal component behaves as a monotone transformation of $\ln x_0^k$, preserving the ordering of the latent factor.

the auxiliary model to match moments of the mean and variance of PC_t^k at each t , and the proportion of PC_t^k below the thresholds of the lower quartile of the distribution of PC_0^k . These moments discipline the mean and dispersion of latent health domains separately.

Identification follows if the mapping from model parameters, θ to auxiliary coefficients, β^A , is injective:

$$\beta^A(\theta_1) = \beta^A(\theta_2) \Rightarrow \theta_1 = \theta_2.$$

In practice, a local version of this condition can be verified numerically by checking the Jacobian $\partial\beta^A(\theta)/\partial\theta$ has full rank, following the indirect-inference framework of Gourieroux et al. (1993).

Next, conditional on the identified latent dynamics, the remaining behavioral equations (health investment, employment, mortality, and attrition) are identified from their conditional relationships with the PCA indices and observed covariates, because the indices are known monotone transformations of the latent health factors.

Finally, the covariance matrix of idiosyncratic shocks, Σ_ϵ , is identified using the estimated residuals of the auxiliary regressions described above, where the PCA indices serve as proxies for the latent health states. Because measurement errors are independent across domains conditional on the latent factors, any residual covariance across equations is attributable solely to correlation among the idiosyncratic shocks. The covariance structure of these auxiliary residuals therefore provides a set of moment conditions that, together with the known rescaling implied by the proxy mapping, uniquely pins down the off-diagonal elements of the idiosyncratic error covariance matrix. Further details of this identification argument can be found in Appendix B.2.

2.4 Estimation

We estimate the model parameters using a two-step procedure.¹⁷ In the first step, we recover the distributional parameters of the initial latent endowments and the parameters of the measurement system using data from the first observed period. We employ maximum simulated likelihood to estimate the mixture distribution governing w_0 and the parameters of the measurement equations. This step anchors the scale and dependence structure of the latent factors and fully characterizes the mapping from latent states to observed measures. Details of the likelihood function and the simulation procedure are provided in the Appendix D.

In the second step, conditional on the parameters identified in the first step, we use minimum-distance

¹⁷The two-step approach offers computational advantages by allowing subsets of parameters to be estimated separately in each step although this comes at the cost of efficiency relative to jointly estimating all parameters.

moment matching to estimate the remaining model parameters. The key moments are empirical analogues of the model’s production functions where the latent factors are replaced by domain-specific indices that are obtained from principal components applied to blocks of measures that load exclusively on a single latent factor. Substituting these indices into the transition equations yields non-structural relationships whose conditional means and variances are functions of the underlying production-function parameters and shock variances. Intuitively, the indices act as informative summaries of the latent factors. Because they preserve the ordering and dependence structure of the latent states, their conditional moments capture the same information about the transition dynamics as the latent factors themselves.

In practice, the moments we match correspond to the OLS estimates of equation (8), where the latent variables are replaced by the corresponding principal-component indices. The same indices are also used to construct empirical analogues of the behavioral equations governing investment, employment, mortality, and attrition. For investment, we match the OLS estimates of equation (10), again replacing latent health variables with the principal-component indices. For employment, mortality, and attrition, we match OLS estimates from linear probability models, specified analogously to equations (9), (11), and (12), respectively, with the principal-component indices in place of the latent health domains. Residuals are obtained from each of these regressions, and the model matches the covariance matrix of the full set of residuals.

Finally, we include moments corresponding to the age-specific means of each observed health measure, employment, investment, mortality, and attrition. As previously stated, we match the mean and variance of PC_t^k , $\forall k \in \{p, c, mh\}$, $t \in \{1, \dots, \bar{T}\}$, as well as the share of PC_t^k below the threshold of the lower-quartile of the distribution of PC_0^k , $\forall t \in \{1, \dots, \bar{T}\}$, where $\bar{T}$ is the maximum number of periods an individual can be in the sample. These moments help ensure the model reproduces the relevant marginal distributions in the data. Further details on estimation can be found in Appendix Section D. The fit of empirical moments to their data counterparts is presented in Appendix Section D.5

3 Health and Retirement Study

We estimate the technology of health formation using the Health and Retirement Study (HRS). HRS is a biennial panel survey of noninstitutionalized individuals and their spouses living in the United States. HRS includes an abundant set of measures of one’s health, along with variables related to income, demographics, and other economic conditions. HRS began in 1992, and respondents entered the survey between the ages of 51 and 61. We focus on white male respondents aged 55 or 56 who are observed for at least two total periods.

We exclusively use waves 3-14 of the survey, as there are nontrivial differences in the questionnaire starting in the third wave.¹⁸ Our sample is an unbalanced panel of 2,101 individuals (12,998 total observations).

Table 1: Sample Counts

Age Group	Observed	Dead	Attrition	Right Censoring
55/56	2,101	0	0	0
57/58	2,099	2	0	0
59/60	1,904	9	128	60
61/62	1,626	25	222	228
63/64	1,373	33	319	376
65/66	1,123	49	394	535
67/68	895	62	473	671
69/70	692	84	536	789
71/72	497	93	604	907
73/74	353	102	640	1,006
75/76	243	107	675	1,076
77/78	92	112	711	1,186
Totals:	12,998	678	4,702	6,834

Table 1 reports the number of observations by age group. Sample size declines considerably with age, driven primarily by attrition and right censoring, where the latter refers to respondents who entered the survey in later waves and are therefore not observed at older ages despite remaining in the sample.

Recovering the distribution of the latent health domains requires mapping observable health measures to each domain. We describe the indicators used for each domain in turn, noting how they relate to the underlying construct.

Mental Health

Mental health is measured using a set of questions derived from the Center for Epidemiological Studies Depression (CESD) scale. This scale is based on the sum of eight indicators. Six of these indicators are “negative,” where respondents report “yes” or “no” to experiencing the following sentiments all or most of the time: depression, everything is an effort, restless sleep, feeling alone, feeling sad, and not being able to get going. The remaining two indicators are “positive,” where respondents report “yes” or “no” to experiencing the following sentiments all or most of the time: feeling happy and enjoying life. The standard CESD score

¹⁸The 15th wave of the HRS corresponds to 2020, which we omit due to the COVID-19 pandemic.

aggregates these responses after reverse-coding the positive items, giving a score that ranges from 0 to 8, where higher scores indicating worse mental health. In our framework, we treat each indicator separately.¹⁹

Cognitive Functioning

Cognitive functioning is measured using seven indices collected consistently across all survey waves. These measures comprise scores from tests administered to survey respondents and self-reported functional limitations. These tests are not intended as measures of cognitive skill per-se. Rather, they are designed to capture aspects of cognitive health that tend to deteriorate with age, such as memory, basic arithmetic ability, clarity of thought, and fluid intelligence.

First, we use two measures of fluid intelligence (the capacity to think logically and solve problems in novel situations, independent of acquired knowledge). Prior research suggests that fluid intelligence is strongly associated with labor market outcomes (Heineck and Anger, 2010).²⁰ To measure fluid intelligence, respondents complete two standardized word recall tests: immediate and delayed word recall. The immediate word recall score captures the number of words out of ten that are correctly recalled immediately after presentation, while the delayed word recall score reflects the number of words recalled correctly following a delay of approximately five minutes.

Second, we incorporate two measures derived from tests of basic arithmetic ability. The first is the serial sevens test, in which respondents are asked to sequentially subtract 7 from 100 across five trials. Scores range from 0 to 5, based on the number of correct subtractions. The second measure involves backward counting, where respondents are instructed to count backward from 20 and from 86 for ten consecutive numbers. Scoring is based on performance: a score of 2 is assigned if the respondent completes the task correctly on the first attempt, 1 if correct on the second attempt, and 0 if unsuccessful in both attempts.

Finally, we include three subjective measures of functional limitations related to cognition. Respondents are asked whether they have difficulty using a map and whether they have difficulty managing money. Additionally, they self-report their memory on a five-point scale (scale is from poor to excellent). These questions reflect everyday cognitive functioning limitations that may not be fully captured by formal test scores.

¹⁹The mental health measures capture feelings of depression, anxiety, and stress. They do not directly measure objective mental health conditions, such as schizophrenia.

²⁰Fluid intelligence is distinguished from crystallized intelligence, which relies more on retrieving information from long-term memory.

Physical Functioning

Physical functioning is assessed using a set of responses to questions about difficulty performing specific everyday tasks. For each task, respondents report whether they experience difficulty that is expected to last at least three months. Responses are coded as 1 for difficulty and 0 for no difficulty.

The HRS functional limitations divide into two broad areas of physical limitation, based on similarity of the nature of limitation. First, mobility limitations include: walking several blocks, walking one block, climbing several flights of stairs, and climbing one flight of stairs. Second, large muscle limitation includes five tasks: getting up from a chair, stooping, kneeling or crouching, picking up a dime, lifting a ten pound object, and pushing or pulling a large object.

Employment and Health Investment

We model the health transition process as a function of employment status. We derive a indicator for whether an individual is employed from their reported labor force status. An individual is flagged as employed if they report working either part time or full time.²¹

HRS respondents are surveyed on a set of questions regarding health-related behaviors. Respondents are asked how often they engage in vigorous physical activity, such as aerobics, running, swimming, or bicycling. Individuals are flagged as engaging in regular exercise if they report participating in such activities three or more times per week. In addition, respondents are asked whether they consume alcohol, and if so, how frequently and how many drinks they typically consume. We use this to construct a binary indicator for the extensive margin of drinking, and another indicator relating to excessive drinking, which is defined as drinking more than 4 alcoholic drinks per day when the respondent drinks. Last, we have an indicator variable of whether the respondent currently smokes cigarettes. We use these four measures to construct a composite measure of health investment. We extract the first principal component of these four measures to construct an index, that is treated as an observable measure of overall healthy investment. The mapping is reported in Section C.3 of the Appendix.

Measures of Latent Heterogeneity

Respondents enter the sample at age 55, implying their initial observed health reflects a lifetime of prior choices and experiences. These accumulated factors may influence both the formation of health and other

²¹If the respondent is working full-time or part-time and there is no mention of retirement in the previous two years, they are considered employed.

individual decisions, such as engagement in healthy behaviors or employment. Importantly, unobserved characteristics, such as education and early-life socioeconomic status, may be jointly correlated with health outcomes and behavioral choices. For instance, individuals from disadvantaged backgrounds may have received fewer investments in health-promoting behaviors (e.g., regular exercise) or in the development of skills that enhance workplace productivity.

To account for this, we recover the latent health productivity type, b_0 , based on information collected prior to age 55. This variable summarizes pre-existing individual differences at the start of the observation window that are plausibly related to both subsequent health and employment trajectories (Currie and Moretti, 2003; Case et al., 2005; Currie, 2009; Conti et al., 2010; Currie et al., 2010; Case and Paxson, 2010; Lundborg et al., 2014; Almond et al., 2018; Adhvaryu et al., 2019). The pre-55 variables are used as observed measures in the factor analytic framework to recover the distribution of the initial heterogeneity, b_0 .

First, we include the respondent’s own education level and his mother’s education level. These variables relate to the efficient producer hypothesis, which posits that more educated individuals are more efficient at producing and maintaining health (Grossman, 1972). Second, we incorporate a measure of childhood socioeconomic status (SES) based on the respondent’s self-reported assessment of their family’s financial situation during childhood (categorized as “well off,” “about average,” or “poor”). This variable reflects early-life resource constraints and social environment, consistent with the early-life adversity or health capital frameworks, which emphasize the long-term effects of early deprivation. We also include the respondent’s self-rated health at age 16 (reported on a five-point scale: excellent, very good, good, fair, or poor), which captures early-life health endowments that may shape later-life health outcomes and labor market capacity. Finally, we include indicators of early-life risk exposure, such as whether the respondent ever smoked prior to age 50 and whether either parent used alcohol or drugs to an extent that caused problems in the family. These variables proxy for exposure to adverse family environments and behavioral risk factors that may have long-run impacts on both health and socioeconomic outcomes.

Additional Health Measures

Our model incorporates a set of additional health measures as functions of all the latent health domains. Including these measures enables us to examine how each latent health dimension relates to commonly used indicators of health status. First, we include self-reported health, a widely used summary measure in empirical research on health and its economic consequences. Self-reported health is a simple and readily available measure of health that has been shown to predict mortality (Idler and Benyamini, 1997) and labor

supply (Bound, 1989; Stern, 1989), and to correlate strongly with other health measures.²² In our analysis, we use respondents’ self-rated health, measured on a five-point categorical scale where “5” indicates excellent health.

Second, we incorporate a set of objective health indicators, capturing whether the respondent has ever been diagnosed with a set of chronic health conditions. The conditions are 1) high blood pressure or hypertension; 2) diabetes or high blood sugar; 3) cancer or a malignant tumor of any kind except skin cancer; 4) chronic lung disease except asthma such as chronic bronchitis or emphysema; 5) heart attack, coronary heart disease, angina, congestive heart failure, or other heart problems; 6) stroke or transient ischemic attack; 7) emotional, nervous, or psychiatric problems; and 8) arthritis or rheumatism. Linking these diagnoses to the latent health domains reveals how specific conditions relate to the latent health domains. The estimated factor loadings on these objective conditions indicate how each latent health domain maps onto particular diagnoses.

3.1 Descriptive Statistics

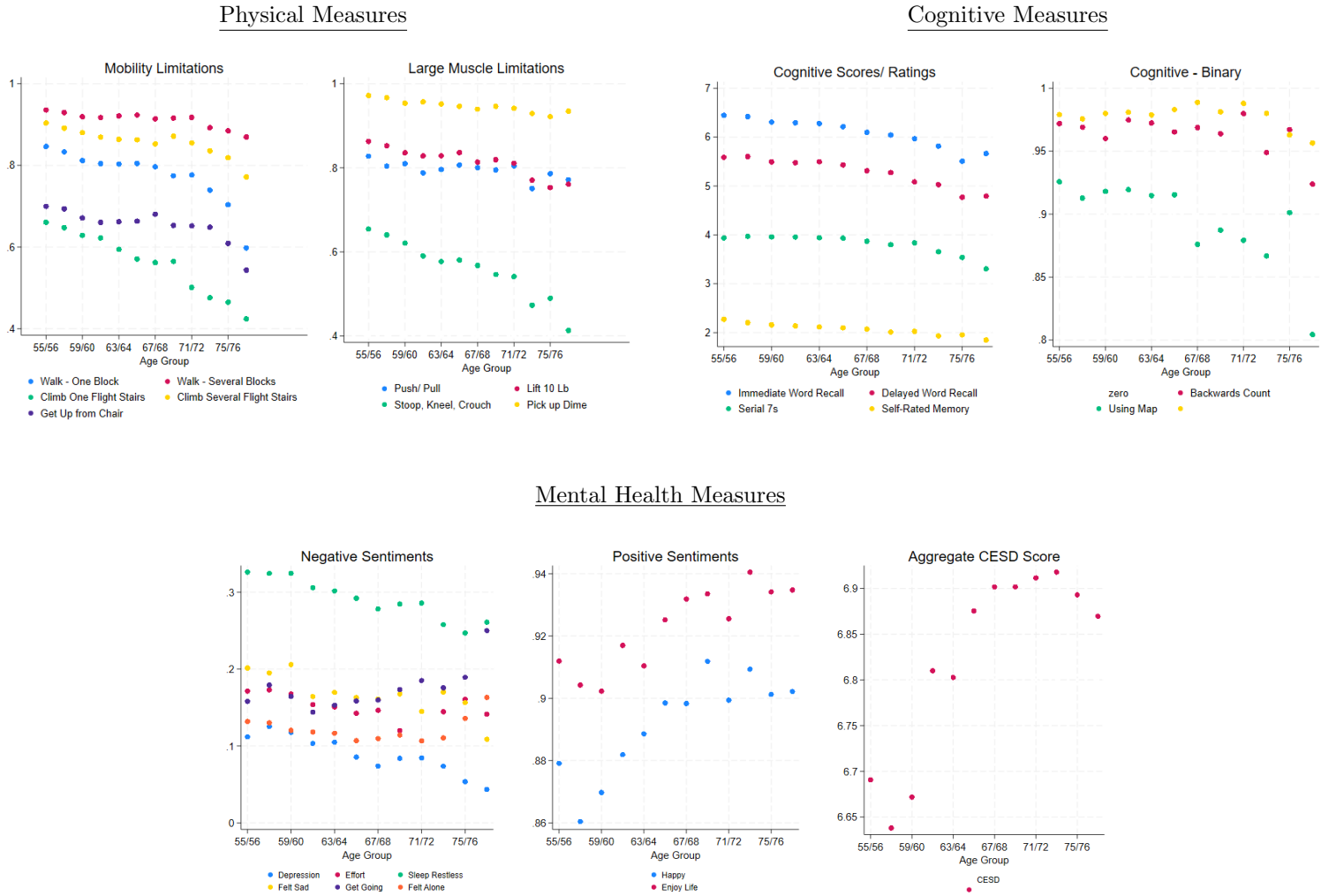
This section describes the dynamics of the observed health measures and their joint correlation structure. Figure 1 displays the averages of physical health measures (left) and cognitive health measures (right). The x-axis reports age in two-year intervals (e.g., 55–56, 57–58, etc.). The top two panels of Figure 1 illustrate that the mean profiles of physical and cognitive health deteriorate with age.

Next, the bottom panel of Figure 1 shows the age trend for the binary measures of mental health, distinguishing negative sentiments (left graph) and positive sentiments (middle graph). In contrast to the age profiles for physical and cognitive health, the mental health measures gradually improve over age, an exception being “could not get going.” The aggregate CESD, reported in the bottom-right panel, is signed such that higher values correspond to better mental health and shows a clear improvement in the aggregated measure. The improvement in mental health prior to retirement is consistent with evidence that the life cycle profile of mental health is U-shaped, remaining relatively high early in working life, declining through midlife, and improving again at older ages (Dijk and Mierau, 2023). However, the subsequent decline may reflect the accumulation of health deficits at older ages. These patterns suggest that mental health follows a distinct life-cycle trajectory relative to physical and cognitive health, raising the question of how changes in mental health interact with other health domains in shaping economic behavior at older ages

Despite the seemingly different age trajectories across health domains observed in Figure 1, Table 2

²²See White (2023) and Blundell et al. (2023) for further discussion on the use of self-reported health in economics research.

Figure 1: Mean of Standardized Health Measures by Age



Note: Each panel reports the sample average of the indicated health measure by two-year age bin for white males in the Health and Retirement Study.

Table 2: Average Pairwise Correlations of Measures Within and Between Each Health Domain.

	Physical	Cognitive	Mental Health
Physical	0.385 [0.160, 0.600]	0.125 [0.019, 0.228]	0.194 [0.056, 0.327]
Cognitive	0.125 [0.019, 0.228]	0.167 [0.038, 0.739]	0.114 [0.003, 0.217]
Mental Health	0.194 [0.056, 0.327]	0.114 [0.003, 0.217]	0.391 [0.238, 0.657]
Employment	0.145 [0.102, 0.177]	0.066 [0.026, 0.116]	0.089 [0.022, 0.147]
Mortality	-0.071 [-0.104, -0.028]	-0.040 [-0.066, -0.016]	-0.035 [-0.052, -0.021]
Investment	0.143 [0.046, 0.191]	0.107 [0.066, 0.161]	0.092 [0.034, 0.128]

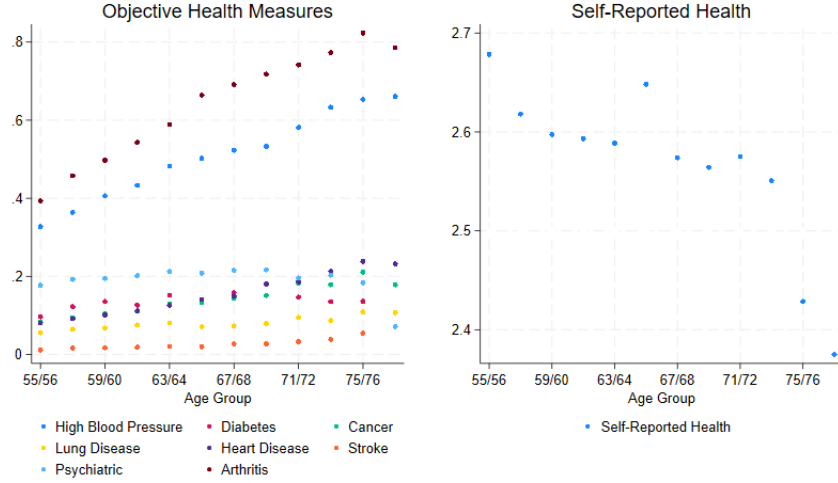
Average pairwise correlations are reported for each combination of health domains, computed using the sample pooled across all age groups. For example, the mean entry in row 1, column 1 is calculated by computing all pairwise correlations among the physical health measures and then taking their average. The entry in row 1, column 2 is calculated by computing the correlations between each physical health measure and each cognitive measure, and then averaging across all such correlations. Brackets report the minimum and maximum pairwise correlations within each cell. For instance, among the physical health measures, the smallest pairwise correlation is 0.160 and the largest is 0.600.

reports the average pairwise correlations of measures within and between each domain in the first three columns. Specifically, the first row reports the average correlation across physical health measures (column 1), the average correlation between physical and cognitive measures (column 2), and the average correlation between physical and mental health measures (column 3). The brackets report the minimum and maximum pairwise correlations for each set of measures.

The table reveals a positive association across all health domains, suggesting that individuals experiencing declines in one dimension of health tend to have worse health in others. In particular, worse physical or cognitive health tend to coincide with worse mental health. These patterns motivate a model that allows for dependencies across domains. At the same time, the range of correlations indicates that the measures are not perfectly collinear, so each contributes distinct information about the underlying domains.

The last three columns of Table 2 report the average correlations between measures within each health domain and employment, mortality, and health investment. All health measures are positively correlated with employment and health investment and negatively correlated with mortality. The strongest correlations for each outcome arise with the physical health measures, consistent with evidence linking physical health to labor market outcomes (Bound, 1989; Currie and Madrian, 1999), to mortality risk (Adams et al., 2003; Smith, 1999), and to health-related behaviors and investments (Grossman, 1972; Cutler and Lleras-Muney,

Figure 2: Mean of Objective Health Measures and Self-Reported Health by Age



Note: Each panel reports the sample average of the indicated health measure by two-year age bin for white males in the Health and Retirement Study. The objective health conditions are 1) high blood pressure or hypertension; 2) diabetes or high blood sugar; 3) cancer or a malignant tumor of any kind except skin cancer; 4) chronic lung disease except asthma such as chronic bronchitis or emphysema; 5) heart attack, coronary heart disease, angina, congestive heart failure, or other heart problems; 6) stroke or transient ischemic attack; 7) emotional, nervous, or psychiatric problems; and 8) arthritis or rheumatism.

2010).

Next, Figure 2 shows the average probability of self-reporting an objective health condition (left panel) and the evolution of standardized self-reported health (right panel). The objective and self-reported health measures display age patterns similar to those observed for the physical and cognitive measures. The prevalence of most objective health conditions rises with age, with the exception of psychiatric conditions. In addition, individuals' self-assessments of their health deteriorate on average as they age. Together, these figures are consistent with the cumulative decline in health as individuals age into and beyond retirement.

4 Estimation Results

This section presents the parameter estimates from the empirical model and discusses their implications for health dynamics and economic behavior. The fit of the empirical moments to their data counterparts is presented in Appendix Section D.5.

Table 3 presents the estimated correlations among the initial endowments, $\ln x_0^c$, $\ln x_0^{mh}$, $\ln x_0^p$, and b_0 . The endowments of the three health domains are positively correlated, with the strongest association between physical and mental health. Moreover, initial heterogeneity positively correlates with each health

Table 3: Estimated Correlation Matrix of Initial Endowments

	$\ln x_0^c$	$\ln x_0^{mh}$	$\ln x_0^p$	b_0
$\ln x_0^c$	1.000	0.185*** (0.043)	0.249*** (0.023)	0.608*** (0.023)
$\ln x_0^{mh}$		1.000	0.543*** (0.029)	0.535*** (0.048)
$\ln x_0^p$			1.000	0.524*** (0.042)
b_0				1.000

Note: Standard errors, reported in parentheses, are obtained from the inverse of the information matrix evaluated at the simulated maximum likelihood estimates. Standard errors for structural parameters are computed using the delta method to account for the nonlinear transformation from the estimated parameters to the structural objects of interest. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

domain, indicating that individuals with higher endowments of b_0 tend to exhibit better health outcomes. These correlations indicate that favorable initial conditions tend to span multiple health domains, consistent with the presence of common underlying determinants of health at older ages.

The results for initial endowments support the efficient-producer hypothesis (Grossman, 1972), which asserts that early investments in human capital, encompassing both personal and parental investments, improve the efficiency of health production in later life. Furthermore, they align with findings on the long-term health consequences related to adverse childhood environments and risky adolescent behaviors (Case and Paxson, 2010; Conti et al., 2010).²³

Table 4 reports estimates of the health formation process. The columns correspond to the cognitive, mental, and physical health equations, respectively. Recalling that the latent health domains are normalized in the initial period, the first row reports the intercept, indicating that cognitive and physical health experience a consistent period-by-period decline, while no such trend is estimated for mental health. Further, the age coefficient (last row) accelerates the rate of decline trend as people get older. Rows 2-4 capture the first order own-lag and cross-lag effects. The diagonal elements show that each health domain exhibits substantial persistence, with the strongest autocorrelation observed in physical (0.868) and cognitive health (0.862), while mental health is somewhat less persistent (0.519). That is, shocks to mental health dissipate relatively quickly, approximately half of any deviation from trend is corrected within a single period.

The cross-lag estimates reveal an asymmetric pattern of interdependence. Physical health exerts a positive and sizable effect on subsequent mental health (0.180), identifying it as an important determinant

²³Table 9 in Appendix C describes how b_0 maps to individual characteristics, providing intuition for the variation it captures.

Table 4: Model Estimates: Health Formation Process

	(1) Cognitive Health $\ln x_{t+1}^c$	(2) Mental Health $\ln x_{t+1}^{mh}$	(3) Physical Health $\ln x_{t+1}^p$
Intercept	-0.062* (0.033)	0.000 (0.045)	-0.046** (0.019)
$\ln x_t^c$	0.862*** (0.026)	0.001 (0.037)	0.071*** (0.015)
$\ln x_t^{mh}$	-0.013 (0.020)	0.519*** (0.024)	0.021 (0.018)
$\ln x_t^p$	-0.044*** (0.015)	0.180*** (0.028)	0.868*** (0.010)
b_0	0.112** (0.049)	0.125* (0.074)	0.016 (0.026)
$(\ln x_t^c)^2$	-0.024 (0.015)	0.002 (0.030)	-0.002 (0.016)
$(\ln x_t^{mh})^2$	0.003 (0.020)	0.045*** (0.014)	0.019 (0.013)
$(\ln x_t^p)^2$	0.000 (0.018)	0.067*** (0.022)	0.000 (0.009)
$(b_0)^2$	0.006 (0.055)	0.000 (0.094)	0.000 (0.032)
L_t	0.122*** (0.018)	0.046* (0.023)	0.049*** (0.013)
$\ln x_t^c \times \ln x_t^{mh}$	0.001 (0.027)	-0.040 (0.030)	0.017 (0.016)
$\ln x_t^c \times \ln x_t^p$	-0.002 (0.025)	-0.006 (0.035)	-0.003 (0.010)
$\ln x_t^{mh} \times \ln x_t^p$	0.008 (0.023)	0.000 (0.024)	0.010 (0.016)
$b_0 \times \ln x_t^c$	-0.001 (0.053)	0.012 (0.100)	0.011 (0.044)
$b_0 \times \ln x_t^{mh}$	-0.002 (0.048)	-0.025 (0.068)	0.000 (0.034)
$b_0 \times \ln x_t^p$	0.000 (0.054)	-0.025 (0.084)	0.001 (0.031)
I_t	-0.005 (0.015)	-0.001 (0.025)	0.013 (0.017)
$I_t \times \ln x_t^c$	0.003 (0.023)	0.019 (0.030)	0.019 (0.018)
$I_t \times \ln x_t^{mh}$	-0.001 (0.019)	-0.031 (0.020)	0.001 (0.010)
$I_t \times \ln x_t^p$	0.011 (0.021)	0.003 (0.024)	-0.057*** (0.018)
$I_t \times b_0$	-0.015 (0.037)	-0.004 (0.059)	0.002 (0.030)
Age	-0.002* (0.001)	-0.001*** (0.000)	-0.004*** (0.001)

Notes: Standard errors are reported in parentheses below the parameter estimates. They are computed using the analytical formula of Gouriéroux et al. (1993) with the optimal weighting matrix. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of mental well-being (García-Gómez et al., 2013; Ohrnberger et al., 2017). By contrast, physical health negatively affects cognitive health (-0.044). This result could reflect substitutability in health-preserving effort. For example, as mobility deteriorates, individuals may shift time towards cognitively stimulating activities, such as reading or puzzles, which can help preserve their cognitive functioning.²⁴ Appendix C provides suggestive evidence to support this mechanism. A decline in one’s physical health increases their time spent reading newspapers and magazines, reading books, solving puzzles, and spending waking hours using ones mind, albeit with weak statistical power.

Conversely, lagged cognitive health has a positive and significant effect (0.071) on subsequent physical health. A natural interpretation is that cognitive capacity supports behaviors, such as the maintenance of healthy routines, that sustain physical health. Together, these asymmetric cross-domain effects suggest that, while cognitive health promotes physical health, the converse does not hold and physical health may even modestly crowd out improved cognitive function.

Latent heterogeneity b_0 positively affects the formation of cognitive (0.112) and mental health (0.125), indicating that individuals with favorable early-life endowments experience a slower rate of decline in these domains (De Nardi et al., 2024). The effect on physical health formation is small and statistically insignificant (0.016), suggesting that the contemporaneous influence of early-life factors operates primarily through cognitive and mental health channels in this age range. However, the covariance structure of endowments reveals a strong correlation between $\ln x_0^p$ and b_0 , indicating that early-life factors remain important for physical health production earlier in the life cycle, even if operating through mental and cognitive health.

The quadratic terms reveal nonlinear dynamics in health formation that alter the effective persistence of shocks at different levels of lagged health stocks. As a result, the same individual can exhibit qualitatively different dynamics depending on their current level of health. The cross-domain interaction effects (cognitive $\times$ mental, cognitive $\times$ physical, etc.) are small and statistically indistinguishable from zero, suggesting that the nonlinearities in health formation are primarily own-domain phenomena.

Three distinct patterns in these higher order effects arise. First, the positive coefficient on $(\ln x_t^{mh})^2$ (0.045) in column two suggests that mental health exhibits increasing returns in reinforcing its own formation, such that individuals in good mental health are better able to maintain it. For those experiencing mild distress, the dynamics imply quicker return to the baseline trend than for those in better mental health.²⁵

²⁴Such activities have been shown to yield positive effects on preservation of cognitive health in the psychological literature (Hertzog et al., 2008). In a human-capital framework, Mazzonna and Peracchi (2012) model age-related cognitive decline as partly offset by investment in “cognitive repair” activities, such as reading newspapers and social or cultural engagement.

²⁵There is a lower threshold at which the marginal effect changes sign. As a rough approximation, ignoring interaction terms, $\partial \ln x_{t+1}^{mh} / \partial \ln x_t^{mh} \approx 0.519 + 0.045 \ln x_t^{mh}$ which implies a threshold of approximately -6.105. This value is an extreme outlier in the distribution of simulated mental health stocks.

These results reflect average tendencies in the population, but the model can generate chronically poor mental health in two ways. Persistently poor mental health can arise when negative shocks arrive repeatedly before recovery can take hold, trapping individuals away from their baseline not because of any structural shift in dynamics but because adverse circumstances continuously offset the system’s natural mean-reversion. Further, the coefficient on b_0 shifts the long-run level around which mental health fluctuates. For individuals with a high endowment, this anchors the system near a healthy baseline. For those with a negative endowment, however, the long-run equilibrium is pulled into persistently negative territory. This provides a structural account of chronic depression resulting from a stable equilibrium at a low level of mental health determined by enduring individual characteristics such as early-life conditions, genetics, or accumulated disadvantage.

Second, the positive coefficient on $(\ln x_t^p)^2$ (0.067) in the mental health equation indicates that the effect of physical health on mental health is convex. The marginal effect of physical health on subsequent mental health is increasing in the level of physical health. For someone in good physical health, the marginal benefit is increasingly self-reinforcing in its mental health benefits, and maintaining high physical function carries particularly large returns for psychological well-being. However, for individuals in sufficiently poor physical health, physical deterioration begins to actively harm mental health.²⁶

Third, the negative coefficient on $(\ln x_t^c)^2$ (-0.024) in column one reveals an important asymmetry in cognitive health dynamics, although is marginally insignificant ($p \approx 0.11$). For individuals with high cognitive health, the quadratic term dampens persistence, so shocks dissipate more quickly for those at the top of the distribution. This is consistent with diminishing returns, where very high cognitive health is stable but offers less self-reinforcement. In contrast, the dynamics work in the opposite direction for individuals in worse cognitive health. Modest declines are slow to correct on their own due to the already-high baseline persistence, while severe declines risk crossing into a self-reinforcing trajectory that the system’s natural dynamics have increasing difficulty reversing without external intervention. The individual endowment b_0 (0.112) determines where within this landscape a person’s long-run equilibrium sits, with low-endowment individuals anchored closer to the tipping point and therefore more exposed to the risk of irreversible decline.

The effect of lagged employment is complementary to the formation of cognitive (0.122), mental (0.046), and physical health (0.049). This result aligns with the related literature, which finds that the rate of health decline accelerates upon labor force exit (Rohwedder and Willis, 2010; Fitzpatrick and Moore, 2018; Black

²⁶There is a lower threshold at which the marginal effect changes sign. As a rough approximation, ignoring interaction terms, $\partial \ln x_{t+1}^{mh} / \partial \ln x_t^p \approx 0.180 + 0.067 \ln x_t^p$ which implies a threshold of approximately -2.68. This value corresponds to the bottom 2% of the distribution of simulated physical health stocks.

et al., 2018). Surprisingly, the first-order effects of lagged health investment are not significant determinants of the formation of any health domain, consistent with health trajectories become less responsive to behaviors captured in I_t at older ages, especially once early-life endowments are controlled for. However, the negative coefficient on lagged investment interacted with lagged physical health (-0.057) implies that investment is more effective at improving physical health for individuals with a lower current stock, implying decreasing returns to the health stock in translating investment into physical health gains.

Table 5: Covariance Matrix of Idiosyncratic Shocks to Production

	ϵ^c	ϵ^{mh}	ϵ^p
ϵ^c	0.179*** (0.017)	0.096*** (0.018)	0.025*** (0.008)
ϵ^{mh}		0.628*** (0.042)	0.089*** (0.007)
ϵ^p			0.207*** (0.007)

Notes: Standard errors are reported in parentheses below the parameter estimates. They are computed using the analytical formula of Gouriéroux et al. (1993) with the optimal weighting matrix. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 reports the estimated covariance matrix of idiosyncratic production shocks. Mental health is the most volatile domain (0.628), more than three times as volatile as cognitive (0.179) or physical health (0.207). The positive off-diagonal covariances indicate that unobserved shocks tend to move all three domains together. Among the cross-domain covariances, the cognitive-physical pair is the smallest (0.025), while cognitive-mental (0.096) and mental-physical (0.089) are similar and larger, suggesting that mental health shocks are tied to disruptions in the other domains.

Effects of Health on Employment, Mortality, and Investment.

Despite an extensive literature on the effects of health on employment, a lack of consensus on the magnitude of these effects and the specific dimensions of health most predictive of particular outcomes remains (French and Jones, 2017; Blundell et al., 2023). This lack of consensus may, in part, arise from the variety of empirical methodologies and datasets used to evaluate these effects. Table 6 reports estimates of the behavioral equations governing employment, health investment, mortality, and attrition. Across these equations, the latent health domains enter with the expected signs, such that better health is associated with higher employment, greater health investment, and lower mortality.

Table 6: Estimates of Determinants of Employment, Mortality, Health Investment, and Attrition

	Employment (L_t)	Investment (I_t)	Mortality (d_t)	Attrition (a_t)
Intercept	0.555*** (0.172)	0.009 (0.015)	-2.935*** (0.032)	-1.884*** (0.003)
$\ln x_t^c$	0.030 (0.116)	0.026 (0.024)	0.003 (0.041)	0.000 (0.025)
$\ln x_t^{mh}$	0.181** (0.072)	0.058 (0.040)	-0.179*** (0.056)	0.100*** (0.032)
$\ln x_t^p$	0.240*** (0.078)	0.048* (0.025)	-0.156*** (0.041)	-0.004 (0.024)
b_0	0.007 (0.196)	0.226*** (0.028)	0.000 (0.080)	0.031 (0.034)
t	-0.305*** (0.013)	0.012* (0.006)	0.107*** (0.031)	0.089*** (0.003)
t^2			-0.001 (0.002)	
Wealth	-0.177 (0.120)	0.086*** (0.009)		
Experience	0.025*** (0.000)			
Married	-0.301 (0.270)	0.048** (0.024)		

Notes: Standard errors are reported in parentheses below the parameter estimates. They are computed using the analytical formula of Gourieroux et al. (1993) with the optimal weighting matrix. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Physical health is the dominant predictor of employment (0.240), although mental health plays a sizable role as well (0.181). The relative magnitudes suggest that physical capacity is the primary constraint on labor supply near retirement, but that mental well-being constitutes a quantitatively important secondary channel. The effect of cognitive health on employment is small in magnitude and statistically insignificant. This finding is consistent with Blundell et al. (2023), who document modest cognitive effects on employment once self-reported health is controlled for. These results support the use of physical health as the primary indicator when analyzing the effects of health on employment, while also highlighting mental health as a meaningful secondary channel. These estimates also speak to the documented link between physical and cognitive health, whereby adverse physical health shocks depress cognitive functioning (Schiele and Schmitz, 2023). Our direct production estimate has physical health lowering cognitive formation, but an indirect, positive channel operates through employment. That is, negative physical health shocks reduce labor market participation, which in turn lowers cognitive functioning.

In the health investment equation (column 2), b_0 has the largest coefficient (0.226) among the included regressors, exceeding the effects of any current health stocks. This suggests that investment behavior is more strongly shaped by an individual’s health productivity type, capturing persistent early-life factors, such as education, socioeconomic background, and health habits, than by current health status, consistent with related evidence (Sawyer et al., 2012). Among the current health stocks, physical health has a significant positive effect (0.048), consistent with physical capacity enabling health-promoting behaviors such as exercise.

In the mortality equation, mental health has the largest effect (-0.179), followed by physical health (-0.156), while cognitive health is statistically insignificant. Mortality risk increases significantly with age (0.107). The insignificance of b_0 suggests that, conditional on current health stocks and age, early-life endowments do not exert an independent direct influence on mortality risk. Instead, the influence of b_0 appears to operate primarily through its effect on the latent health stocks themselves, rather than a separate pathway to mortality.

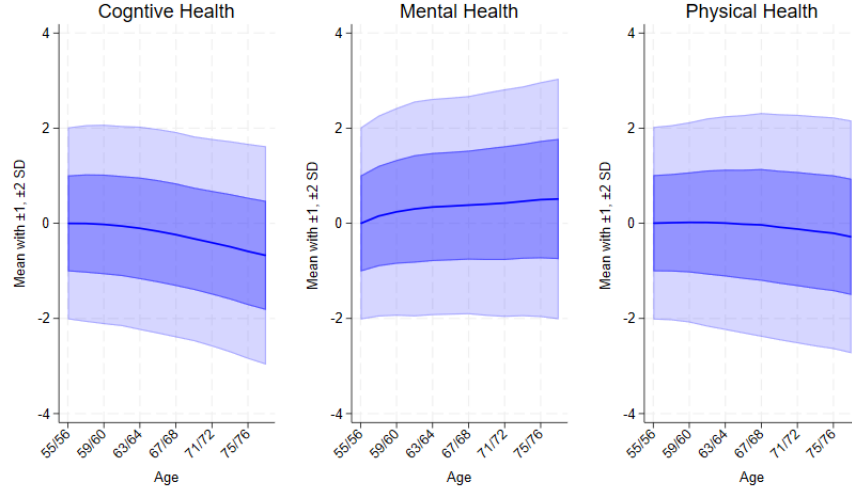
Turning to the attrition equation (column 4), both mental health (0.100) and age (0.089) are positive and highly significant predictors of sample exit. The positive association between mental health and attrition may reflect selective departure by healthier individuals, for instance, voluntary survey exit among those with sufficient well-being and resources to do so. These patterns warrant attention when interpreting the other equations, as non-random attrition could introduce selection effects into the estimation sample.

Simulated Latent Health Dynamics

Figure 3 plots the mean paths of the health domains using data simulated from the estimated model. The simulated paths for cognitive and physical health follow age trends similar to those observed in the raw health measures discussed in Section 3.1. Likewise, the model reproduces the positive age trend in mental health observed in the CESD indicator variables in Figure 1.

Turning to dispersion, physical and mental health become more dispersed with age, while cognitive health remains relatively stable. The dispersion in mental health increases because it is subject to much larger idiosyncratic shocks than the other health dimensions. Physical health also becomes more dispersed because it experiences larger and slightly more persistent shocks than cognitive health, while also accumulating additional variation transmitted from cognitive health through its positive effect on physical health. In contrast, the effect of physical health on cognition is negative, so shocks to physical health are not propagated back to cognitive health, leaving the dispersion in cognitive health largely unchanged.

Figure 3: Simulated Path in Latent Health Domains



Note: Figures plot mean path in simulated latent health factors over time. The figures also show the first and second standard deviations from the mean in shaded regions.

4.1 Average Marginal Effects Across Health Domains

An important policy question concerns the extent to which health domains reinforce or substitute one another over time, and through what channels these cross-domain effects operate. The answer has direct implications for the targeting and timing of health interventions. If cross-domain effects are large and concentrated at particular points in the health distribution or at particular ages, policies that improve one domain may generate multiplier effects across others.

This section uses the estimated model to simulate average marginal effects (AME) of each health domain on the others. The total marginal effect of domain k on the next-period stock of domain j is, by the chain rule, the sum of the direct effect through the production function and two indirect effects operating through the behavioral responses of investment and employment; i.e.,

$$\frac{\partial \ln x_{t+1}^j}{\partial \ln x_t^k} = \underbrace{\frac{\partial f_t^j}{\partial \ln x_t^k}}_{\text{direct}} + \underbrace{\frac{\partial f_t^j}{\partial I_t} \cdot \frac{\partial I_t}{\partial \ln x_t^k}}_{\text{investment}} + \underbrace{\frac{\partial f_t^j}{\partial L_t} \cdot \frac{\partial L_t}{\partial \ln x_t^k}}_{\text{employment}}. \quad (13)$$

The rich interactions in the health production technology and the endogenous investment and employment functions make these marginal depend on an individual's current state. That is, these marginal effects are not constant, they depend on the individual's current health stocks, investment level, employment, and draw

of b_0 .²⁷ Aggregating these individual-level marginal effects and tracing them over time therefore reveals two things simultaneously. It shows the structural curvature of the production technology and how the evolving distribution of population health shapes the average returns to each domain.

Figure 4 plots the simulated AMEs for each health domain by age bin. The AME of $\ln x_t^k$ on $\ln x_{t+1}^j$ is obtained by evaluating equation (13) for each simulated individual and averaging over the full sample. In Figure 5 below, we plot the AMEs for individuals in the lower quartile of b_0 , illustrating how effects differ for those with worse economic circumstances and poorer health. The top, middle, and bottom panels correspond to cognitive, mental, and physical health, respectively, in the numerator of equation (13). The denominator of the partial derivative corresponds to lagged cognitive health in the left figures, lagged mental health in the middle figures, and lagged physical health in the right figures.

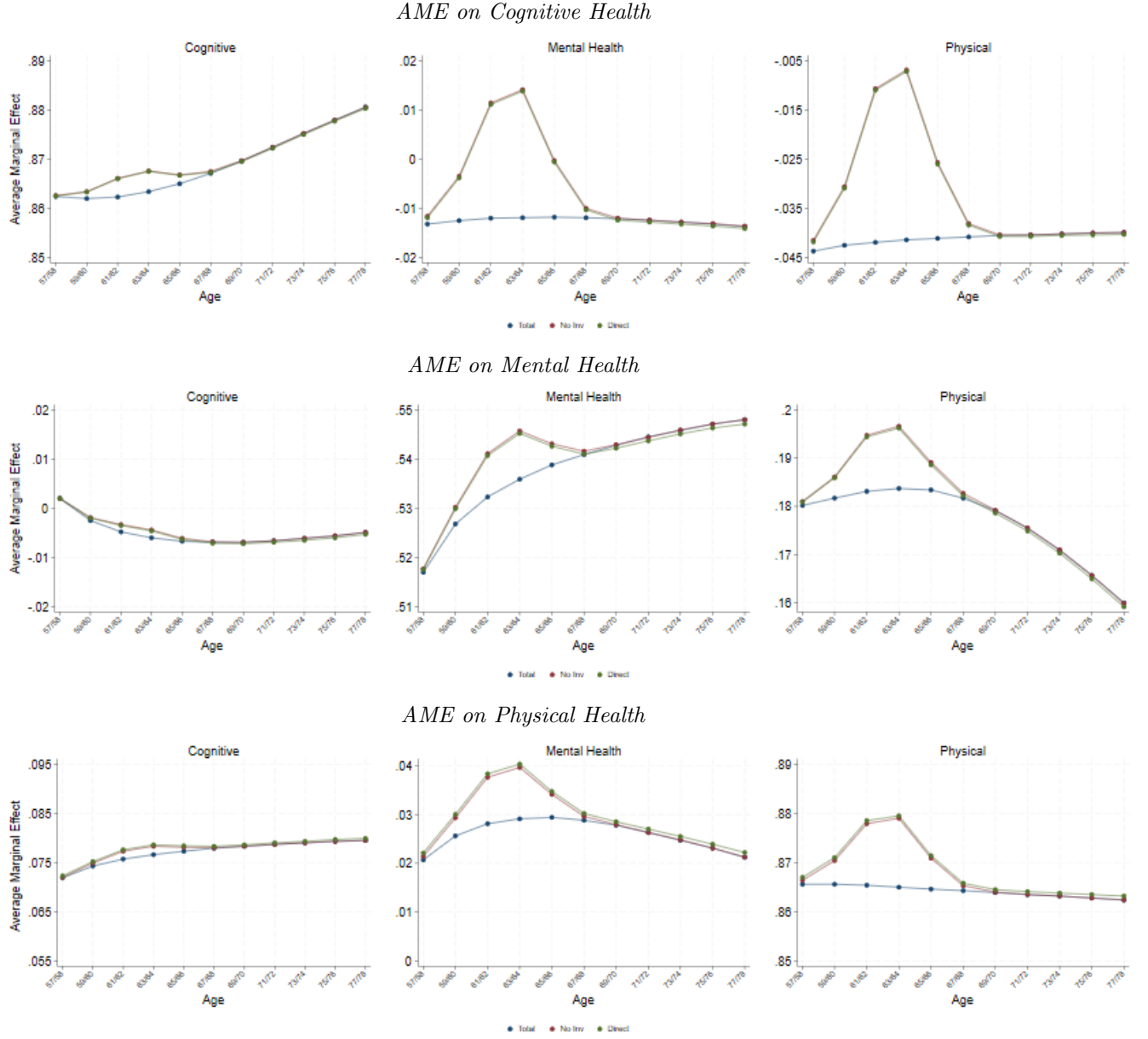
Each figure reports three time trends. The green line corresponds to the total AME from Equation (13). The red line drops the indirect investment channel from the AME, so the gap between the green and red lines measures the investment response. The blue line reports the direct effect only, so the gap between the red and blue lines measures the indirect employment channel.

Considering first to the top panel (AMEs on cognitive health), we note the hump-shaped pattern in the earlier periods visible across all three figures in this panel. This hump reflects the indirect employment channel. The hump shape arises as the employment channel transmits a health shock into a more substantial change in employment probability where the probit slope is steep. When nearly everyone is already employed (early periods) or has already exited the labor force (late periods), the probit lies in a flat tail and a marginal change in health does not substantially shift employment probability. Thus, the indirect employment effect peaking around age bin 63/64 indicates that the indirect health-on-health pathway via employment is strongest during the mass retirement window, which is precisely when the employment margin is most responsive to health shocks.

The blue line in the leftmost figure depicts the own partial derivative ($\frac{\partial \ln x_{t+1}^c}{\partial \ln x_t^c}$), which captures the persistence of cognitive health from one period to the next. The own AME for cognitive health rises with age, indicating that cognitive health becomes increasingly reinforcing over the life cycle. That is, mean reversion weakens on average as individuals age, and one's prior cognitive stock becomes an increasingly powerful determinant of future cognitive health. This result follows from the estimated concavity of the production

²⁷In the full model, an idiosyncratic shock to one health domain today would also propagate through correlated shocks to the other model equations. Moreover, a change to health today affects mortality probability and selection. To focus on the dynamic propagation through the three channels of equation (13), the partial derivative to calculate the average marginal effects is taken holding the shocks to the other model equations unchanged and assumes individuals at t survive and do not attrit from the sample in period $t + 1$, shutting down selection channels in the measured AME.

Figure 4: Average Marginal Effects (AME) of Health Domains



Note: The green line corresponds to the average marginal effects using formula (13). The red lines correspond to the partial derivative of the production function ignoring the indirect effect on the investment. The blue line is the partial of the production function ignoring the indirect effects on investment and employment (direct effect of the production function exclusively).

function with respect to cognitive health and from the fact that cognitive health levels decrease with age on average in the simulated sample.

In the middle panel, the direct effect of lagged mental health on cognitive health is negative, small

in magnitude, and varies little across ages, although cross-domain estimates from lagged mental health are statistically insignificant determinants of cognitive health. However, the indirect effect of mental health via the employment channel overcomes this negative direct effect in the earlier periods, yielding a positive total AME prior to retirement. This result implies that policies aimed at improving mental health generate positive spillovers on cognitive health through the employment channel during working ages, albeit small in magnitude.

The total AME of lagged physical health on cognitive health is negative at all ages. While the indirect employment channel generates positive marginal effects from physical health improvements, these are not large enough to reverse the sign of the overall AME. The direct effect of physical health on cognitive health trends toward zero with age, suggesting that any health complementarity between the two domains weakens in later life. Taken together, these findings indicate that policies targeting physical health do not generate positive spillovers to cognitive health in the population on average.

The middle three panels report the AMEs on mental health. In the leftmost graph, the marginal effect of lagged cognitive health on mental health is small and slightly negative at all ages. The middle graph finds that mental health is increasingly self-reinforcing over the life cycle, though at a decreasing rate. This suggests that individuals become slightly more susceptible to mental health shocks as they age, as mean reversion weakens. In the age range around retirement, the indirect effect via employment enhances AME of mental health shocks, such that 54-55% of an innovation to mental health persist into the the next period.

In the rightmost panel, the direct effect of lagged physical health on mental health has the largest cross-domain effect at approximately one-third the magnitude of the own mental-health direct effect. The total AME from lagged physical health is largest around the retirement window, before declining sharply after age bin 63/64. As individuals grow older, the own-domain persistence effect dominates and cross-domain spillovers from physical health interventions become less relevant. However, the sizable indirect effect via employment suggests that physical health interventions around the retirement window are a fruitful time to target physical health in order to generate spillovers to mental health.

Finally, the bottom three panels consider AMEs for physical health. The own AME of lagged physical health (rightmost panel) is the largest of all AMEs in the retirement window but modestly declines with age. The physical health process exhibits only a negligible decline in persistence, and shocks to physical health remain a major determinant of an individual's physical health trajectory. Additionally, the returns to a marginal improvement in lagged physical health are amplified around the retirement window through the employment channel, with modest additional gains through the investment margin. The small investment

channel reflects the estimated interaction coefficient in the health production function ($I_t \times \ln x_t^p = -0.057$). That is, a marginal improvement in physical health raises health investment, which in turn improves physical health for individuals with low physical health stocks. Physical health interventions near retirement age thus generate significant gains to physical health and have sizable positive spillovers to mental health, but this comes at a tradeoff with cognitive health, although these negative effects are mitigated by the indirect employment channel.

The AME from lagged cognitive health on future physical health increases with age, indicating that better cognitive health contributes to better physical health more strongly near the end of life. The AME of lagged mental health on physical health is positive and concave, though modest in overall magnitude. At earlier ages, it generates meaningful spillovers through the employment channel and marginal gains through the investment channel.

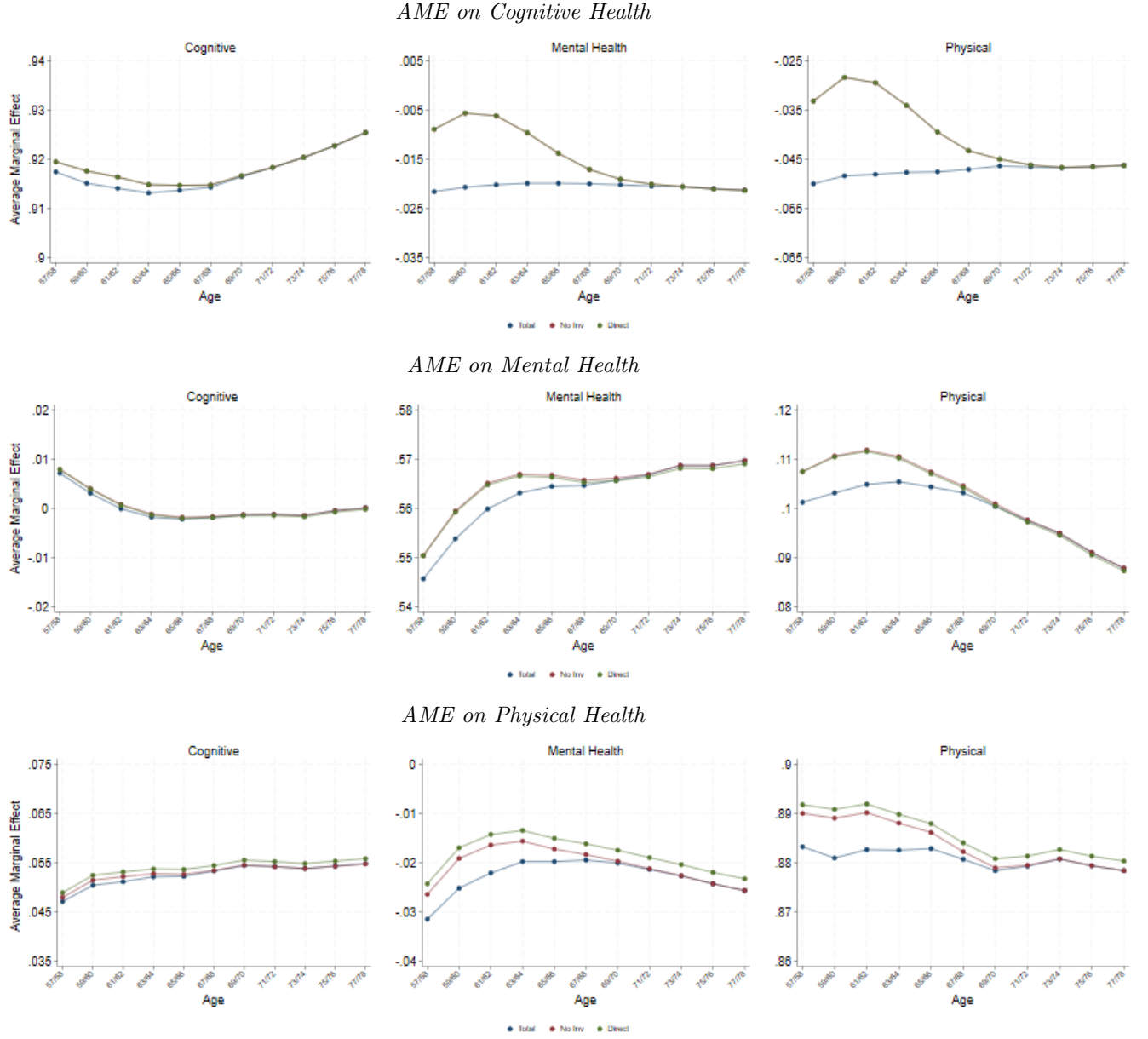
The AME exercises above are calculated for the full simulated sample. Next, we consider the AME conditional on simulated individuals in the lower quartile of the b_0 distribution. These individuals are more likely to be economically disadvantaged in various ways. These individuals have lower education, worse childhood SES, worse health in childhood.²⁸ Moreover, Table 3 implies this group is more likely to enter the model with lower health stocks across all domains. The nonlinearities in the model cause marginal effects to be functions of the level of health stocks, implying AMEs may produce very different effects for these individuals. Such differences must be considered for policy interventions that are targeting more economically disadvantaged demographic groups.

Figure 5 reports the cross-domain AMEs for individuals in the lower quartile of the b_0 distribution at each age bin. In the top panels (where cognitive health is in the numerator) we find that self-reinforcement is larger compared to the AME for the full simulated sample, reflecting the estimated diminishing returns to lagged cognitive health in its own production function. Moreover, the spillovers from lagged physical and mental health are always negative, indicating that positive spillover effects do not generate meaningful improvements in cognitive health for this group. Hence, policymakers with the goal of improving cognitive health for economically disadvantaged older adults are constrained in interventions that target cognitive health directly.

In the middle three panels, mental health exhibits a similar increasing reinforcement in itself over time. The indirect effect via employment is smaller compared to the indirect effect calculated for the population on average. Individuals with lower b_0 have lower labor market attachment than the population on average,

²⁸See Table 9 in Appendix C.

Figure 5: Average Marginal Effects (AME) of Health Domains for the Bottom Quartile of b_0



Note: The green line corresponds to the average marginal effects using formula (12). The red lines correspond to the partial derivative of the production function ignoring the indirect effect on the investment. The blue line is the partial of the production function ignoring the indirect effects on investment and employment (direct effect of the production function exclusively).

and hence are less sensitive to marginal changes in health on their employment decision. Further, the direct spillover effect from lagged physical health is also weaker in this group relative to the overall population. Again, these individuals are endowed with a lower level of physical health, and do not reap the high return

observed for individuals with high stocks of physical.

For physical health, we first note that the indirect effect via investment is stronger. That is, individuals in this group enter the model with lower stocks of physical health on average, and therefore receive modest gains from healthy investment, such as quitting smoking or engaging in exercise. The AME from lagged mental health becomes negative for this group, but remains small, and physical health shows slightly stronger persistence in itself.

In sum, these results highlight three broad policy implications. First, health interventions are most effective within their own domain, making direct interventions the most reliable policy lever for improving health. This is particularly true for cognitive health, where indirect improvements through physical and mental health are more negative for more economically disadvantaged individuals. We also find more meaningful spillovers from cognitive to physical health, and from physical to mental health, though the latter comes at a trade-off with negative effects on cognitive health.

Second, the years surrounding retirement are a critical period for health intervention. Health shocks affect labor force participation, which in turn reinforces health formation. Because labor supply is most responsive around ages 63-65, indirect effects operating through employment are strongest during this age range. These employment gains are largest from physical health interventions. However, these indirect effects largely disappear once individuals have exited the labor force.

Third, the effectiveness of indirect policy channels differs importantly across socioeconomic groups. Spillovers across health domains are generally weaker for the economically disadvantaged, but not uniformly so. Moreover, the gains from the indirect employment channel are lower for the economically disadvantaged, who are more weakly attached to the labor market. This creates a compounding dynamic in which poor health is both more persistent and less responsive to indirect interventions. A notable exception is health investment, which emerges as a potential source of indirect benefits for individuals in poor physical health. Initiatives to promote health investment, through exercising or reducing smoking and drinking, have a relatively greater return for disadvantaged populations. Nevertheless, improvements in cognitive and mental health for these populations appear to require more direct intervention rather than relying on cross-domain spillovers.

4.2 Objective Chronic Conditions and Self-Reported Health

The final set of results examines the predictive power of the latent health domains for the set of objective chronic conditions described in Section 3, and for an individual's self-assessed overall health. The estimated

parameters reveal the extent to which each latent health domain contributes to both types of outcome. For objective conditions, we estimate how the probability of each diagnosis varies with the latent health stocks; for self-reported health, we estimate how each domain shapes an individual’s overall self-assessment.

These results serve two interpretive purposes. First, the domains’ ability to predict objective clinical diagnoses indicate they capture genuine, measurable dimensions of health. Second, if the domains reflect aspects of health that people themselves recognize and care about, they should predict subjective overall assessment of health. Together, the two outcomes also offer a mechanistic account of why treatment or policy effects can diverge across these commonly used health measures.

Table 7: Self-Reported Health

$\ln x^c$	0.259*** (0.015)
$\ln x^{mh}$	0.186*** (0.012)
$\ln x^p$	0.615*** (0.012)
b_0	-0.416*** (0.022)
Age	0.011*** (0.002)
Intercept	2.552*** (0.01)

Notes: Self-reported health is modeled as a continuous measure. Standard errors are reported in parentheses below the parameter estimates. They are computed using the analytical formula of Gourieroux et al. (1993) with the optimal weighting matrix. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Table 7, we examine the relationship between latent health domains and self-reported health, which we model as a continuous measure for simplicity of exposition. The loading associated with physical health is larger than that of cognitive and mental health, indicating that respondents tend to prioritize somatic limitations in their overall health assessments. Nonetheless, all domains significantly influence self-reported health, indicating that individuals with varying perceptions of latent health domains may still self-report a comparable overall rating of their health.

Taken together, these findings demonstrate that self-reported health is a weighted aggregate of distinct latent dimensions rather than a sufficient statistic for any single domain. While self-reported health is widely used in applied research and correlates strongly with mortality and future health outcomes (French, 2005; Pijoan-Mas and Ríos-Rull, 2014; Capatina, 2015; De Nardi et al., 2024), its aggregated nature obscures the

domain-specific mechanisms through which health shapes economic behavior. This matters because different health domains carry distinct economic implications. For instance, physical health is the primary driver of labor supply, while cognitive health plays a negligible role. Conflating these pathways into a single index makes it difficult to identify which dimension is driving an observed effect.

Table 8: Objective Health Conditions

	1. High Blood Pressure	2. Diabetes	3. Cancer	4. Lung Disease
$\ln x^c$	-0.093*** (0.011)	-0.094** (0.043)	0.074* (0.044)	-0.116 (0.354)
$\ln x^{mh}$	-0.006*** (0.000)	-0.010 (0.031)	0.037 (0.033)	-0.112** (0.044)
$\ln x^p$	-0.189*** (0.010)	-0.310*** (0.000)	-0.119*** (0.038)	-0.458* (0.257)
b_0	0.109*** (0.000)	0.036 (0.085)	-0.025 (0.089)	-0.002 (0.888)
Age	0.030*** (0.000)	0.013** (0.006)	0.059*** (0.006)	0.013 (0.041)
Intercept	-0.205*** (0.007)	-0.919*** (0.036)	-1.079*** (0.028)	-1.474*** (0.224)
	5. Heart Problem	6. Stroke	7. Psychiatric	8. Arthritis
$\ln x^c$	0.000 (0.046)	-0.272 (0.386)	-0.101 (0.163)	0.003 (0.056)
$\ln x^{mh}$	-0.063** (0.029)	-0.129 (0.08)	-0.254*** (0.031)	0.019 (0.025)
$\ln x^p$	-0.258*** (0.039)	-0.339 (0.252)	-0.179 (0.116)	-0.443*** (0.052)
b_0	0.038 (0.078)	0.076 (0.974)	0.065 (0.366)	0.127 (0.117)
Age	0.060*** (0.007)	0.034 (0.054)	0.001 (0.015)	0.079*** (0.004)
Intercept	-1.118*** (0.034)	-2.247*** (0.292)	-0.599*** (0.051)	-0.231*** (0.004)

Notes: Standard errors are reported in parentheses below the parameter estimates. They are computed using the analytical formula of Gouriéroux et al. (1993) with the optimal weighting matrix. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The outcome variables are binary indicators for whether a doctor has told the respondent if they have a broad set of chronic conditions. The conditions are 1) high blood pressure or hypertension; 2) diabetes or high blood sugar; 3) cancer or a malignant tumor of any kind except skin cancer; 4) chronic lung disease except asthma such as chronic bronchitis or emphysema; 5) heart attack, coronary heart disease, angina, congestive heart failure, or other heart problems; 6) stroke or transient ischemic attack; 7) emotional, nervous, or psychiatric problems; and 8) arthritis or rheumatism.

Table 8 presents the estimated latent-domain coefficients for the eight doctor-diagnosed chronic conditions.

The probability of each chronic condition tends to increase with age, with significant age gradients for high blood pressure (0.030), diabetes (0.013), cancer (0.059), heart problems (0.060), and arthritis (0.079). Interestingly, latent heterogeneity does not significantly predict most conditions, with the sole exception of high blood pressure (0.109). This suggests that, conditional on the three latent health stocks, unobserved individual heterogeneity provides little additional predictive power for chronic disease, consistent with the health domains capturing the economically relevant variation in health capital.

Physical health is the strongest and most consistently significant predictor across conditions, consistent with functional physical capacity being closely linked to chronic disease burden (Crimmins and Beltrán-Sánchez, 2011). The association is largest for lung disease (-0.458) and arthritis (-0.443), both of which impose substantial physical impairment. Respiratory limitations from lung disease directly reduce the capacity to perform physical activities, while joint inflammation from arthritis severely restricts mobility. Mental health is the only statistically significant predictor of psychiatric conditions (-0.254), suggesting that psychiatric disorders and conditions are closely linked to the measures of depression, anxiety, and stress used to identify the mental health factor in the model.

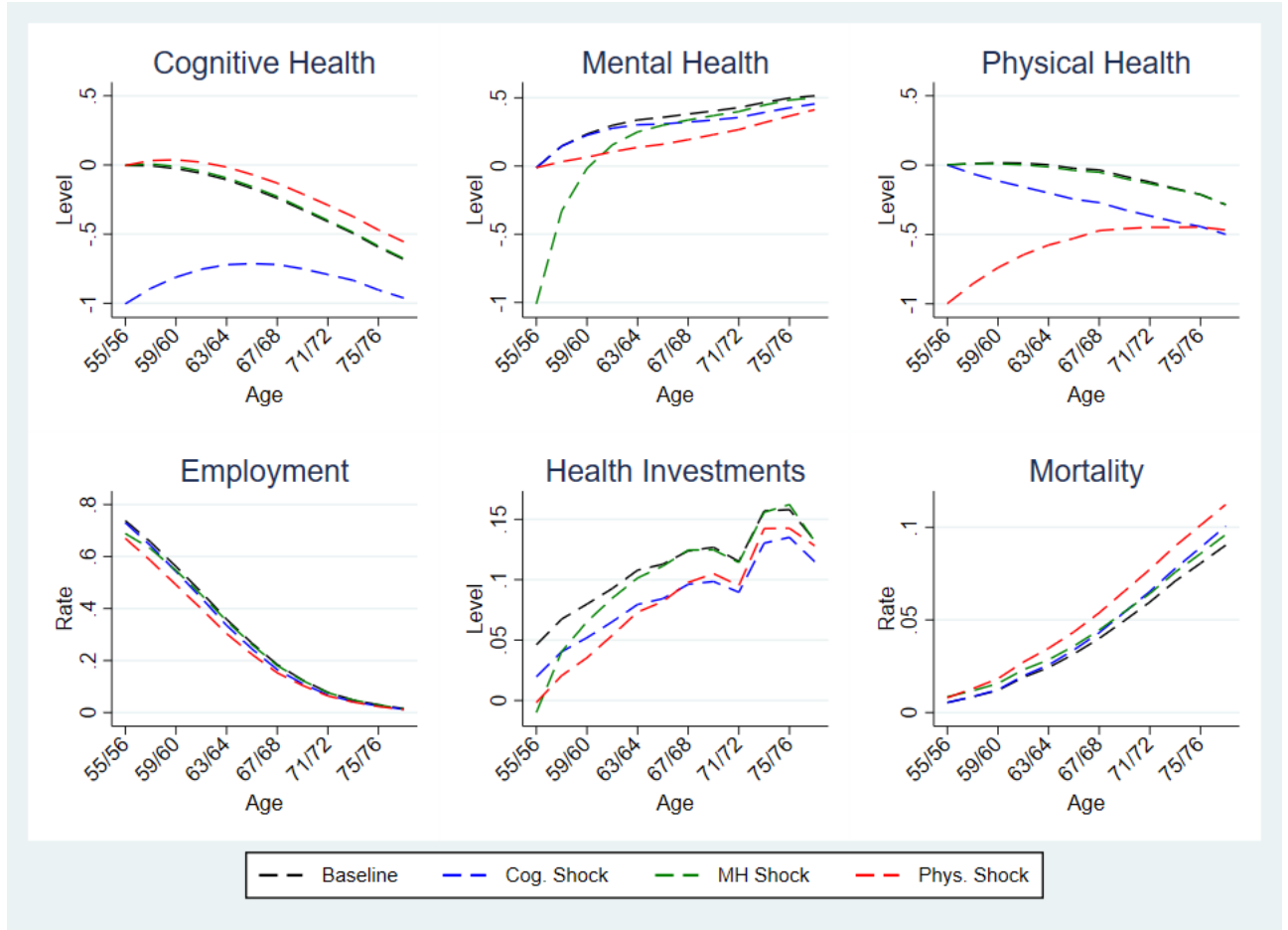
5 Experiments Using the Model: Shocks to Health Domains

In this final section, we use the estimated model to simulate how shocks to the health domains propagate through the model. Unlike AMEs, which capture period-by-period impacts holding the sample fixed, these counterfactual simulations account for endogenous selection through mortality and trace how a shock propagates dynamically across all subsequent periods through the full system of health, employment, and investment.

Figure 6 reports three counterfactual exercises, each imposing a one-standard-deviation negative shock to one health domain. The figure shows simulated paths for each of the three health domains (top three panels), and for employment, health investment, and mortality (bottom three panels). Each graph plots the dynamic responses in the baseline model (black), and the simulated paths following a one standard deviation shock to cognitive health (blue), mental health (green), and physical health (red).

A negative shock to physical health generates the strongest cross-domain spillovers. Cognitive health improves slightly and while physical health only partially recovers, with its average remaining persistently below baseline. Mental health declines immediately and stays persistently below baseline. These health losses translate into economically meaningful consequences. Employment falls sharply on impact ($\approx 9\%$)

Figure 6: Effects of a -1 SD Shock to Health Domains at $t = 1$



Notes: Each panel displays three counterfactual simulations relative to the baseline (black line). The red, green, and blue lines correspond to scenarios in which physical health, mental health, and cognitive health, respectively, are subjected to a one-standard-deviation negative shock in the first period.

and remains steadily lower, consistent with worse physical health inducing early retirement. Mortality rises considerably ($\approx 24\%$ by age bin 77/78), which partially accounts for reversion to the baseline trend of physical health, since those in worse physical standing are more likely to die. Health investment also remains steadily below the baseline level as individuals respond to their worsened health trajectory. These results underscore the central role of physical health. Shocks originating in this domain propagate broadly across other health domains and generate sustained economic consequences.

A shock to cognitive health also generates a meaningful spillover to physical health, which progressively worsens as the cognitive domain fails to return to baseline. This decline in physical health indirectly affects mental health, which begins to noticeably deviate from its baseline trend 4-5 years after the initial shock. Cognitive health itself only partially recovers, leaving a persistent deficit. These spillovers to physical and

mental health produce downstream effects on mortality and employment. Health investment also shifts persistently downward following the cognitive shock, by slightly more than under the physical shock.

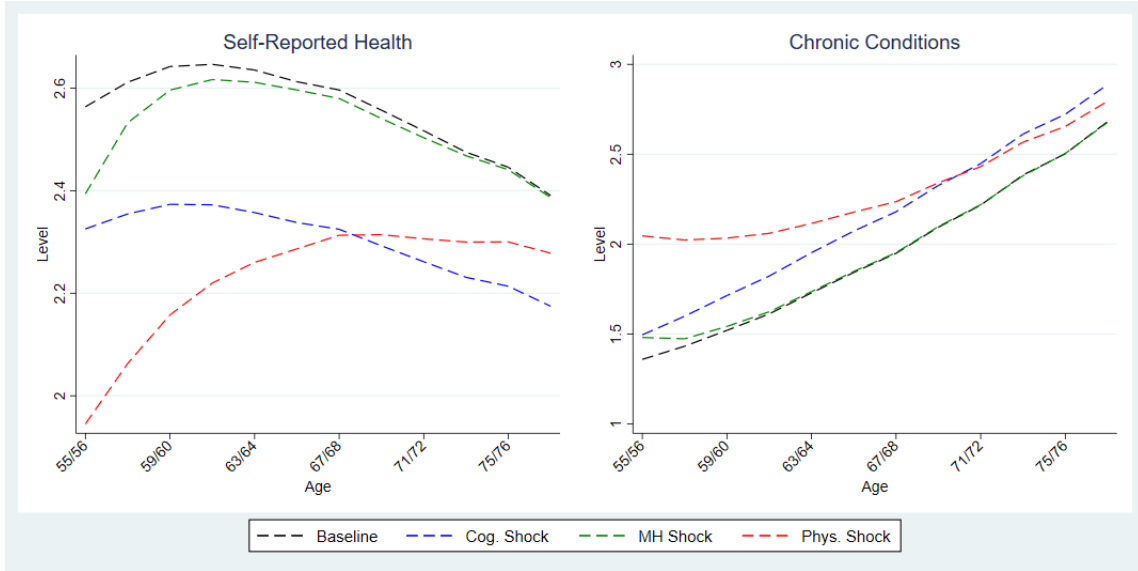
A shock to mental health has little impact on physical or cognitive health. Unlike the other domains, the mental health shock is largely transitory. Mental health converges back toward its baseline trajectory over time. Employment and health investment deviate initially but recover relatively quickly as mental health returns to baseline, with the effects buffered by the absence of any response in physical and cognitive health. Mortality rises modestly on impact, culminating in approximately a 6% increase by age bin 77/78.

Taken together, these counterfactuals reveal a clear asymmetry in the health production process. Physical health plays a central, system-wide role, as shocks in this domain propagate broadly and durably across other health dimensions and economic outcomes. Cognitive health occupies an intermediate position, with meaningful spillovers operating primarily through physical health. Mental health shocks are more localized and transitory relative to shocks to physical or cognitive health. In other words, mental health is more dependent on stocks of physical and cognitive health than the reverse. This highlights the importance of modeling the dependence of mental health on the other health domains, while suggesting that the reverse relationship, in which mental health feeds back into physical or cognitive capital, may be of secondary importance for individuals in this age range. These dynamics underscore the importance of accounting for cross-domain complementarities in health policy design.

Figure 7 presents the dynamic effects of each health shock on self-reported health (left panel) and the expected number of chronic conditions (right panel), where the latter is computed as the expected sum of objective chronic condition measures. In the baseline scenario (black line), self-reported health declines gradually over the life cycle. The physical health shock generates the largest initial deviation from baseline, while shocks to cognitive and mental health produce comparable initial deviations. The mental health shock, however, dissipates relatively quickly, whereas the cognitive health shock exhibits no recovery and, by age bin 67–68, generates a larger deviation from baseline than the physical health shock. This crossing reflects two offsetting forces. The physical health shock produces positive spillovers to cognitive health, partially attenuating the decline in self-reported health, and selective mortality operates such that individuals in worse physical health exit the sample at higher rates, mechanically raising mean self-reported health among survivors.

A similar pattern emerges for the chronic disease index. Mental health shocks are largely transitory, with the expected number of chronic conditions returning to baseline within two model periods. Cognitive health shocks produce an initial effect of comparable magnitude, but unlike mental health shocks, the effect

Figure 7: Effects of a -1 SD Shock to Health Domains at $t = 1$ on Self-Reported Health and Expected Chronic Conditions.



Note: The black line in each figure represents the baseline simulated scenario. The red line represents a scenario where cognitive health is shocked by one standard deviation in the first period. The green line represents a scenario where mental health is shocked by one standard deviation in the first period. The blue line represents a scenario where physical health is shocked by one standard deviation in the first period.

is persistent, remaining roughly parallel to the baseline throughout the horizon. Physical health shocks elicit the largest increase in the expected number of chronic conditions, though the trajectory crosses that of the cognitive shock near age bin 69–70, driven by the same forces of positive spillovers and selective mortality discussed above.

6 Conclusion

This paper investigates the formation and dynamics of health in older adults, focusing on the interplay among physical, cognitive, and mental health. We develop and estimate a model using data from the Health and Retirement Study to examine how these health dimensions evolve over time and influence economic behavior, including employment and health investment decisions. Our findings suggest important complementarities across health domains. Notably, physical functioning exhibits strong persistence and has a substantial influence on mental health. The analysis also examines how the latent health domains relate to commonly used health measures and how well they predict outcomes such as employment.

Overall, the results indicate that physical health is the most important determinant of employment. It is also central to health investment and mortality, although it is not the single dominant factor in either

case. Health investment is driven most strongly by early-life type, b_0 , while the largest per-period mortality coefficient is associated with mental health. In contrast, physical health dominates cumulatively once its higher persistence and propagation into other health domains are taken into account. However, the joint analysis reveals a nontrivial role for mental health, and, to a lesser extent, cognitive health. In particular, the estimated health production function highlights strong linkages between physical and mental health. Accounting for multiple health domains and their interdependence is therefore crucial for understanding the mechanisms through which health shapes economic behavior and outcomes at older ages. Finally, each latent health domain contributes to individuals' self-assessed health, providing insight into what this commonly used measure of self-reported health captures in empirical work.

This paper provides a foundation for understanding health formation and its role in shaping health-related economic inequality at older ages. However, as the focus of this paper is on the health formation technology, the remaining model elements are interpreted non-structurally and provide only descriptive evidence of how the health formation process affects economic outcomes. Future work building on this paper will embed the estimated health formation technology within the theoretical model presented in Appendix A. Doing so will structurally link health dynamics with economic behavior and enable counterfactual policy analysis, such as evaluating the impacts of retirement-age reforms.

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A Motivating Framework for the Technology of Health Formation

This section presents a dynamic model to illustrate the many ways individuals' choices and circumstances affect, and in turn are affected by their health. We adapt the stylized model presented in Blundell et al. (2023), expanding the choice set and augmenting the health production technology to depend on individual's decisions. This exercise aids in guiding the empirical analysis and identification strategy for estimating the structural parameters of the health formation process.

The model describes an individual's decision process until their period of death, T_i . At each age, $t \in \{0, \dots, T_i\}$, individuals choose how much to work and how much of their income to save, to consume of a regular consumption good, and to consume of a separate "unhealthy" good. The unhealthy consumption good, such as smoking or drinking, gives positive utility but is a detrimental input to health formation. Choices for individual i are made to maximize the expected discounted value of their current and future utility. Health is a vector of stocks which affects individual's choices through impacting current utility, as well as dynamically by determining future health stocks via the health formation process. In what follows, we consider the problem of a single cohort, so that time and age are interchangeable.

At each age t , individual's receive utility from consumption and leisure. Utility, given labor supply, Y_{it} , and a vector of health capital $H_{it} \in \mathbb{R}^K$, where K is the number of health domains modeled, is

$$U(C_{it}, I_{it}; Y_{it}, H_{it}, \xi_i, \zeta_{it}) = \frac{(C_{it} + \alpha I_{it})^{1-\gamma}}{1-\gamma} - v(Y_{it}, H_{it}, \xi_i, \zeta_{it}), \quad (14)$$

where C_{it} is normal good consumption and I_{it} is unhealthy good consumption. Unhealthy goods are detrimental to health production, so, in order to rationalize interior solutions, there is an additional preference parameter associated with to unhealthy good consumption, $\alpha > 1$, as in (Strulik, 2022). captures the utility costs associated with work and poor health and is assumed to be additively separable from consumption. This assumption simplifies the solution of the model and is commonly adopted in the related literature (Blundell et al., 2023; De Nardi et al., 2024). The utility costs depend on the health stock, reflecting that work and leisure may become more costly during periods of poor health. The utility costs also depend on unobserved tastes heterogeneity, ξ_i , and idiosyncratic cost shock, ζ_{it} .

The individual's decision problem is subject to several dynamic constraints. First, the budget constraint summarizes available resources for consumption and savings as

$$A_{it+1} = (1 + r_t)(A_{it} + Y_{it}W_{it}(1 - \tau_t(Y_{it}W_{it}, H_{it})) + b_t(H_{it}) - C_{it} - p_t^I I_{it}). \quad (15)$$

The budget is determined by the assets available at the start of the period, A_{it} , potential earnings, W_{it} , and transfer payments that may be available for individuals in bad health, such as disability insurance, $b_t(H_{it})$. If working, $Y_{it} = 1$, employment income is subject to the tax and transfer function $\tau(\cdot)$, which summarizes the relationship between employment income and the tax and transfer system, including health-related credits and benefits. The consumption of regular goods, C_{it} is in real terms, and p_t^I reflects price differences in the real cost of unhealthy goods. The interest rate, r_t , determines the per-period return to savings.

When working, the potential earnings of individual i at age t are

$$W_{it} = w(t, H_{it}, \phi_i, v_{it}). \quad (16)$$

Potential earnings combine the price and supply of labor and depend on age, approximating the life-cycle profile of productive human capital accumulation, and vary with the stock of health, representing a disruption of translating productive human capital into output for those in bad health. Earnings depend on an individual fixed effect, ϕ_i , which can be interpreted as heterogeneous productive ability, and v_{it} , which is an idiosyncratic transitory earnings shock.

The technology of health formation for each health domain k is described by a production function,

$$H_{it}^k = h^k(t, H_{it-1}, Y_{it-1}, I_{it}, \psi_i, \epsilon_{it}). \quad (17)$$

Health formation is influenced by individuals' prior labor supply decisions. For instance, at older ages, physically demanding tasks or sustained workplace stress can negatively affect health (Strulik, 2022; Jolivet and Postel-Vinay, 2020). Conversely, continued labor force participation may have beneficial effects, such as providing a sense of purpose or preserving cognitive and physical functioning, which is consistent with evidence that retirement can lead to health deterioration or increased mortality risk (Black et al., 2018). Moreover, lifestyle choices, such as consumption of unhealthy goods like drinking alcohol and smoking cigarettes, also enter directly into the health production function.

Heterogeneity in health formation is captured by an individual-specific parameter, ψ_i , reflecting differences in health productivity. Because health evolves dynamically, early-life endowments and prior choices have long-term consequences for current health. This heterogeneity, which can be thought of as a health productivity "type," accounts for differences in health outcomes arising from factors such as initial health endowments, genetic predispositions, past health shocks, and life-course decisions, including education and work history (De Nardi et al., 2024). Finally, health formation is subject to unobserved transitory shocks,

ϵ_{it} , which introduce additional variation over time.

Finally, an individual's stock of health determines their expected lifespan. We define the probability that an individual alive at age t survives to $t + 1$ as

$$S(t, H_{it}). \quad (18)$$

The survival probability captures a key tension in the decision problem. Working generates labor income that supports current consumption and retirement savings, but entails a direct utility cost and affects future health through the health production process, with the direction of this effect an empirical question. Separately, unhealthy good consumption raises current utility but deteriorates health, reducing the value of leisure in retirement and shortening the expected planning horizon.

A.1 Structure of Unobserved Components

The model allows for unobserved fixed heterogeneity in health, earnings, and the preferences for work, (ϕ_i, ψ_i, ξ_i) . We allow for an arbitrary correlation between these three dimensions. For example, individuals of lower socioeconomic status when growing up may have been relatively deprived in early-life investments that foster good health formation and human capital. Similarly, education may simultaneously impact health productivity, earnings, and preferences for work. These are distinct from transitory shocks to health, earnings, and preferences $(\zeta_{it}, v_{it}, \epsilon_{it})$, which are assumed serially uncorrelated, mutually independent, and independent of unobserved heterogeneity components.

A.2 Individual's Problem

The solution to the individual's problem at age t can be represented in the Bellman formulation. For the set of state variables, $\Omega_{it} = (t, H_{it}, A_{it}, \phi_i, \xi_i, \zeta_{it})$, the individual's choices satisfy,

$$\begin{aligned} V_t(\Omega_{it}) &= \max_{C_{it}, I_{it}, Y_{it}} \left\{ U(C_{it}, I_{it}, Y_{it}; H_{it}, \xi_i, \zeta_{it}) + \beta S(t, H_{it}) \mathbb{E} V_{t+1}(\Omega_{it+1}) \right\} \\ \text{subject to } A_{it+1} &= (1 + r_t)(A_{it} + Y_{it}W_{it}(1 - \tau_t(Y_{it}W_{it}, H_{it})) + b_t(H_{it}) - C_{it} - p_t^I I_{it}), \\ H_{it}^k &= h^k(t, H_{i,t-1}, I_{it}, Y_{it}, \psi_i, \epsilon_{it}), \quad \forall k \in \{1, \dots, K\} \\ W_{it} &= w(t, H_{it}, \phi_i, v_{it}). \end{aligned}$$

Consider an interior solution for consumption. Conditional on $Y_{it} = y$, the first order conditions are used to derive the optimal policy functions for the normal and unhealthy good consumption. The first order conditions are

$$\frac{\partial}{\partial C} : U_C - \beta(1 + r_t)S(t, H_t)\mathbb{E}V_{A,t+1} = 0 \quad (19)$$

$$\frac{\partial}{\partial I} : U_I - \beta(1 + r_t)S(t, H_t)\mathbb{E}(p_t^I V_{A,t+1} + V_{H,t+1}H_I) = 0, \quad (20)$$

where $V_{x,t+1}$ denotes the partial derivative of V_{t+1} with respect to x and H_x denotes the partial derivative of H with respect to x . From equation (19), we obtain the usual relation between marginal utility of consumption and the discounted marginal value of assets, where discounting accounts for mortality risk. Then, combining the first order conditions, we obtain

$$U_I = p_t^I U_C + \beta S(t, H_t)\mathbb{E}V_{H,t+1}H_I. \quad (21)$$

The optimal level of unhealthy good consumption, I^* , is such that the marginal utility of consuming unhealthy good equates marginal utility of consumption priced at the unhealthy good plus a term capturing the effect of the unhealthy good on future health times the marginal value of higher health in the future.

Denote Ω_{t+1}^y as next period's state variables conditional on $Y_{it} = y$, for $y \in \{0, 1\}$. Then, the decision rule for employment is

$$Y_{it}^* = \mathbb{1} \left[\max_{C_t, I_t} \left\{ U(C_{it}, I_{it}; Y_{it} = 1, H_{it}, \xi_i, \zeta_{it}) + \beta S(t, H_{it})\mathbb{E}V_{t+1}(\Omega_{it+1}^1) \right\} \right. \\ \left. - \max_{C_t, I_t} \left\{ U(C_{it}, I_{it}; Y_{it} = 0, H_{it}, \xi_i, \zeta_{it}) + \beta S(t, H_{it})\mathbb{E}V_{t+1}(\Omega_{it+1}^0) \right\} \geq 0 \right]. \quad (22)$$

Denote (C_t^y, I_t^y) as solutions to equations (19) and (20). Then we can express this using a variable capturing the latent value working,

$$Y_t^l = \frac{(C_{it}^1 + \alpha I_{it}^1)^{1-\gamma} - (C_{it}^0 + \alpha I_{it}^0)^{1-\gamma}}{1-\gamma} - v(Y_{it}, H_{it}, \xi_i, \zeta_{it}) + \beta S(t, H_{it}) \left(\mathbb{E}V_{t+1}(\Omega_{it+1}^1) - \mathbb{E}V_{t+1}(\Omega_{it+1}^0) \right),$$

and the optimal policy for employment is

$$Y_{it}^* = \mathbb{1}(Y_t^l > 0). \quad (23)$$

We can express the policy functions characterizing unhealthy good demand and labor supply as

$$I_{it}^* = I(H_{it}, W_{it}, A_{it}, t, \xi_i, \phi_i, \psi_i, \zeta_{it} | Y_{it}, \theta), \quad (24)$$

$$Y_{it}^* = Y(H_{it}, W_{it}, A_{it}, t, \xi_i, \phi_i, \psi_i, \zeta_{it} | \theta), \quad (25)$$

where θ is the set of all parameters in equations (14) - (18). This expression is useful to illustrate the determinants of health formation. Substituting equations (24) and (25) into equation (26) yields

$$H_{it+1}^k = h^k(t, H_{it}, I_{it}^*, Y_{it}^*, \psi_i, \epsilon_{it+1}), \quad (26)$$

which clearly illustrates the endogeneity present when estimating (26) using observed data. The unobserved input to health formation, ψ_i , is correlated with observed optimal employment and demand for unhealthy goods. The correlation works through ψ_i directly and through its covariance with individual preference heterogeneity, ξ_i , and heterogeneity in ability, ϕ_i . Hence, controlling for ψ_i is needed to obtain unbiased estimates of the health transition process.

B Further Details on Identification

B.1 Identification of Initial Distributional Parameters and the Measurement System

This section establishes identification of the measurement system and the joint distribution of initial latent endowments. Throughout, m indexes observed measures and k indexes latent factors. Because all measurement parameters are assumed to be age-invariant, the initial period identifies both the measurement parameters and the distribution of initial latent states. We therefore suppress the time index for the remainder of this section.

Let $w_0 = (\ln x_0^p, \ln x_0^c, \ln x_0^{mh}, b_0)'$ denote the vector of initial latent factors. The joint distribution of w_0 is approximated by a two-component Gaussian mixture,

$$F_{w_0} = \pi \Phi_4(\omega^1, \Sigma^1) + (1 - \pi) \Phi_4(\omega^2, \Sigma^2),$$

where $\Phi_k(\omega^j, \Sigma^j)$ denotes a multivariate normal CDF with mean ω^j and covariance matrix Σ^j . The location and scale are normalized by $E(w_0) = 0$ and $\text{diag}(\text{Var}(w_0)) = 1$. These restrictions anchor the latent scale

and resolve the observational equivalence between measurement loadings and the variance of latent factors. The off-diagonal elements of Σ^j are unrestricted and capture dependence across initial endowments.

We proceed in three steps. First, we establish identification of a single latent domain, k , when it is measured exclusively by binary proxies, the most restrictive case. A binary indicator reveals only whether the factor lies above or below a threshold, so each proxy is coarse, and identification must be established from how these indicators move together.²⁹ We show that the measurement parameters (loadings and thresholds) and the distribution of the latent factor can be recovered from a set of co-occurrence probabilities of the observed binary proxies. This recovery is generic (holds for *almost every* parameter value, except possibly a small, exceptional set) and local. We argue the map from parameters to moments is locally invertible for almost every parameter value, with failures confined to a negligible set that the model's assumptions rule out. Second, we establish identification when the set of proxies for k contains a mixture of binary and continuous measures, showing how to recover the location and scale parameters and measurement error variances of the continuous measures. Third, we recover the off-diagonal elements of the component covariance matrices from cross-domain proxy comovement, which completes the recovery of the joint distribution of initial endowments.

B.1.1 Identification Using Binary Measures

Fix a latent domain k . The log latent factor, $\ln x_0^k$, follows a two-component Gaussian mixture,

$$\ln x_0^k \sim \pi \mathcal{N}(\omega_k^1, (s_k^1)^2) + (1 - \pi) \mathcal{N}(\omega_k^2, (s_k^2)^2).$$

The normalizations $E(\ln x_0^k) = 0$ and $\text{Var}(\ln x_0^k) = 1$ impose two restrictions on the five mixture parameters, leaving three free. Solving the two restrictions yields the second component in closed form,

$$\omega_k^2 = -\frac{\pi}{1 - \pi} \omega_k^1, \quad (s_k^2)^2 = \frac{1 - \pi[(s_k^1)^2 + (\omega_k^1)^2] - (1 - \pi)(\omega_k^2)^2}{1 - \pi}, \quad (27)$$

so the free mixture parameters are (π, ω_k^1, s_k^1) . The factor is measured by $M \geq 4$ binary proxies,

$$z_m^k = \mathbf{1}[\lambda_m^k \ln x_0^k + v_m^k \geq \tau_m^k], \quad v_m^k \sim N(0, 1),$$

where λ_m^k is the measure-specific loading and τ_m^k is a threshold. Because $\text{Var}(\ln x_0^k) = 1$ anchors the scale of the factor, no further normalization on the loadings is needed.

²⁹For examples of identification of latent factor models using continuous indicators, see Cunha et al. (2010); Agostinelli and Wiswall (2025).

Identifying assumptions. We make the following assumptions on the measurement errors:

- (V1) $v_{t,m}^k$ are mean zero for all $k \in \{p, c, mh, b\}$, $t \in \{1, \dots, T_i\}$, and $m \in \{1, \dots, M^k\}$. The variance of measurement errors for discrete and categorical measures is normalized to one.
- (V2) $v_{t,m}^k$ is independent of $(x_\tau^p, x_\tau^c, x_\tau^{mh}, b_0)$ for all $t, \tau \in \{1, \dots, T_i\}$; $m \in \{1, \dots, M^k\}$; and $k \in \{p, c, mh, b\}$.
- (V3) $v_{t,m}^k$ is independent of $v_{\tau,n}^l$ whenever $(k, m, t) \neq (l, n, \tau)$, for $t, \tau \in \{1, \dots, T_i\}$; $m \in \{1, \dots, M^k\}$; $n \in \{1, \dots, M^l\}$; and $k, l \in \{p, c, mh, b\}$.

In addition to the measurement-error restrictions, we impose:

- (C1) *Relevance.* $\lambda_m^k \neq 0$ for all m , so every proxy is informative about the factor.
- (C2) *Distinctness.* The pairs (λ_m^k, τ_m^k) is unique, ensuring no two measurement functions are identical.
- (C3) *Sign normalization.* $\lambda_1^k > 0$, fixing the orientation of the latent state.
- (C4) *Label order.* $\omega_k^1 < \omega_k^2$, resolving the label-swap ambiguity inherent to mixtures.
- (C5) *Non-degeneracy.* $\pi \in (0, 1)$ and $(\omega_k^1, s_k^1) \neq (\omega_k^2, s_k^2)$, so the two components are genuinely distinct; otherwise the mixture collapses to a single Gaussian, and the component-specific parameters are not separately recoverable.

Conditional on $\ln x_0^k$, the measurement-error restrictions, (V1)-(V3), imply that each binary proxy is an independent Bernoulli draw with a conditional probability function that is strictly monotone in $\ln x_0^k$:

$$g_m(\ln x_0^k) = \Pr(z_m^k = 1 | \ln x_0^k) = \Phi(\lambda_m^k \ln x_0^k - \tau_m^k).$$

The core identifying logic is that all co-movement among proxies must be attributed to the shared latent factor, $\ln x_0^k$, as measurement errors are independent by restrictions (V1)-(V3). The observable co-occurrence probabilities are therefore integrals of products of conditional probability functions against the mixing distribution G :

$$P_S = \int \prod_{m \in S} \Phi(\lambda_m^k \ln x_0^k - \tau_m^k) dG(\ln x_0^k), \quad G \sim \pi \mathcal{N}(\omega_k^1, (s_k^1)^2) + (1 - \pi) \mathcal{N}(\omega_k^2, (s_k^2)^2),$$

where S is a subset of the binary proxy indices of size $|S| \leq 3$, representing the specific combination of measures being evaluated. These probabilities are the identifying movements we use.

(M1) **Marginal probabilities.** For each proxy m ,

$$P_m = \pi \Phi(A_m^1) + (1 - \pi) \Phi(A_m^2), \quad A_m^c = \frac{\lambda_m^k \omega_k^c - \tau_m^k}{\sqrt{(\lambda_m^k)^2 (s_k^c)^2 + 1}}$$

where $c \in \{1, 2\}$ indexes the mixture component. This yields M moment conditions.

(M2) **Pairwise probabilities.** For any two proxies, j and l with $j \neq l$, of the same latent domain k , conditional independence implies that all dependence between z_j^k and z_l^k flows through the shared factor. Within mixture component c , the pair $(\lambda_j^k \ln x_0^k + v_j^k, \lambda_l^k \ln x_0^k + v_l^k)$ is bivariate normal with within-component correlation,

$$\rho_{jl}^c = \frac{\lambda_j^k \lambda_l^k (s_k^c)^2}{\sqrt{[(\lambda_j^k)^2 (s_k^c)^2 + 1][(\lambda_l^k)^2 (s_k^c)^2 + 1]}}. \quad (28)$$

The joint probability

$$\Pr(z_j^k = 1, z_l^k = 1) = \pi \Phi_2(A_j^1, A_l^1; \rho_{jl}^1) + (1 - \pi) \Phi_2(A_j^2, A_l^2; \rho_{jl}^2),$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the bivariate standard normal CDF with correlation ρ . This gives $\binom{M}{2}$ moment conditions.

(M3) **Trivariate Probabilities:** Analogously, the joint probability that any three proxies j, l, n , where $j \neq l \neq n$, is

$$\Pr(z_j^k = z_l^k = z_n^k = 1) = \pi \Phi_3(A_j^1, A_l^1, A_n^1; \rho_{jl}^1, \rho_{jn}^1, \rho_{ln}^1) + (1 - \pi) \Phi_3(A_j^2, A_l^2, A_n^2; \rho_{jl}^2, \rho_{jn}^2, \rho_{ln}^2), \quad (29)$$

where Φ_3 is the trivariate standard normal CDF. This gives $\binom{M}{3}$ additional moment conditions.

The structural unknowns per domain are the M loadings (λ_m^k) , M thresholds (τ_m^k) , and the three free mixture parameters (π, ω_k^1, s_k^1) , for $2M + 3$ parameters in total. Together, moments (M1)-(M3) supply $M + \binom{M}{2} + \binom{M}{3}$ equations. The system is identified when $M \geq 4$, which yields 14 equations for 11 unknowns.³⁰ Marginals and pairwise moments alone supply $M + \binom{M}{2} = 10$ equations at $M = 4$, one short of the 11 unknowns, so identification is impossible without at least one third-order moment. Order counting is necessary but not sufficient; the lemma below supplies the recovery argument.

³⁰ $M=4$ is the minimum necessary number of binary indicators to identify the parameters. At $M=3$, we have 9 free parameters and only 7 equations, so the system is underidentified.

(*Intuition for what moments are doing the work*). We first show the mechanism in a more restrictive single Gaussian benchmark, in which $\ln x_0^k \sim N(0, 1)$.³¹ Define the objects,

$$a_m \equiv \frac{\tau_m^k}{\sqrt{(\lambda_m^k)^2 + 1}}, \quad c_m \equiv \frac{\lambda_m^k}{\sqrt{(\lambda_m^k)^2 + 1}} \in (-1, 1).$$

Then the marginal probability is $P_m = \Phi(-a_m)$, which implies that $a_m = -\Phi^{-1}(P_m)$, and P_m is recovered directly from the data. The pairwise probability $P_{jl} = \Phi_2(-a_j, -a_l; \rho_{jl})$ is strictly increasing in ρ_{jl} , and ρ_{jl} is identified given a_j, a_l , and P_{jl} . The key insight is that, using equation (28), the within-component correlation between any two binary proxies can be factored into $\rho_{jl} = c_j \cdot c_l$, the product of the signal to noise ratios of proxy j and l .³² This structure means that any three proxies, (j, l, n) , recovers a mapping from the within-component correlation to c_j via

$$c_j = \sqrt{\rho_{jl} \cdot \rho_{jn} / \rho_{ln}}.$$

With $M \geq 4$ we can recover c_m for every $m \in \{1, \dots, M\}$ given ρ_{jl} are identified for each $i, j \in \{1, \dots, M\}$ where $i \neq j$. The structural parameters are then recovered directly by

$$\lambda_m^k = \frac{c_m}{\sqrt{1 - c_m^2}}, \quad \text{and} \quad \tau_m^k = \frac{a_m}{\sqrt{1 - c_m^2}}.$$

Under the two-component mixture, the same intuition applies, but each pairwise moment now reflects a weighted average of the joint probabilities from the two latent components. As a result, the observed correlations combine information about the loadings, thresholds, mixture weights, and $(\rho_{jl}^1, \rho_{jl}^2)$. Consequently, moments (M1)–(M2) alone are no longer sufficient to separately identify these parameters, as the number of parameters now exceed the number of equations provided by moments (M1) and (M2), and the higher order moments (M3) are required. Thus, the single Gaussian benchmark example illustrates both which features of the data relate to the measurement parameters (pairwise co-movements fix loadings and marginals fix thresholds) and why an additional free dimension (the second mixture component) requires adding the third order-moments.

Lemma 1 (Single-domain identification). *Fix a domain k with $M \geq 4$ binary proxies satisfying the measurement-*

³¹Formally, this is the boundary of the parameter space where the two components collapse into a single component Gaussian distribution. This benchmark case is ruled out by condition (C5).

³²The signal to noise ratio captures the correlation between proxy m 's latent index (z_m^{k*}) and the underlying factor ($\ln x_0^k$). This factorization conveniently decomposes the within-component correlation into two distinct parts, one depending solely on j and the other solely on l . This decomposition is valid because, under the assumption of mutually independent measurement errors, all covariance between the measures is attributable to the shared latent factor.

error restrictions (V1)–(V3) and conditions (C1)–(C5). Then $(\pi, \omega_k^1, s_k^1, \{\lambda_m^k, \tau_m^k\})$ are generically locally identified from moments (M1)–(M3) under the stated normalizations and (ω_k^2, s_k^2) are identified via equation (27).

Before proving Lemma 1, we describe some useful preliminaries on real-analytic functions that underpin the identification argument.

Definition 1 (Real analyticity). *A function $f : U \rightarrow \mathbb{R}$ is real analytic on a connected open set $U \subset \mathbb{R}^n$ if, for every point $x_0 \in U$, there exists an open neighborhood of x_0 on which f is represented by a convergent Taylor series*

$$f(x) = \sum_{\alpha=0}^{\infty} \frac{D^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha,$$

where D^α denotes the corresponding partial derivative, and the series converges to $f(x)$ for all x sufficiently close to x_0 .³³

Property 1 (Zero-set of a real-analytic function). *Let f be a real analytic and not identically zero on a connected open set U . Then the zero set $\{x \in U : f(x) = 0\}$ has Lebesgue measure zero. Equivalently, if f vanishes on any open subset of U , then $f \equiv 0$ on all of U (Krantz and Parks, 2002, Corollary 1.2.6).*

These two ingredients underpin our moment-based identification strategy. By establishing that the Jacobian matrix of the moment map has full rank at some parameter value θ_0 , the Inverse Function Theorem guarantees that the moment map is locally invertible near that point, so the parameters are locally identified there. The difficulty is that verifying full rank across the entire parameter space is analytically intractable. This is where real-analyticity has bite. Property 1 implies that a real-analytic function that is nonzero at a single point can only vanish on a set of Lebesgue measure zero, and therefore cannot be identically zero. By showing the moment equations are real analytic in the parameters, it follows that their Jacobian determinant is also real analytic. Hence, a single point at which this determinant is nonzero establishes that identification fails only on a negligible exceptional set. Conditions (C1)–(C5) are designed to exclude this exceptional set, ensuring the parameters are identified across the entire admissible parameter space.

Proof of Lemma 1: The argument proceeds in three steps:

1. (*Reduction*). By the conditional-independence restrictions (V1)–(V3), the proxies are mutually independent given $\ln x_0^k$. The joint distribution of $(z_1^k, \dots, z_M^k)$ is therefore a one-factor latent-variable

³³Unlike a smooth (C^∞) function, whose Taylor series may fail to converge or may converge to a different function, a real analytic function is completely determined locally by all of its derivatives. It's Taylor series reproduces the function exactly in a neighborhood of every point in its domain.

measurement model with $M \geq 4$ conditionally independent binary proxies. Each proxy is a strictly monotone probit probability function $g_m(x) = \Phi(\lambda_m^k x - \tau_m^k)$.

2. (*Generic local identification via the Jacobian*) Let $\theta = (\pi, \omega_k^1, s_k^1, \{\lambda_m^k, \tau_m^k\})$ collect all free parameters and let $\Psi : \theta \rightarrow \{P_S\}_{|S| \leq 3}$ be the mapping from the parameters to moments defined by moments (M1)-(M3). Each moment, P_S , is an integral of a product of standard normal CDFs against a Gaussian probability density, which makes Ψ a smooth *real-analytic* function in θ on the open set $\{s_k^1 > 0, \pi \in (0, 1)\}$.³⁴

We apply this property to the determinant of an 11-equation subsystem of the Jacobian of Ψ , $D\Psi$, which inherits real analyticity from Ψ .³⁵ Identification requires only that $D\Psi$ has full column rank (i.e., 11 independent moment conditions and the remaining three are overidentifying).³⁶ Verifying that $\det D\Psi(\theta_0) \neq 0$ at a single parameter value, θ_0 , establishes that $\det D\Psi$ is not zero everywhere. The real-analyticity of $\det D\Psi$ then implies that its zero-set (where $\det D\Psi(\theta_0) = 0$ and the parameters are not identified) is a thin subset of the parameter space with Lebesgue measure zero. Hence, local identification fails only on this exceptional set. Everywhere else, the Jacobian has full rank and is locally invertible. Thus, local identification holds generically, with any violations confined to a set of parameter values that can, in principle, be characterized explicitly.

Parameter sets where local identification fails are knife edge cases. The single-Gaussian benchmark provides a guide for places in the parameter space where $\det D\Psi = 0$, conditions (C1) - (C5) rule out such cases. Condition (C5) excludes the boundary where the mixture collapses, so the true parameter lies in the interior where local injectivity holds; (C4) selects a unique representative among the two label permutations that the moments cannot distinguish; and the normalizations $\mathbb{E}(\ln x_0^k) = 0$, $\text{Var}(\ln x_0^k) = 1$ together with (C3) eliminate the location, scale, and sign freedoms; conditions (C1)-(C2) rule out two proxies with identical probability functions and proxies that are either perfectly correlated or uncorrelated ($|\rho_{jl}| \in \{0, 1\}$)

3. Thus, given the normalizations and the generic local result, θ is generically locally identified on an

³⁴ $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is real analytic as the composition and products of exponential and polynomial functions are real analytic. Φ , the standard normal CDF, is real analytic since the antiderivative of a real-analytic function is real analytic (Krantz and Parks, 2002). Ψ is real analytic because each moment P_S is a multivariate normal CDF that is real-analytic in its arguments (A_j^c, ρ_{jl}^c) (Plackett, 1954), and these are themselves real analytic in θ .

³⁵Each entry of the Jacobian ΨD is a first partial derivative of a component of Ψ , and partial derivatives of real-analytic functions are real-analytic (Krantz and Parks, 2002); hence every entry of $D\Psi$ is real-analytic. The determinant is a polynomial in the entries of $D\Psi$, and sums and products of real-analytic functions are real-analytic; hence $\det D\Psi$ is real-analytic (Krantz and Parks, 2002)

³⁶ $D\Psi$ is a 14×11 matrix, which has full column rank iff at least one of its 11×11 minors is nonzero, so it suffices to work with a single such subsystem.

admissible parameter set Θ , where $\pi \in (0, 1)$, $s_k^1 > 0$, (C1-C5) hold, and where the moments are informative (e.g., where the proxies neither perfectly collinear nor uncorrelated, $|\rho_{jl}| \in (0, 1)$, or where the marginals do not degenerate, $|a_m| < \infty$). Identification of (ω_k^2, s_k^2) follows from equation (27). We verify full rank of the Jacobian at the estimated parameter value to show we are not within the rank deficient set that may not be ruled out by (C1)-(C5).

B.1.2 Identification with Mixed Binary and Continuous Measures

Consider a latent factor measured by two continuous measures and $M \geq 4$ binary indicators. The three mixture parameters, thresholds and scales of binary measures, and parameters of the continuous measures must be identified. As established in Section 2.1, the $M \geq 4$ binary measures are sufficient to identify the mixture distribution and all thresholds. We now recover the continuous measurement parameters $(\mu_m^k, \lambda_m^k, \sigma_{v,km}^2)$.

The means of the continuous measures identify the intercepts;

$$E(z_m^k) = \mu_m^k + \lambda_m^k E(\ln x_0^k) = \mu_m^k$$

because $E(\ln x_0^k) = 0$ by normalization.

The covariance between the two continuous measures identifies the product of loadings;

$$\text{Cov}(z_1^k, z_2^k) = \lambda_1^k \lambda_2^k \text{Var}(\ln x_0^k) = \lambda_1^k \lambda_2^k,$$

where the last equality uses $\text{Var}(\ln x_0^k) = 1$.

The joint distribution of (z_1^k, z_2^k) conditional on component j is bivariate normal with component-specific means and variances that are functions of $(\mu_m^k, \lambda_m^k, \omega_k^j, s_k^j)$. Provided that the two mixture components are separated in the latent factor ($\omega_k^1 \neq \omega_k^2$, which holds under a non-degeneracy condition that the two components have distinct means), the loadings are identified by

$$\lambda_1^k = \frac{\omega_{z1}^1 - \omega_{z1}^2}{\omega_k^1 - \omega_k^2}, \quad \lambda_2^k = \frac{\omega_{z2}^1 - \omega_{z2}^2}{\omega_k^1 - \omega_k^2},$$

where ω_{zm}^j denotes the component- j mean of continuous measure m , which is identified from the mixture. Finally, the measurement error variance is recovered from

$$\text{Var}(z_m^k) = (\lambda_m^k)^2 + \sigma_{v,km}^2.$$

B.1.3 Identification of Cross-Domain Covariances

It remains to identify the off-diagonal elements of Σ^j , which capture the dependence among initial health stocks and latent heterogeneity. Consider two distinct latent factors, x^k and x^l ($k \neq l$), each observed by at least one binary or continuous proxy. Under the conditional independence assumption, measurement errors are independent across domains given w_0 . Hence all cross-domain co-movement in the data is attributable to the correlation of (x^k, x^l) within each mixture component.

Formally, for a binary proxy of x^k and a binary proxy of x^l , the cross-domain pairwise probability satisfies a bivariate normal mixture in which the within-component correlation is a function of the off-diagonal element ρ_{kl}^j . Given identified marginal mixture distributions and thresholds (from Steps 1 and 2), the cross-domain pairwise probabilities provide moment equations in ρ_{kl}^j . With sufficient cross-domain proxies, these equations identify all off-diagonal elements of Σ^j .

B.2 Identification of the Transition Process and Remaining Model Parameters

Taking as given the measurement parameters and joint distribution of initial latent endowments identified in Step 1, we establish identification of the health formation technology and the remaining behavioral equations. In line with the estimation strategy, the identification strategy uses an auxiliary model based on principal components of the observed binary measures. Because the latent states are not directly observed, we work with a mapping from structural parameters to the moments of an observable auxiliary statistic, following the logic of indirect inference.

B.2.1 The PCA Index as an Auxiliary Statistic

Let $x_t = (x_t^p, x_t^c, x_t^{mh})'$ denote the latent health domains and let b_0 denote time-invariant heterogeneity. For each health domain $k \in \{p, c, mh\}$, we observe a vector of measures $z_t^k = (z_{t,1}^k, \dots, z_{t,M_k}^k)'$ that are exclusively generated by x_t^k through the measurement system.

Fix a latent domain x_t^k and suppose it is measured by M binary proxies

$$z_{t,m}^k = \mathbb{1}[\lambda_m^k \ln x_t^k + v_{t,m}^k \geq \tau_m^k], \quad v_{t,m}^k \sim N(0, 1).$$

Conditional on x^k , each observed indicator satisfies

$$P(z_{t,m}^k = 1 \mid x_t^k) = \Phi(\lambda_m^k \ln x_t^k - \tau_m^k),$$

where Φ is the Standard Normal CDF. This conditional probability is a strictly increasing function of $\ln x_t^k$. Under the conditional independence of measurement errors, all co-movement across binary indicators is driven by x_t^k .

Let

$$PC_t^k = \sum_m g_m^k z_{t,m}^k$$

denote the first principal component of the observed binary vector. Because PCA is applied to nonlinear threshold variables rather than to linear measurements of $\ln x_t^k$, the first principal component does not recover the latent factor exactly. Instead, its conditional expectation satisfies

$$E(PC_t^k | x_t^k) = \sum_{m=1}^M g_m^k \Phi(\lambda_m^k \ln x_t^k - \tau_m^k) \equiv h(x^k), \quad (30)$$

where $h(\cdot)$ is a known, strictly increasing function once the PCA weights $\{g_m^k\}$ and the identified thresholds $\{\tau_m^k\}$ are fixed. Thus PC_t^k is a monotone but nonlinear proxy for $\ln x_t^k$. We use PC_t^k as an auxiliary statistic for the latent state, not as a direct estimate of it.

When measures are continuous, the mapping from the principal component to the respective latent factor is clearly one-to-one and injective;

$$PC_t^k = \sum_m g_m^k z_{t,m}^k = \sum_m g_m^k (\mu_m^k + \lambda_m^k \ln x_t^k + v_{t,m}^k). \quad (31)$$

We focus this section on a case more similar to our setting in which we have discrete measures exclusively for many latent health domains. For identification using continuous linear measures, see Cunha et al. (2010); Agostinelli and Wiswall (2025).

B.2.2 Identification of Transition Parameters via Indirect Inference

The structural parameters are identified through the unique reduced-form implications they generate for the PCA-based auxiliary moments. For each domain k , define the auxiliary transition equation as

$$PC_{t+1}^k = \tilde{a}_0^k + \tilde{w}_t^\top \tilde{\beta}^k + \frac{1}{2} \tilde{w}_t^\top \tilde{B}^k \tilde{w}_t + \tilde{c}_e^k e_t + \tilde{v}_{t+1}^k,$$

where

$$\tilde{w}_t = (PC_t^p, PC_t^c, PC_t^{mh}, PC_0^b, I_t)'$$

collects the principal component indices and observed state variables (I_t), e_t denotes the observed employment status, and $\tilde{\nu}_{t+1}^k$ is a mean-zero innovation independent of $\tilde{w}_t$.

Estimating this auxiliary model by OLS yields a reduced-form parameter vector $\beta^A(\theta)$, which is a function of the structural parameters θ . The structural parameters are identified if the auxiliary mapping $\theta \rightarrow \beta^A(\theta)$ is injective over the admissible parameter space; i.e.,

$$\beta^A(\theta_1) = \beta^A(\theta_2) \Rightarrow \theta_1 = \theta_2.$$

The economic content of this condition is as follows. The structural parameter a_0^k shifts the unconditional location of the latent process, altering the average probability of positive responses across indicators and hence the mean of PC_t^k . The own-lag coefficient in β^k changes the persistence of x_t^k , which maps directly into the autocorrelation of the binary indicators and the serial correlation of PC_t^k . The innovation variance $\sigma_{\nu^k}^2$ affects the cross-sectional dispersion and intertemporal movement of the latent process, which are reflected in the variance and transition moments of PC_t^k . Because these three features, location, persistence, and dispersion, affect distinct moments of the observable auxiliary statistic, perturbations in the structural parameters generate distinguishable changes in $\beta^A(\theta)$.

B.2.3 Identification of Behavioral Equations

Conditional on the identified latent dynamics, the remaining model parameters governing health investment, employment, mortality, and attrition are identified from the conditional distributions of observed outcomes given the current state vector. Because each outcome depends on the current latent factors only through

$$w_t = (\ln x_t^p, \ln x_t^c, \ln x_t^{mh}, b_0, I_t)',$$

and because the PCA index, $h(x_t^k)$, is a monotone transformation of each latent factor with known parameters, the coefficients in the reduced-form regressions of observed outcomes on

$$(PC_t^p, PC_t^c, PC_t^{mh}, PC_0^b)$$

bear a one-to-one relationship to the underlying structural parameters, up to the rescaling implied by the measurement system.

B.2.4 Identification of the Structural Error Covariance Matrix

The structural model allows the idiosyncratic shocks $(\epsilon_t^p, \epsilon_t^c, \epsilon_t^{mh}, \epsilon_t^I, \epsilon_t^L, \epsilon_t^d, \epsilon_t^a)$ to be freely correlated, introducing a 7×7 covariance matrix Σ_ϵ as additional parameters. We identify Σ_ϵ using the covariance structure of the auxiliary residuals recovered from the regressions in Sections B.2.3 and B.2.2.

Let $\tilde{\nu}_{t+1}^k$ denote the residual from the auxiliary PCA regression for domain k (Section B.2.3). Because measurement errors are independent across domains (V3), all cross-domain covariance in $(\tilde{\nu}^p, \tilde{\nu}^c, \tilde{\nu}^{mh})$ is attributable solely to correlations among the structural health shocks $(\epsilon^p, \epsilon^c, \epsilon^{mh})$. Specifically,

$$\text{Cov}(\tilde{\nu}_{t+1}^k, \tilde{\nu}_{t+1}^l) = \delta_k \delta_l \text{Cov}(\epsilon_{t+1}^k, \epsilon_{t+1}^l), \quad (32)$$

where δ_k is the known scaling factor implied by the proxy mappings (31) and (30).³⁷ The three pairwise cross-covariances of auxiliary residuals therefore identify the off-diagonal elements of the $(\epsilon^p, \epsilon^c, \epsilon^{mh})$ block of Σ_ϵ . The diagonal elements follow by subtracting the measurement noise contribution,

$$\text{Var}(\epsilon^k) = \frac{1}{\delta_k^2} \left(\text{Var}(\tilde{\nu}^k) - \sum_m (g_m^k)^2 \sigma_{v,km}^2 \right), \quad (33)$$

where $\sum_m (g_m^k)^2 \sigma_{v,km}^2$ is a known constant.

The residual from the auxiliary regression of I_t on the PC indices is a direct proxy for ϵ_t^I . Its sample cross-covariance with each $\tilde{\nu}_{t+1}^k$ identifies $\text{Cov}(\epsilon^k, \epsilon^I)$ up to the same rescaling δ_k .

For employment, mortality, and attrition the latent-index residuals are not directly observed. Identification proceeds through the joint distribution of each binary outcome and the PC health indices. For instance, consider the parameters determining employment, L_t . Conditional on the latent state, the joint distribution of (L_t, PC_{t+1}^k) is a bivariate normal mixture whose within-component cross-correlation is a function of $\text{Cov}(\epsilon^L, \epsilon^k)$. Since the marginal distributions of L_t and PC_{t+1}^k are already identified, the cross-covariance parameters $\text{Cov}(\epsilon^L, \epsilon^k)$ and $\text{Cov}(\epsilon^d, \epsilon^k)$ are identified from the additional pairwise moments. The same argument identifies $\text{Cov}(\epsilon^L, \epsilon^d)$, $\text{Cov}(\epsilon^L, \epsilon^a)$, and $\text{Cov}(\epsilon^d, \epsilon^a)$ from the joint distribution of (L_t, d_t) , (L_t, a_t) , and (d_t, a_t) given the PC indices.

Together, these moment conditions provide a mapping from Σ_ϵ to the covariance structure of observable auxiliary residuals and outcome pairs.

³⁷For continuous measures, the PCA index is exactly linear in $\ln x^k$, so the scaling factor is simply $\delta_k = \sum_m g_m^k \lambda_m^k$. For binary measures, the relationship is nonlinear. Differentiating the conditional expectation of PC^k with respect to $\ln x^k$ and evaluating at the mean gives: $\delta_k = \sum_m g_m^k \lambda_m^k \Phi(\lambda_m^k E(\ln x^k) - \tau_m^k)$.

C Additional Results

C.1 How b_0 relates to its measures

Table 9: Simulated Conditional Mean of Latent Heterogeneity, b_0

Education		Mother's Education		SES	
None	-1.229	None	-0.629	Poor	-0.837
Above Bachelors	1.223	College +	1.676	Rich	1.186
Self Reported Health in Childhood		Ever-Smoked		Parents Drink/ Drug	
Poor	-1.779	Yes	-0.514	Yes	-0.925
Excellent	0.412	No	0.524	No	0.265

Note: Table reported the average simulated value of b_0 conditional on the simulated realizations of the measures used to recover it, where the unconditional distribution of b_0 is normalized to be mean zero variance one.

The latent distribution of b_0 is estimated using retrospective information on individuals and their parents that relates to health productivity over the life cycle. To provide intuition for how b_0 maps into individual characteristics, Table 9 reports the mean of b_0 conditional on selected realizations of the variables used to recover its distribution. The statistics indicate that b_0 is positively correlated with both the individual's and their mother's education, as well as with childhood socioeconomic status and self-reported health during childhood. In contrast, b_0 is lower among individuals who have smoked before the age of 55 and among those whose parents engaged in excessive drinking or drug use. Taken together, these correlations suggest that b_0 captures persistent heterogeneity related to early-life human capital and health environments.

C.2 Information Content of Measures

This section reports the information content of measures and the extent of measurement error. Our measurement system contains three distinct types of measures: continuous, (ordered) categorical, and discrete. For log-linear continuous measures, the variance can be decomposed as

$$Var(z_{t,m}^k) = (\lambda_t^k)^2 Var(\ln x_t^k) + Var(v_{t,m}^k).$$

The fraction of the variance of $z_{t,m}^k$ coming from the latent factor is

$$s_{t,m}^k = \frac{(\lambda_t^k)^2 \text{Var}(\ln x_t^k)}{(\lambda_t^k)^2 \text{Var}(\ln x_t^k) + \text{Var}(v_{t,m}^k)}.$$

Similarly, binary and categorical measures are generated by a latent index $z_{t,m}^{k*} = \lambda_m^k \ln x_t^k + v_{t,m}^k$, where $\text{Var}(v_{t,m}^k) = 1$. The fraction of the variance of $z_{t,m}^k$ coming from the latent factor is

$$s_{t,m}^k = \frac{(\lambda_t^k)^2 \text{Var}(\ln x_t^k)}{(\lambda_t^k)^2 \text{Var}(\ln x_t^k) + 1}$$

The proportion of the variance in measures attributed to the latent factor, i.e., the information content of the measures in the initial period, is reported in Table 10. We report for the initial period as this is precisely where the variance of the latent factors are normalized to one, and is the period in which we identify and estimate the parameters of the measurement system.

Table 10: Signal and Noise Content of Measurement Variables at t=0

Measure	Signal	Noise	Measure	Signal	Noise
<i>Cognitive Measures</i>			<i>Mental Health Measures</i>		
Immediate Word Recall	0.884	0.116	Was Happy	0.747	0.253
Delayed Word Recall	0.779	0.221	Felt Depressed	0.802	0.198
Serial 7's	0.095	0.905	Everything an Effort	0.654	0.346
Backwards Count	0.162	0.838	Restless Sleep	0.392	0.608
Difficulty Using a Map	0.194	0.806	Enjoyed Life	0.787	0.213
Difficulty Managing Money	0.045	0.955	Felt Sad	0.762	0.238
Self-Rated Memory	0.105	0.895	Could Not Get Going	0.545	0.455
			Felt Lonely	0.657	0.343
<i>Physical Health Measures</i>			<i>Initial Condition Measures</i>		
Diff. Walk Several Blocks	0.873	0.127	Own Education	0.263	0.737
Diff. Walk One Block	0.862	0.138	Mother's Education	0.162	0.838
Diff. Climb Several Flights	0.725	0.275	SES at Age 16	0.053	0.947
Diff. Climb One Flight	0.769	0.231	Self-Reported Health at 16	0.119	0.881
Diff. Get Up From Chair	0.625	0.375	Ever Smoked	0.026	0.974
Diff. Push/Pull Large Object	0.645	0.355	Parents Drink/Drugs	0.011	0.989
Diff. Lift/Carry 10 lbs	0.613	0.387			
Diff. Pick Up a Dime	0.379	0.621			
Diff. Stoop/Kneel/Crouch	0.660	0.340			

Notes: Signal is defined as the proportion of total variance in a measure attributable to the latent factor, $\lambda_m^2 \text{Var}(x)/(\lambda_m^2 \text{Var}(x) + \sigma_{v_m}^2)$. Noise is the complementary share attributable to measurement error.

C.3 Deriving Investment Measure

HRS respondents are surveyed on a set of health behavior questions. Respondents report the frequency of vigorous physical activity, such as aerobics, running, swimming, or cycling, and are classified as regular

exercisers if they engage in such activities three or more times per week. Respondents also report whether they consume alcohol and, if so, the frequency and typical quantity consumed. From these responses we construct a binary indicator for mild alcohol consumption and a separate indicator for excessive drinking, defined as consuming more than four drinks per day on drinking occasions. A final binary indicator records current cigarette smoking.

We aggregate these four measures into a composite health investment index using the first principal component. Table 11 reports the component loadings. Alcohol consumption and vigorous exercise load positively, while excessive drinking and smoking load negatively, so the index is increasing in health-promoting behaviors and decreasing in harmful ones.

Table 11: Principal Component Loadings for the Health Investment Index

Variable	Loading
Excessive Drink	−0.618
Drink	0.599
Vigorous Exercise	0.259
Smokes	−0.439

C.4 Time allocation Regressions

This section provides descriptive evidence on a possible mechanism underlying the main-text finding that lagged physical health is negatively associated with subsequent cognitive-health formation. One interpretation of this negative cross-domain relationship is compensatory time allocation. That is, as physical health deteriorates, individuals may substitute away from physically demanding activities and toward activities that help preserve cognitive health. We examine this possibility using the Consumption and Activities Mail Survey (CAMS), a supplemental questionnaire administered to a subsample of HRS respondents. CAMS collects information on respondents' time use and expenditures. We focus on the activities module, which records time spent on a range of leisure and daily activities.

Table 12 reports OLS estimates relating changes in cognitively stimulating activities to changes in physical health. We consider three continuous outcomes: hours spent in the previous week reading newspapers or magazines, reading books, and playing cards or solving puzzles. We also consider one ordinal outcome measuring how often respondents report using their mind in daily activities: rarely, sometimes, often, or almost all of the time. For each activity, the dependent variable is the change in the outcome from the

previous survey wave. The key explanatory variable is the change in the first principal component extracted from the dedicated physical-health measures,

$$\Delta PC_{it}^p = PC_{it}^p - PC_{i,t-1}^p,$$

where higher values of PC^p indicate better physical health.

Table 12: Changes in Cognitively Stimulating Activities and Physical Health

	Reading Papers/Magazines		Reading Books		Cards/Puzzles		Often Use Mind	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔPC^p	-0.081 (0.091)	-0.091 (0.092)	-0.036 (0.090)	-0.027 (0.092)	-0.106 (0.283)	-0.135 (0.288)	-0.006 (0.009)	-0.004 (0.009)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
N	2,898	2,873	2,885	2,860	2,874	2,849	2,813	2,788

Notes: Standard errors are reported in parentheses below the estimates. Controls include the first principal component extracted from the cognitive-health measures, the first principal component extracted from the mental-health measures, respondent's education, mother's education, socioeconomic status at age 16, self-reported health at age 16, smoking history, whether the respondent's parents drank or used drugs to an extent that caused problems for the family, marital status, total non-housing wealth at age bin 55/56, total work experience at age bin 55/56, and age.

The point estimates are consistent with the compensatory-time-allocation interpretation. In all specifications, improvements in physical health are associated with reductions in cognitively stimulating activities; equivalently, declines in physical health are associated with increases in these activities. The signs are stable after adding controls for other health domains, early-life conditions, socioeconomic characteristics, marital status, wealth, work experience, and age. The estimates are imprecise, however, and should therefore be interpreted as suggestive at best. Taken together, the results provide descriptive support for the idea that physical-health deterioration may induce behavioral responses that partially offset cognitive decline.

D Estimation Details

D.1 Step 1: Distribution of Initial Endowments and Measurement Parameters

Recall that the distribution of initial conditions is

$$F_{w_0} = \pi \Phi(\omega^1, \Sigma^1) + (1 - \pi) \Phi(\omega^2, \Sigma^2),$$

$$\text{where } \omega^j = \begin{bmatrix} \omega_1^j \\ \omega_2^j \\ \omega_3^j \\ \omega_b^j \end{bmatrix} \text{ and } \Sigma^j = \begin{bmatrix} (s_1^j)^2 & \sigma_{12}^j & \sigma_{13}^j & \sigma_{1b}^j \\ \sigma_{12}^j & (s_2^j)^2 & \sigma_{23}^j & \sigma_{2b}^j \\ \sigma_{13}^j & \sigma_{23}^j & (s_3^j)^2 & \sigma_{3b}^j \\ \sigma_{1b}^j & \sigma_{2b}^j & \sigma_{3b}^j & (s_b^j)^2 \end{bmatrix}.$$

Let

$$\sigma^j = (\sigma_{12}^j, \sigma_{13}^j, \sigma_{1b}^j, \sigma_{23}^j, \sigma_{2b}^j, \sigma_{3b}^j) \text{ and } s^j = (s_1^j, s_2^j, s_3^j, s_b^j), \text{ and } \omega^j = (\omega_1^j, \omega_2^j, \omega_3^j, \omega_b^j).$$

We denote the vector of parameters to estimate in the initial step by

$$\theta_0 = \{\{\omega^j, \sigma^j, s^j\}_{j=1,2}, \{\mu_m^k, \lambda_m^k, \tau_{j,m}^k, \sigma_{v_m^k}\}_{k \in \{p,c,mh,b\}, m \in 1, \dots, M_k}\}.$$

We denote the set of observed measures in period 0 as

$$z_0 = (z_{0,1}^p, \dots, z_{0,M^p}^p, z_{0,1}^{mh}, \dots, z_{0,M^{mh}}^{mh}, z_{0,1}^c, \dots, z_{0,M^c}^c, z_1^b, \dots, z_{M^b}^b).$$

The likelihood function is constructed from the joint distribution of initial heterogeneity and the latent health domains, w_0 . As w_0 is latent, we write the joint density of observed measures at $t = 0$ conditional on w_0 as

$$f(z_0|w_0, \theta_0) = \prod_{j=1}^{M^b} f_{z_{0,j}^b}(z_{0,j}^b|b_0, \theta_0) \prod_{k \in \{p,mh,c\}} \prod_{j=1}^{M^k} f_{z_{0,j}^k}(z_{0,j}^k|x_0^k, \theta_0),$$

where $f_{z_{0,j}^k}(z_{0,j}^k|b_0, \theta_0)$ is the density of initial latent heterogeneity measure $z_{0,j}^b$ for $j = 1, \dots, M^b$. The joint density of initial endowments, $w_0 = (\ln x_0^p, \ln x_0^{mh}, \ln x_0^c, b_0)$ given θ_0 is $f_{w_0}(w_0)$.

The likelihood contribution for individual i , given parameter values θ_0 can be written as

$$L_i(\theta_0) = \int \cdots \int f_{w_0}(w_0) f(z_0|w_0, \theta_0) dw_0.$$

Initial heterogeneity and the health domains are latent and need to be integrated out of the likelihood. The integrals are taken over the latent initial heterogeneity, b_0 , and latent health vector $x_0 = (x_0^p, x_0^c, x_0^{mh})$.

We estimate $\hat{\theta}_0$ by maximum simulated likelihood. The observed-data likelihood integrates over the period-0 latent health stocks, $\{x_0^p, x_0^c, x_0^{mh}\}$, and initial heterogeneity, b_0 . We approximate this integral with Monte Carlo draws from the model-implied joint distribution of latent states. The following steps simulate the latent states and evaluate the likelihood for a single individual.

1. *Simulate Latent variables:* For a candidate guess of parameters $\tilde{\theta}_0$, draw R independent replications of the latent state,

$$w_0^{(r)} = (\tilde{x}_0^p, \tilde{x}_0^c, \tilde{x}_0^{mh}, \tilde{b}_0)' \sim F_{w_0}(\tilde{\theta}_0), \quad r = 1, \dots, R.$$

2. *Evaluate the Likelihood Contribution of a Path:* For each replication r , evaluate the conditional density of the observed measures given the simulated state,

$$L_i^{(r)}(\tilde{\theta}_0) = f(z_0|w_0^{(r)}, \tilde{\theta}_0)$$

3. *Approximate the Likelihood:* The likelihood contribution of individual i is then approximated by averaging over the R simulated paths,

$$\hat{L}_i(\tilde{\theta}_0) \approx \frac{1}{R} \sum_{r=1}^R L_i^{(r)}(\tilde{\theta}_0).$$

We use antithetic acceleration in simulating for variance reduction (Geweke, 1988) and we maximize the approximated log-likelihood with respect to parameters θ_0 using a combination of Nelder-Mead and the Broyden-Fletcher-Goldfarb-Shanno algorithms.

D.2 Step 2: Transition, mortality, employment, investments

Fixing the parameters of the measurement system and joint distribution of initial endowments, we use minimum distance moment matching to pin down the remaining model parameters. To construct the empirical moments used in estimation, we first extract principal components corresponding to each latent health domain using the observed health measures in the data. Specifically, we perform principal component analysis (PCA) separately on groups of measures that load exclusively on a single latent health domain. These correspond to cognitive health, mental health, physical health, and a time-invariant heterogeneity component. Categorical measures are first converted into binary indicators equal to one when the corresponding category is selected. Applying PCA to these binary measures over all time periods yields a set of principal components for each individual i and period t , denoted

$$PC_{it} = \{PC_{it}^p, PC_{it}^c, PC_{it}^{mh}, PC_i^b\}$$

where PC_i^b captures persistent heterogeneity and therefore does not vary over time.

These principal components serve as empirical proxies for the latent health factors and allow us to construct moments that correspond to the model's transition equations. In particular, define $\tilde{w}_t = \{PC_{it}^p, PC_{it}^c, PC_{it}^{mh}, PC_i^b, I_t\}$. We estimate reduced-form analogues of the health production functions of the form:

$$PC_{t+1}^k = a_0^k + \tilde{w}_t' \beta^k + \frac{1}{2} \tilde{w}_t' B^k \tilde{w}_t + c_e^k L_t + \nu_t^k.$$

The coefficients from these regressions, together with the variance of the residuals, form the key moments used to identify the parameters governing the latent health transition process.

A key feature of the procedure is that the mapping from observed measures to principal components is fixed using the data. PCA is performed once using the observed outcomes, producing a vector of loadings, centering constants, and scaling constants. These quantities define a linear transformation from the observed measures z_{it}^{k*} to the principal components, where z_{it}^{k*} refers to the observed set of measures after converting categorical measures into a set of binary indicators. When simulating the model, the same transformation is applied to the simulated measures so that the simulated indices are constructed using the identical mapping,

$$PC_{it}^k = \sum_{m=1}^{M^{k*}} v_m \frac{z_{im}^{k*} - \mu_m}{\sigma_m},$$

where v_m is the calculated PCA weight, and M^{k*} is the number of measures for health domain k , after converting categorical measures to binary indicators. That is, we fix the mapping from the data, and calculate the principal components from the simulated model by

$$PC_{it}^{k,sim} = \sum_{m=1}^{M^{k*}} v_m \frac{\bar{z}_i^{k*,sim} - \mu_m}{\sigma_m}$$

where $\bar{y}_{ik}^{sim}$ is the predicted probability that the simulated binary measure $y_{ik}^{sim} = 1$.

In addition to the health transition equations, we construct moments from behavioral relationships linking health to investment and labor market outcomes. Using the principal component indices as proxies for the latent health domains, we estimate the investment equation

$$I_t = \beta_0^I + \beta_1^I PC_t^p + \beta_2^I PC_t^c + \beta_3^I PC_t^{mh} + \beta_4^I PC^b + \beta_5^I t + \nu_t^I,$$

$$\nu_t^I \sim iid \mathcal{N}(0, \sigma_I^2),$$

where $\nu_t^I \sim N(0, \sigma_{\nu^I})$. The coefficients from this regression, along with the variance of the residual, are included among the moments matched in estimation. We also estimate linear probability models describing employment, mortality, and attrition. We estimate the residuals from each and all of the regressions, and calculate their covariance structure to at moments to pin down covariance of the structural errors.

Finally, we match unconditional moments of key outcomes across time. These include the period-specific means of each health measure, as well as the average levels of employment, health investment, mortality, and attrition in each period. Matching these moments ensures that the model reproduces both the dynamic relationships between health domains and the empirical distributions of the key outcomes observed in the data. Further, we match the mean and variance of PC_t^k , $\forall k \in \{p, c, mh\}$, $t \in \{1, \dots, T_p\}$, as well as the proportion of PC_t^k below the threshold of the lower quartile of the distribution of PC_0^k , $\forall t \in \{1, \dots, T_p\}$, where T_p is the maximum number of periods an individual can be in the sample. These moments help ensure the model reproduces the relevant marginal distributions in the data.

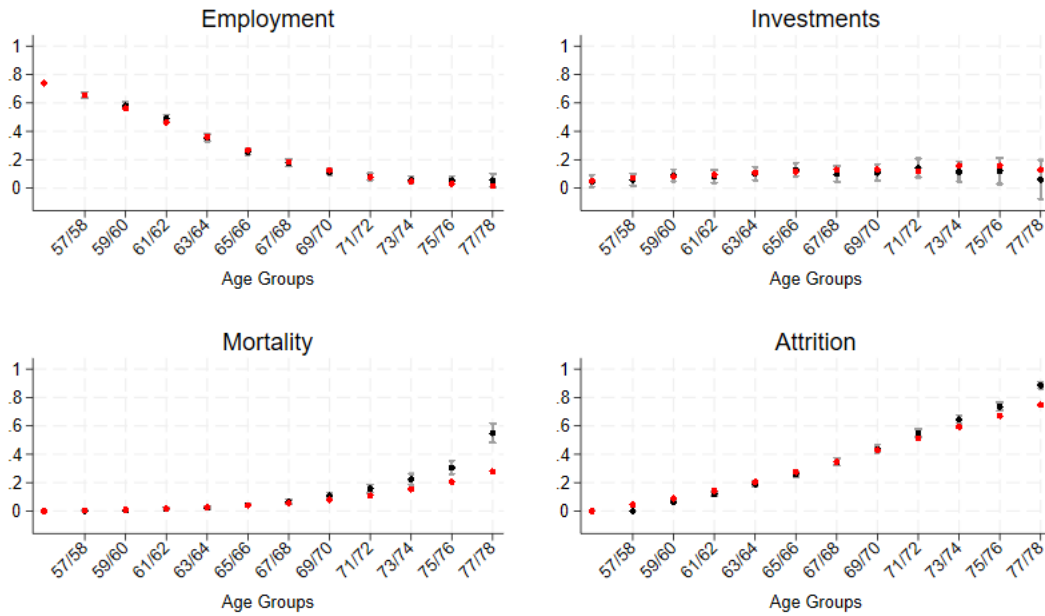
D.3 Smoothed Simulation and Construction of Principal Components

A practical complication arises because the health measures used to construct the principal components are discrete, while the simulated model produces latent states and discrete realizations only after applying threshold rules. If the simulated principal components were constructed directly from simulated binary

outcomes, the resulting objective function would generally be non-smooth in the structural parameters. Small changes in the parameter vector may change whether a simulated latent index crosses a threshold, producing discontinuous changes in the simulated moments. To avoid this problem, we replace simulated discrete outcomes with their simulated conditional probabilities and construct the simulated principal components from these smoothed objects.

D.4 Fit of Employment, Investment, Mortality, and Attrition

Figure 8: Model Fit of Behavioral Outcomes



Note: Note: The x-axis reports age in two-year intervals. Red dots denote predictions from the simulated model, while black dots denote the observed data. The gray lines represent the 95% confidence intervals for the data estimates.

Figure 8 shows how well the predicted behavior simulated from the empirical model fits the average behavior and outcomes observed in the data over age groups. The red lines correspond to the simulated data and the blue lines correspond to the observed data. Overall, we see that the model fits the overall age-trajectories of the observed data fairly well. Although, we note that the simulated data under predicts mortality and attrition at older ages.

D.5 Fit of Moments

Table 13: Model Fit: Health Formation Process - Cognitive

	Simulated	Data	Standard Error	95% CI
$\ln x_t^c$	0.534	0.563	0.008	[0.546, 0.580]
$\ln x_t^{mh}$	0.062	0.050	0.013	[0.025, 0.075]
$\ln x_t^p$	0.037	0.061	0.010	[0.041, 0.081]
b_0	0.121	0.094	0.009	[0.077, 0.111]
$(\ln x_t^c)^2$	-0.014	-0.016	0.004	[-0.024, -0.008]
$(\ln x_t^{mh})^2$	0.005	0.003	0.003	[-0.003, 0.008]
$(\ln x_t^p)^2$	0.000	-0.001	0.003	[-0.007, 0.004]
$(b_0)^2$	0.005	0.002	0.005	[-0.007, 0.011]
L_{t-1}	0.292	0.120	0.023	[0.075, 0.165]
$\ln x_t^c \times \ln x_t^{mh}$	0.000	-0.003	0.004	[-0.012, 0.005]
$\ln x_t^c \times \ln x_t^p$	-0.013	-0.005	0.004	[-0.013, 0.003]
$\ln x_t^{mh} \times \ln x_t^p$	0.002	0.003	0.003	[-0.003, 0.009]
$b_0 \times \ln x_t^c$	-0.007	0.001	0.005	[-0.009, 0.012]
$b_0 \times \ln x_t^{mh}$	0.006	-0.003	0.004	[-0.011, 0.005]
$b_0 \times \ln x_t^p$	0.005	0.004	0.004	[-0.004, 0.012]
I_t	0.046	0.058	0.013	[0.032, 0.083]
$I_t \times \ln x_t^c$	-0.013	-0.010	0.010	[-0.029, 0.008]
$I_t \times \ln x_t^{mh}$	0.001	-0.001	0.007	[-0.015, 0.012]
$I_t \times \ln x_t^p$	0.003	0.011	0.008	[-0.004, 0.027]
$I_t \times b_0$	0.010	-0.006	0.009	[-0.023, 0.012]
Intercept	-0.176	-0.085	0.025	[-0.134, -0.036]
σ_v^2	1.177	1.160	0.008	[1.145, 1.176]

Table 14: Model Fit: Health Formation Process - Mental Health

	Simulated	Data	Standard Error	95% CI
$\ln x_t^c$	0.056	0.079	0.012	[0.057, 0.102]
$\ln x_t^{mh}$	0.460	0.479	0.017	[0.445, 0.513]
$\ln x_t^p$	0.203	0.166	0.014	[0.139, 0.193]
b_0	0.050	0.062	0.012	[0.039, 0.085]
$(\ln x_t^c)^2$	0.005	0.001	0.005	[-0.009, 0.012]
$(\ln x_t^{mh})^2$	0.033	0.017	0.004	[0.009, 0.025]
$(\ln x_t^p)^2$	0.017	0.017	0.004	[0.009, 0.025]
$(b_0)^2$	0.003	0.002	0.006	[-0.010, 0.014]
L_{t-1}	-0.043	-0.075	0.031	[-0.137, -0.014]
$\ln x_t^c \times \ln x_t^{mh}$	-0.020	-0.026	0.006	[-0.037, -0.014]
$\ln x_t^c \times \ln x_t^p$	0.001	-0.006	0.006	[-0.017, 0.005]
$\ln x_t^{mh} \times \ln x_t^p$	-0.010	-0.012	0.004	[-0.020, -0.003]
$b_0 \times \ln x_t^c$	0.002	0.004	0.007	[-0.010, 0.019]
$b_0 \times \ln x_t^{mh}$	-0.010	-0.012	0.005	[-0.023, -0.001]
$b_0 \times \ln x_t^p$	-0.002	-0.011	0.006	[-0.022, 0.000]
I_t	0.085	0.051	0.018	[0.016, 0.087]
$I_t \times \ln x_t^c$	0.024	0.022	0.013	[-0.003, 0.048]
$I_t \times \ln x_t^{mh}$	-0.026	-0.026	0.010	[-0.044, -0.007]
$I_t \times \ln x_t^p$	-0.003	-0.014	0.011	[-0.035, 0.007]
$I_t \times b_0$	-0.008	-0.039	0.012	[-0.063, -0.016]
Intercept	-0.089	-0.025	0.034	[-0.092, 0.042]
σ_v^2	1.584	1.586	0.011	[1.565, 1.607]

Table 15: Model Fit: Health Formation Process - Physical

	Simulated	Data	Standard Error	95% CI
$\ln x_t^c$	0.082	0.061	0.009	[0.044, 0.079]
$\ln x_t^{mh}$	0.093	0.086	0.013	[0.060, 0.113]
$\ln x_t^p$	0.757	0.780	0.011	[0.759, 0.801]
b_0	0.026	0.029	0.009	[0.011, 0.046]
$(\ln x_t^c)^2$	0.001	-0.001	0.004	[-0.009, 0.007]
$(\ln x_t^{mh})^2$	0.002	0.011	0.003	[0.005, 0.017]
$(\ln x_t^p)^2$	0.006	0.001	0.003	[-0.005, 0.007]
$(b_0)^2$	0.005	0.000	0.005	[-0.009, 0.009]
L_{t-1}	0.206	0.076	0.024	[0.029, 0.123]
$\ln x_t^c \times \ln x_t^{mh}$	-0.017	-0.016	0.004	[-0.025, -0.007]
$\ln x_t^c \times \ln x_t^p$	-0.015	0.002	0.004	[-0.006, 0.011]
$\ln x_t^{mh} \times \ln x_t^p$	0.007	0.010	0.003	[0.004, 0.017]
$b_0 \times \ln x_t^c$	0.003	0.003	0.006	[-0.007, 0.014]
$b_0 \times \ln x_t^{mh}$	-0.008	-0.001	0.004	[-0.009, 0.008]
$b_0 \times \ln x_t^p$	-0.006	0.002	0.004	[-0.006, 0.011]
I_t	0.085	0.082	0.014	[0.055, 0.109]
$I_t \times \ln x_t^c$	-0.003	0.008	0.010	[-0.012, 0.027]
$I_t \times \ln x_t^{mh}$	-0.003	-0.005	0.007	[-0.020, 0.009]
$I_t \times \ln x_t^p$	-0.018	-0.029	0.008	[-0.045, -0.012]
$I_t \times b_0$	0.004	0.005	0.009	[-0.013, 0.023]
Intercept	-0.185	-0.166	0.026	[-0.218, -0.115]
σ_v^2	1.225	1.220	0.008	[1.204, 1.236]

Table 16: Fit of Coefficients from Auxiliary Regressions

	Simulated	Data	Standard Error	95% CI
<i>Dependent Variable: Death</i>				
$\ln x_t^c$	-0.001	-0.001	0.001	[-0.002, 0.000]
$\ln x_t^m$	-0.001	-0.001	0.000	[-0.002, 0.000]
$\ln x_t^p$	-0.003	-0.004	0.000	[-0.005, -0.003]
b_0	0.000	0.000	0.001	[-0.001, 0.001]
Age	0.000	0.003	0.001	[0.001, 0.006]
Age ²	0.000	0.000	0.000	[0.000, 0.000]
<i>Dependent Variable: Attrition</i>				
Intercept	0.005	-0.002	0.003	[-0.007, 0.003]
$\ln x_t^c$	0.002	-0.003	0.001	[-0.006, 0.000]
$\ln x_t^m$	0.003	0.002	0.001	[-0.001, 0.004]
$\ln x_t^p$	0.003	0.001	0.001	[-0.001, 0.003]
b_0	0.003	0.004	0.001	[0.002, 0.007]
Age	0.017	0.009	0.001	[0.008, 0.011]
Intercept	0.020	0.017	0.004	[0.010, 0.025]
<i>Dependent Variable: Employment</i>				
$\ln x_t^c$	-0.003	0.009	0.003	[0.004, 0.014]
$\ln x_t^m$	0.011	0.012	0.002	[0.007, 0.016]
$\ln x_t^p$	0.032	0.036	0.002	[0.031, 0.040]
b_0	0.008	0.008	0.003	[0.003, 0.013]
Age	-0.076	-0.079	0.001	[-0.082, -0.076]
Wealth	-0.041	-0.043	0.004	[-0.052, -0.035]
Experience	0.008	0.008	0.000	[0.007, 0.008]
Married	-0.061	-0.065	0.008	[-0.081, -0.050]
Intercept	0.616	0.617	0.013	[0.591, 0.643]
<i>Dependent Variable: Health Investment</i>				
$\ln x_t^c$	0.046	0.046	0.005	[0.035, 0.056]
$\ln x_t^m$	0.016	0.015	0.004	[0.007, 0.024]
$\ln x_t^p$	0.060	0.058	0.004	[0.050, 0.066]
b_0	0.074	0.069	0.005	[0.059, 0.079]
Age	0.016	0.019	0.003	[0.013, 0.024]
Wealth	0.095	0.099	0.009	[0.083, 0.116]
Married	0.093	0.123	0.016	[0.093, 0.154]
Intercept	-0.018	-0.064	0.017	[-0.098, -0.031]
Standard Error	0.862	0.860	0.005	[0.849, 0.870]

Table 17: Fit of Residuals from Auxiliary PCA Regressions

	Simulated	Data	Standard Error	95% CI
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^{mh})$	0.222	0.196	0.018	[0.161, 0.231]
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^p)$	0.111	0.102	0.014	[0.076, 0.129]
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^d)$	0.000	0.002	0.001	[0.000, 0.004]
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^a)$	0.000	-0.001	0.003	[-0.007, 0.004]
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^L)$	-0.015	-0.018	0.005	[-0.027, -0.008]
$Cov(\tilde{\nu}_{t+1}^c, \tilde{\nu}_{t+1}^I)$	-0.064	-0.059	0.009	[-0.077, -0.040]
$Cov(\tilde{\nu}_{t+1}^{mh}, \tilde{\nu}_{t+1}^p)$	0.331	0.341	0.019	[0.304, 0.378]
$Cov(\tilde{\nu}_{t+1}^{mh}, \tilde{\nu}_{t+1}^d)$	0.001	-0.001	0.002	[-0.004, 0.002]
$Cov(\tilde{\nu}_{t+1}^{mh}, \tilde{\nu}_{t+1}^a)$	-0.001	0.001	0.004	[-0.006, 0.008]
$Cov(\tilde{\nu}_{t+1}^{mh}, \tilde{\nu}_{t+1}^L)$	-0.026	-0.023	0.007	[-0.035, -0.010]
$Cov(\tilde{\nu}_{t+1}^{mh}, \tilde{\nu}_{t+1}^I)$	-0.063	-0.044	0.013	[-0.070, -0.019]
$Cov(\tilde{\nu}_{t+1}^p, \tilde{\nu}_{t+1}^d)$	0.001	0.000	0.001	[-0.002, 0.003]
$Cov(\tilde{\nu}_{t+1}^p, \tilde{\nu}_{t+1}^a)$	-0.003	0.002	0.003	[-0.003, 0.008]
$Cov(\tilde{\nu}_{t+1}^p, \tilde{\nu}_{t+1}^L)$	-0.019	-0.039	0.005	[-0.049, -0.029]
$Cov(\tilde{\nu}_{t+1}^p, \tilde{\nu}_{t+1}^I)$	-0.054	-0.072	0.010	[-0.092, -0.053]
$Cov(\tilde{\nu}_{t+1}^d, \tilde{\nu}_{t+1}^a)$	-0.001	-0.001	0.000	[-0.001, 0.000]
$Cov(\tilde{\nu}_{t+1}^d, \tilde{\nu}_{t+1}^L)$	0.000	-0.001	0.000	[-0.002, 0.000]
$Cov(\tilde{\nu}_{t+1}^d, \tilde{\nu}_{t+1}^I)$	-0.001	-0.002	0.001	[-0.004, -0.001]
$Cov(\tilde{\nu}_{t+1}^a, \tilde{\nu}_{t+1}^L)$	0.001	0.002	0.001	[0.000, 0.004]
$Cov(\tilde{\nu}_{t+1}^a, \tilde{\nu}_{t+1}^I)$	0.001	0.001	0.002	[-0.003, 0.005]
$Cov(\tilde{\nu}_{t+1}^L, \tilde{\nu}_{t+1}^I)$	0.002	0.003	0.004	[-0.004, 0.010]

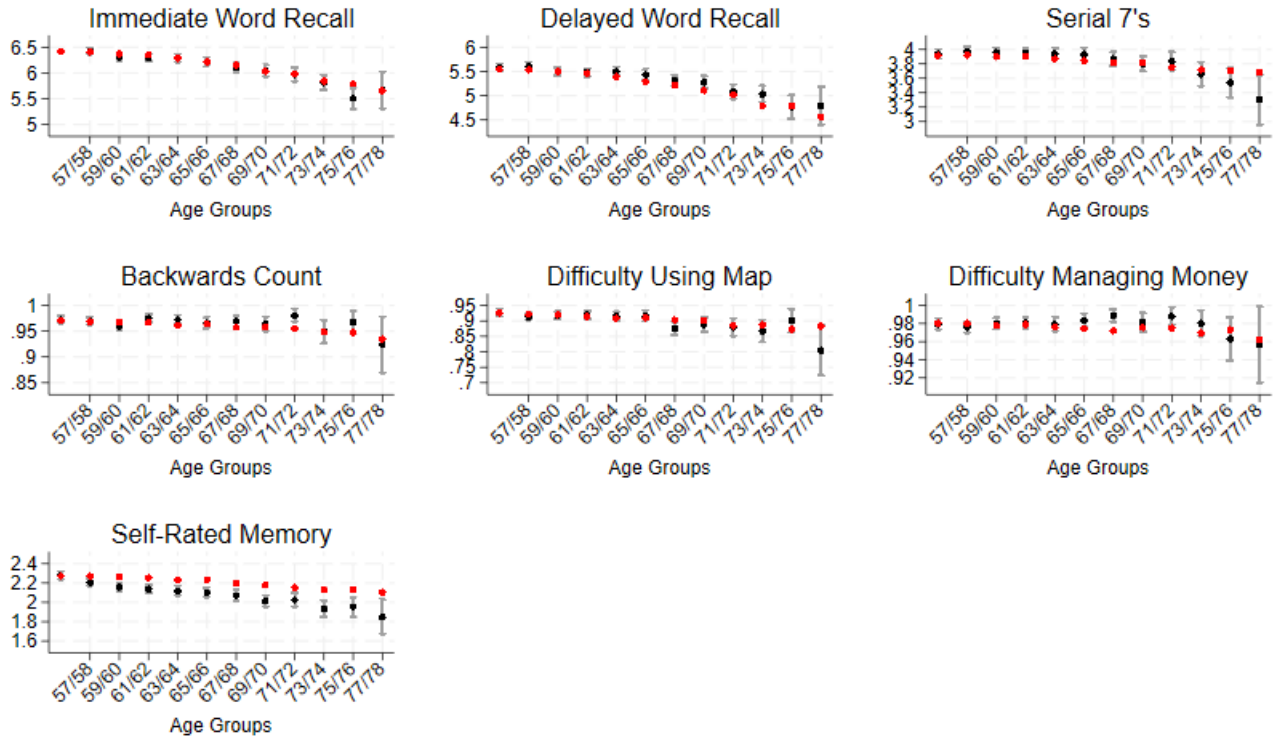
Table 18: Model Fit: Principal Component Trend Regressions

	Simulated	Data	Standard Error	95% CI
<i>Dependent Variable: PC1</i>				
Age	-0.069	-0.070	0.005	[-0.080, -0.060]
Intercept	0.268	0.299	0.025	[0.250, 0.349]
<i>Dependent Variable: PC2</i>				
Age	0.046	0.034	0.006	[0.022, 0.047]
Intercept	-0.256	-0.147	0.032	[-0.210, -0.085]
<i>Dependent Variable: PC3</i>				
Age	-0.059	-0.066	0.007	[-0.079, -0.053]
Intercept	0.164	0.281	0.033	[0.216, 0.347]

Table 19: Fit of Measures of Initial Heterogeneity

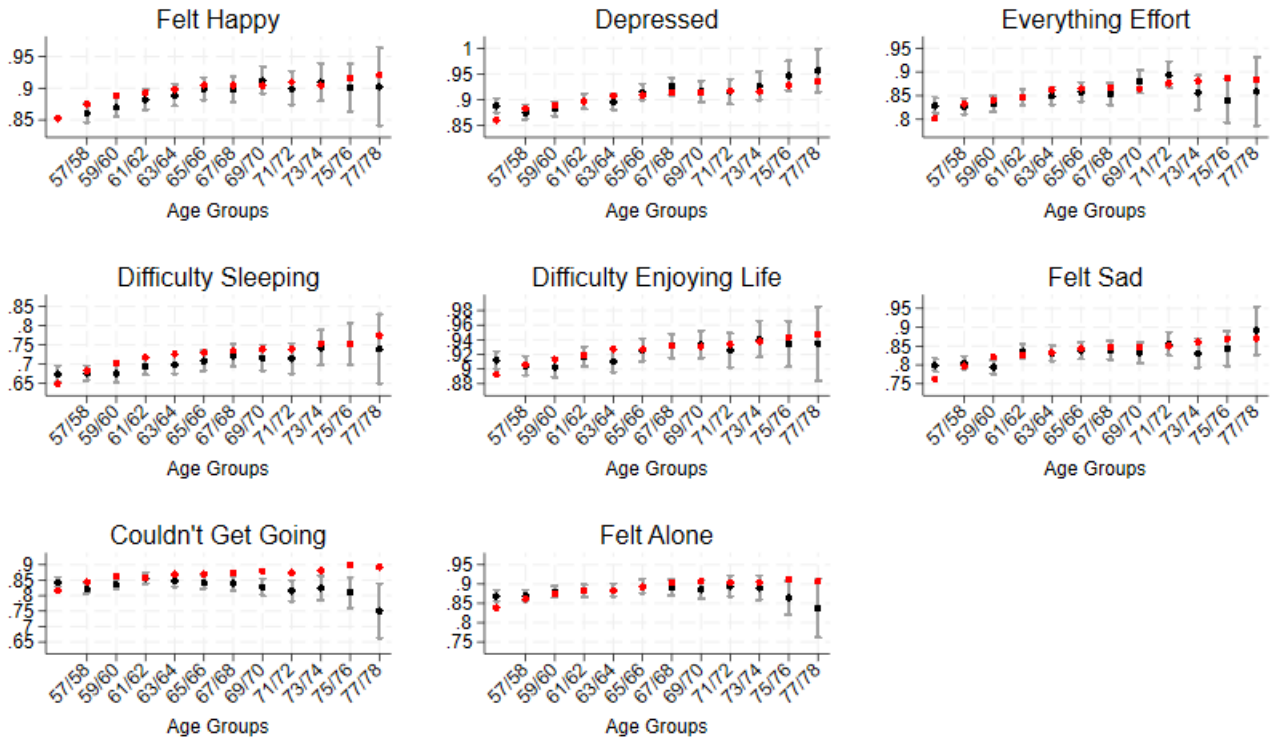
	Simulated	Data	Standard Error	95% CI
Own Education	1.611	1.629	0.025	[1.580, 1.678]
Mothers Education	0.692	0.703	0.015	[0.673, 0.733]
SES in Childhood	0.833	0.840	0.012	[0.817, 0.862]
Self-Reported Health in Childhood	3.294	3.339	0.020	[3.300, 3.379]
Smoke Ever	0.500	0.504	0.011	[0.483, 0.525]
Parents Drink/ Drugs	0.769	0.782	0.009	[0.764, 0.800]

Figure 9: Fit of Measures of Cognitive Health



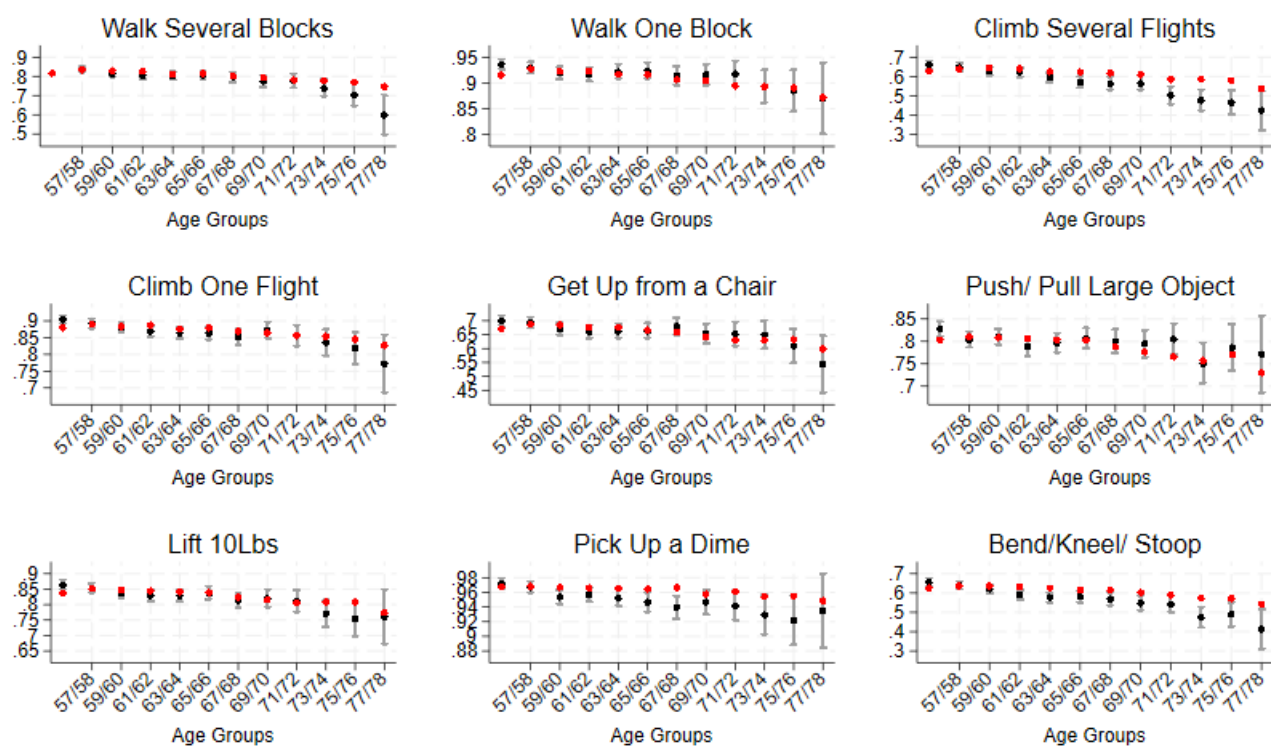
Note: Red dots denote predictions from the simulated model, while black dots denote the observed data. The gray lines represent the 95% confidence intervals for the data estimates.

Figure 10: Fit of Measures of Mental Health



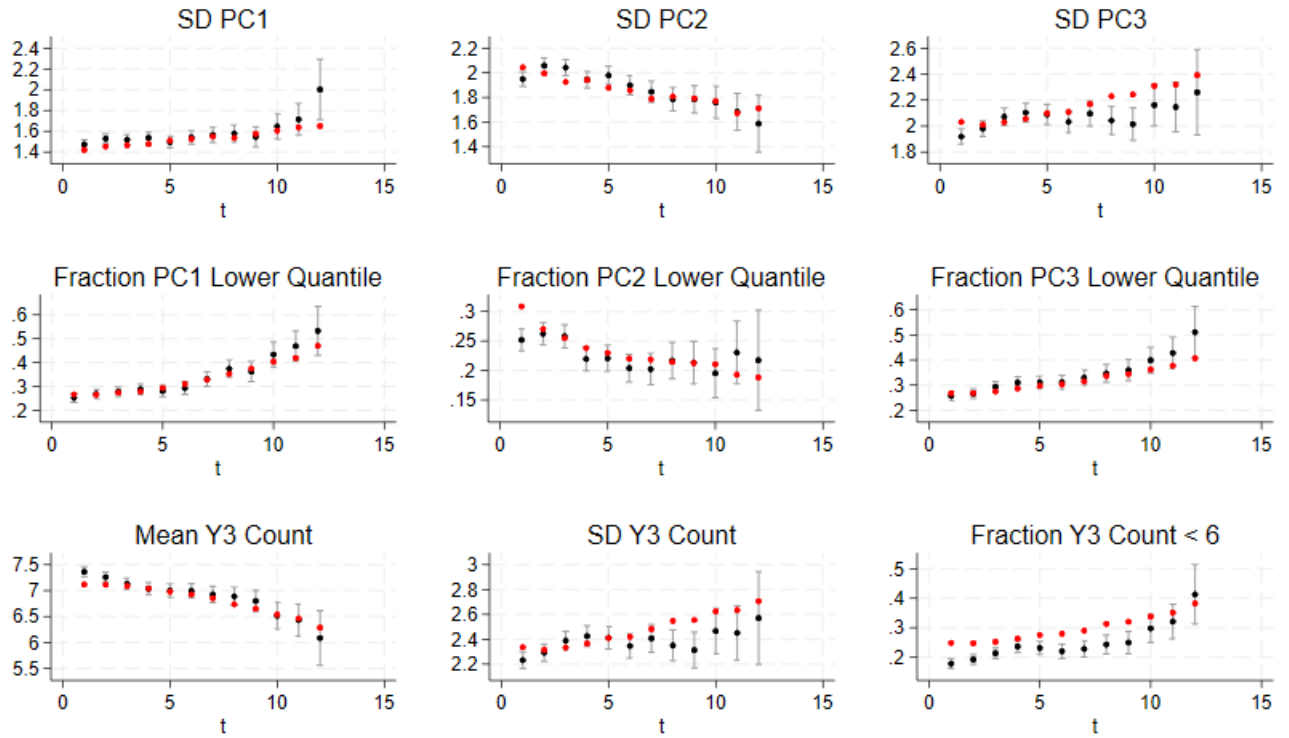
Note: Red dots denote predictions from the simulated model, while black dots denote the observed data. The gray lines represent the 95% confidence intervals for the data estimates.

Figure 11: Fit of Measures of Physical Health



Notes: Red dots denote predictions from the simulated model, while black dots denote the observed data. The gray lines represent the 95% confidence intervals for the data estimates.

Figure 12: Fit of Distributional Moments of Principal Components, and Count Moments for Physical Health



Note: Red dots denote predictions from the simulated model, while black dots denote the observed data. The gray lines represent the 95% confidence intervals for the data estimates.